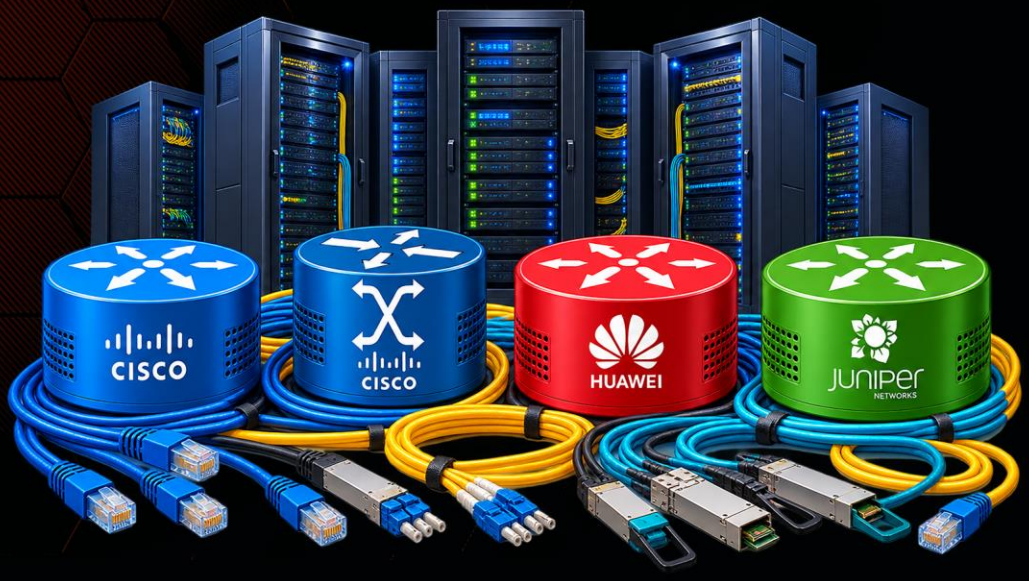
SKYNET
TECHNOLOGIES

SERVICE PROVIDER

>> BUILT TO SURVIVE. DESIGNED TO DOMINATE <<

SKYNET ISP FULL DOCUMENTATION



eve Emulated Virtual Environment Next Generation
JUNIPER NETWORKS
CISCO
HUAWEI
vmware
ubuntu

SKYNET TECHNOLOGIES

PROJECT CONTROL AND FULL AUDIT

COMPANY NAME

SKYNET TECHNOLOGIES

DOCUMENT TYPE

SKYNET ISP FULL AUDIT

DOCUMENT STATUS

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TOPIC INDEX

#	TOPIC	PURPOSE
01	EXECUTIVE ARCHITECTURE SUMMARY	WHAT THE LAB PROVES AND WHY IT MATTERS TO THE CUSTOMER.
02	SOURCE BASELINE AND CONFIGURATION AUDIT	UPDATED SOURCE BASELINE, CONFIGURATION AUDIT SCOPE, AND RELEASE SOURCE-OF-TRUTH NOTES.
03	TOPOLOGY AND ADJACENCY MAP	PHYSICAL/LOGICAL MAPPING INFERRED FROM INTERFACE DESCRIPTIONS AND ADDRESSING.
04	PLATFORM INVENTORY	DEVICE MODELS/APPLIANCES, SOFTWARE VERSIONS AND ROLES.
05	CONTROL-PLANE REPORT	OSPFV2, OSPFV3, IBGPV4, IBGPV6, EBGPV4, EBGPV6 AND ROUTE REFLECTION.
06	SERVICE-PLANE REPORT	MPLS, LDP, TARGETED LDP, LDP/IGP SYNC, L3VPN, PSEUDOWIRE, JUNOS EVC300, EVPN AND DMVPN (v4 and v6)/IPSEC.
07	QOS MODEL	CORE MODEL, SERVICE-EDGE 8-CLASS MODEL, DSCP TREATMENT AND VENDOR TRANSLATION.
08	THREE-ISP FALLBACK AND SPARKLE SIMPLE EBGP EDGE	TATA PRIMARY, VERIZON SECONDARY, HE LAST RESORT; SPARKLE ISOLATED AS GLOBAL EBGP EDGE.
09	SECURITY AND OPERATIONS	SSH HARDENING, PREFIX FILTERING, GRACEFUL RESTART, LLDP, NTP AND OPERATIONAL CONTROLS.
10	FEATURE MATRIX AND DEVICE ADDRESSING INVENTORY	PER-DEVICE SERVICE CAPABILITY MATRIX PLUS IPV4/IPV6 INTERFACE INVENTORY.
11	DEVICE ADDRESS INVENTORY	THE ADDRESS INVENTORY LISTS THE CONFIGURED IPV4 AND IPV6 INTERFACES PER DEVICE.
12	CUSTOMER VALUE STORYLINE	THIS ENVIRONMENT GIVES THE CUSTOMER A SAFE PLACE TO VALIDATE FAILURE BEHAVIOR, VENDOR INTEROPERABILITY, SERVICE PROVISIONING AND OPERATIONAL PROCEDURES.
13	PPDIOO IMPLEMENTATION REFERENCE	MAPS THE SKYNET LAB IMPLEMENTATION TO THE PREPARE, PLAN, DESIGN, IMPLEMENT, OPERATE AND OPTIMIZE LIFECYCLE SO THE CUSTOMER CAN UNDERSTAND THE PROJECT METHODOLOGY.
14	BGP COMMUNITY CONTROL AND EDGE ENFORCEMENT CENTER	DOCUMENTS CUSTOMER ROUTE INTENT, PROVIDER ORIGIN TAGGING, RTBH GUARDRAILS, INTERNAL-ONLY POLICY, NO-EXPORT COMMUNITIES AND ISP-EDGE ENFORCEMENT ACROSS CISCO, JUNOS AND IOS-XR.
15	EVC300 ETHERNET VIRTUAL CIRCUIT SERVICE	DETAILED EVC EXPLANATION, VLAN300, JUNOS L2CIRCUIT SIGNALING, CUSTOMER ENDPOINTS, VALIDATION, AND THE VPN V4/V6 GRE OVER IKEV2/IPSEC BETWEEN EVC-DC-01 AND EVC-DC-02.
16	BGP MD5 AUTHENTICATION	ENABLED MD5 PASSWORD EVELAB@2025 ON ALL EXTERNAL BGP SESSIONS (ISPS AND CE ROUTERS) FOR SECURE PEERING.
17	COPP (CONTROL PLANE POLICING)	DEPLOYED CPU PROTECTION POLICIES ON ALL 10 CORE DEVICES (CISCO IOSV, JUNIPER VMX, IOS-XRV) TO PREVENT CONTROL-PLANE OVERLOAD.
18	VRF MIGRATION (PE01/PE02)	CONVERTED IPV4-ONLY IP VRF TO DUAL-STACK VRF DEFINITION FOR THE INTER-AS-L3VPN VRF.
19	6VPE (IPV6 L3VPN)	ACTIVATED VPNV6 ADDRESS FAMILY ON THE ROUTE REFLECTOR (P01) AND PE ROUTERS (PE01, PE02); ESTABLISHED IPV6 CONNECTIVITY BETWEEN PE LOOPBACKS INSIDE THE VRF.
20	RIPv2 & RIP-NG ON EVC CIRCUITS	RIPv2 AND RIP-NG ENABLED ON EVC ENDPOINTS
21	EIGRP AND EIGRPv6	IMPLEMENTED AMONG EVPN ENDPOINTS FOR IPV4 AND IPV6 ROUTES ADVERTISEMENTS
22	IP SLA AND OBJECT TRACKING	ICMP-ECHO REACHABILITY PROBES AND TRACK OBJECTS THAT DRIVE DETERMINISTIC FAILOVER FOR THE ISP FALLBACK AND DEFAULT ROUTES.

23	**POLICY-BASED NAT / PAT ON THE PROVIDER EDGE**	ROUTE-MAP-DRIVEN, VRF-AWARE PAT OVERLOAD ON PE01-BORDER, PE02-VIOS AND PE07-POP02.
24	**VRRP FIRST-HOP REDUNDANCY**	VRRP GROUP 1 BETWEEN PE04-VMX01 AND PE05-VMX02 PROTECTING THE SHARED AGG.SW GATEWAY 172.16.0.240.
25	**DHCP SERVICES ON THE POP EDGE ROUTERS AND AGG.SW**	LOCAL DHCP POOLS VPN1/VPN2 ON PE06-POP01 AND PE07-POP02 FOR THE IPSEC VPN ENDPOINT SEGMENTS.
26	**MPLS TRAFFIC ENGINEERING (RSVP-TE) READINESS**	RSVP AND OSPF TE EXTENSIONS ENABLED ACROSS CISCO AND JUNIPER CORE/PE NODES ALONGSIDE THE EXISTING LDP TRANSPORT.
27	**LAYER 2 HARDENING ON SKNT-AGG.SW**	RAPID-PVST+ AND PORTFAST EDGE ON ALL AGG.SW TRUNK AND UPLINK PORTS.
28	**UPDATED CONFIGURATION BASELINE**	AUTHORITATIVE JULY 2026 DEVICE ROLES, SOFTWARE, ASN, PROTOCOL/SERVICE FEATURES, AND COMPLETE CONFIGURED INTERFACE ADDRESS INVENTORY.
29	**MPLS OPERATIONS, ADMINISTRATION AND MAINTENANCE (OAM)**	MULTIVENDOR LSP PING/TRACEROUTE IMPLEMENTATION, PLATFORM BEHAVIOR, VALIDATION EVIDENCE, LIMITATIONS, AND OPERATIONAL ACCEPTANCE.
30	**DIRECT CE01-ISP-HE EBGP MIGRATION**	MIGRATES THE 10.207.0.0/30 CE01-HE LINK TO AS301↔AS400 EBGP WITH STRICT PREFIX POLICY, BACKUP PREFERENCE, STATIC-TO-BGP RETURN-PATH CUTOVER, AND FAILURE VALIDATION.
31	**NTP FINE TUNING AND STANDARDIZATION**	STANDARDIZES PREFERRED AND REDUNDANT NTP REFERENCES, LOOPBACK-SOURCED CLIENT BEHAVIOR WHERE SUPPORTED, AND OPERATIONAL TIME CONSISTENCY ACROSS THE LAB.
32	**DUAL-STACK PSEUDOWIRE ROUTING AND FAILURE MONITORING**	DOCUMENTS OSPFV2/OSPFV3 OVER VC-ID 1, NEW IPV4/IPV6 PW LOOPBACKS, VCCV VALIDATION, PW STATUS SIGNALING, SYSLOG MONITORING AND CONTROLLED AC FAILURE RECOVERY.

1.1. EXECUTIVE ARCHITECTURE SUMMARY

SKYNET ISP LAB IS A ROBUST AND RESILIENT PRODUCTION-FLAVORED SERVICE-PROVIDER REFERENCE ARCHITECTURE. IT IS NOT A BASIC ROUTING SANDBOX: IT IS A MULTIVENDOR CARRIER BACKBONE WITH REAL CONTROL-PLANE SEPARATION, MPLS SERVICES, OVERLAY SECURITY, PATH ENGINEERING AND OPERATIONAL GUARDRAILS.

- DUAL-STACK IGP UNDERLAY: OSPFV2 FOR IPV4 AND OSPFV3 FOR IPV6 ACROSS THE INTERNAL TRANSPORT DOMAIN.
- BGP CONTROL PLANE: IBGP/EBGP FOR IPV4, IPV6, VPNV4 AND EVPN SIGNALING, WITH ROUTE-REFLECTION TO AVOID FULL-MESH SCALING PAIN.
- MPLS SERVICE PLANE: LDP TRANSPORT, TARGETED LDP FOR REMOTE LABEL BINDINGS, LDP/IGP SYNC ON CISCO/XRV NODES, AND SERVICE SUPPORT FOR L3VPN AND PSEUDOWIRE.
- MULTIVENDOR REALISM: CISCO IOS/IOL/IOSV, CISCO IOS XRV, JUNIPER VMX AND HUAWEI NE40E COEXIST IN THE SAME SERVICE-PROVIDER SCENARIO.
- CUSTOMER SERVICES: MPLS L3VPN, VPWS/PSEUDOWIRE WITH IPV6 SUPPORT, EVPN DC VLAN EXTENSION AND DUAL-STACK DMVPN/IPSEC OVERLAY VALIDATION.
- CARRIER-GRADE BEHAVIOR: QOS, GRACEFUL RESTART, PREFIX FILTERING, COMMUNITY TAGGING, ROUTE-POLICY CONTROL, SSH HARDENING, LLDP AND NTP.

INFRASTRUCTURE INFORMATION --------- *TOTAL NUMBER OF INFRA NODES: 31*	ADVANCED DISTANCE VECTOR PROTOCOLS --------- *EIGRP V4 AND V6 FOR EVPN ENDPOINTS.* *BOTH RUNNING ON AS 300*	DISTANCE VECTOR PROTOCOLS --------- *RIPv2 AND RIP-NG BOTH RUNNING ON EVC ENDPOINTS*	LINK STATE PROTOCOLS --------- *OSPFv2 DOMAIN AREA 0 AND OSPFv3 DOMAIN AREA 0*	PATH VECTOR PROTOCOLS (BGP) --------- *EXTERNAL PROVIDERS AND iBGP* *(eBGP) 4 ISPs AND iBGP (v4v6) ASN 300*	PSEUDOWIRE MAIN SERVICES --------- *L2VPN, PSEUDOWIRES, VPLS, VPWS, DMVPN FOR IPV4, NHRP FOR IPV6, QOS, IPSECVPN WITH IKEV2*

WHAT MAKES THIS LAB A MASTERPIECE OF SERVICE PROVIDER TECHNOLOGIES?

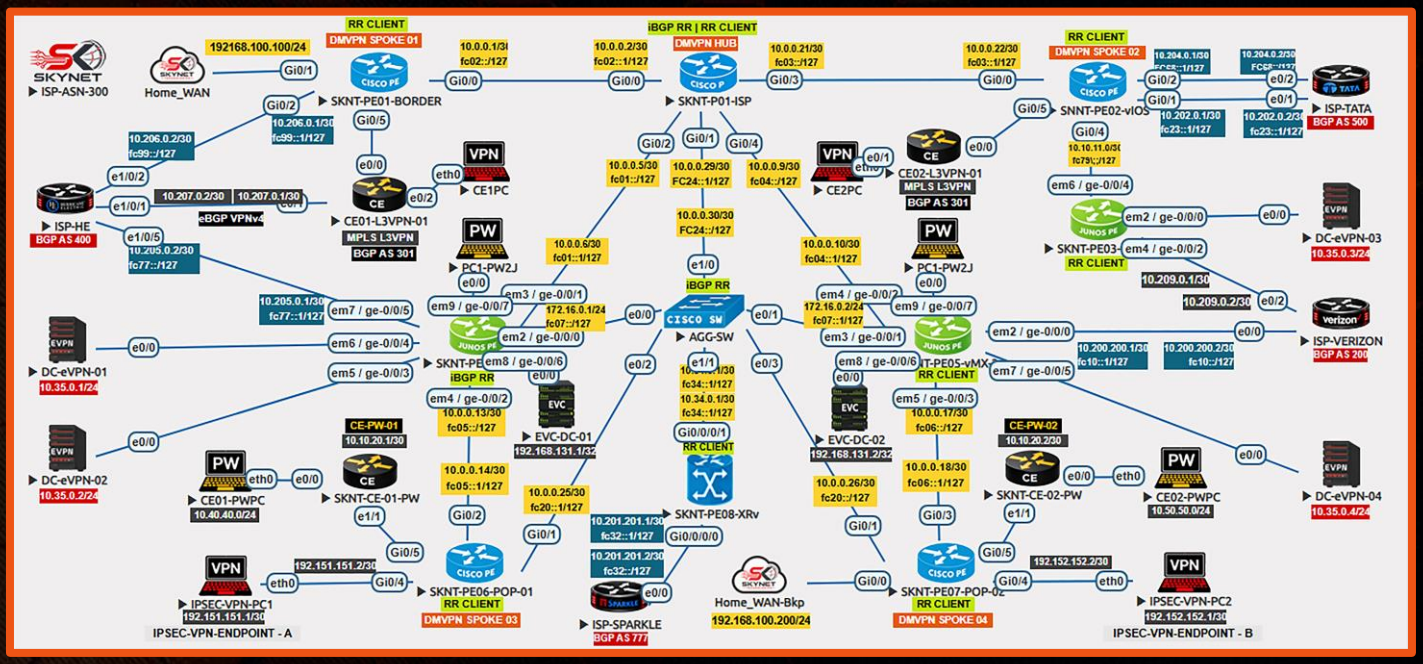
CAPABILITY	CUSTOMER-FACING VALUE
MULTIVENDOR PE/CORE DESIGN	SHOWS THAT THE ARCHITECTURE IS NOT LOCKED TO A SINGLE VENDOR AND CAN MODEL REAL BROWNFIELD PROVIDER NETWORKS.
DUAL-STACK FROM UNDERLAY TO EDGE	DEMONSTRATES IPV6 READINESS WITHOUT ABANDONING THE EXISTING IPV4 OPERATIONAL MODEL.
CONTROLLED ROUTE-REFLECTION	SCALES IBGP WHILE KEEPING CLEAR RR/CLIENT ROLES AND AVOIDING UNCONTROLLED FULL-MESH GROWTH.
MPLS SERVICE PORTFOLIO	L3VPN, PSEUDOWIRE AND EVPN CAN BE DEMONSTRATED FROM THE SAME CARRIER CORE.
MULTI-ISP FALLBACK	UPSTREAM FAILURE TESTS CAN PROVE DETERMINISTIC CUSTOMER REACHABILITY DURING PROVIDER OUTAGES.
QOS ARCHITECTURE	SHOWS TRAFFIC CLASSES, POLICING, PRIORITY TREATMENT AND VENDOR-ALIGNED SERVICE BEHAVIOR BEFORE PRODUCTION DEPLOYMENT.

2. SOURCE BASELINE AND CONFIGURATION AUDIT

THIS UPDATED RELEASE IS BUILT FROM THE LATEST CONFIGURATION FILES ARCHIVES, THE SUPPLIED DOCX REPORT, AND THE SUPPLIED TOPOLOGY DIAGRAM. THE INSPECTION FOCUSED ON LEFTOVERS, CONFIGURATION MISMATCHES, WRONG CONFIGURATIONS, HOSTNAME/DESCRIPTION CONSISTENCY, SERVICE-PLANE ALIGNMENT AND DOCUMENTATION ACCURACY.

3. TOPOLOGY AND ADJACENCY MAP

THE TOPOLOGY IS MAPPED FROM THREE SOURCES: THE SUPPLIED DIAGRAM, INTERFACE DESCRIPTIONS, AND CONFIGURED INTERFACE ADDRESSES. DESCRIPTIONS USING THE STANDARD TO-[HOSTNAME] CONVENTION MAKE THE PHYSICAL AND LOGICAL RELATIONSHIPS DIRECTLY AUDITABLE FROM THE CONFIGS.



INFERRED ADJACENCY INVENTORY

LOCAL DEVICE	INTERFACE	PEER DESCRIPTION	IPV4	IPV6
SKNT-PE01-BORDER	GIGABITETHERNET0/0	SKNT-P01-ISP	10.0.0.1/30	FC02::/127
SKNT-PE01-BORDER	GIGABITETHERNET0/1	HOME-WAN	192.168.100.100/24	-
SKNT-PE01-BORDER	GIGABITETHERNET0/2	BGP-LINK-HE-SECONDARY	10.206.0.1/30	FC99::1/127
SKNT-PE01-BORDER	GIGABITETHERNET0/5	CE01-L3VPN-01 / INTER-AS-L3VPN	192.168.111.1/24	-
SKNT-P01-ISP	GIGABITETHERNET0/0	SKNT-PE01-BORDER	10.0.0.2/30	FC02::1/127
SKNT-P01-ISP	GIGABITETHERNET0/1	SKNT-AGG.SW	10.0.0.29/30	FC24::1/127
SKNT-P01-ISP	GIGABITETHERNET0/2	SKNT-PE04-VMX01	10.0.0.5/30	FC01::/127
SKNT-P01-ISP	GIGABITETHERNET0/3	SKNT-PE02-VIOS	10.0.0.21/30	FC03::/127
SKNT-P01-ISP	GIGABITETHERNET0/4	SKNT-PE05-VMX02	10.0.0.9/30	FC04::/127
SKNT-PE02-VIOS	GIGABITETHERNET0/0	SKNT-P01-ISP	10.0.0.22/30	FC03::1/127
SKNT-PE02-VIOS	GIGABITETHERNET0/1	TATA-LINK1-V4V6	10.202.0.1/30	FC23::1/127
SKNT-PE02-VIOS	GIGABITETHERNET0/2	TATA-LINK2-V4V6	10.204.0.1/30	FC68::1/127
SKNT-PE02-VIOS	GIGABITETHERNET0/4	SKNT-PE03-VMX03	10.10.11.1/30	FC79::1/127

LOCAL DEVICE	INTERFACE	PEER DESCRIPTION	IPV4	IPV6
SKNT-PE02-VIOS	GIGABITETHERNET0/5	CE02-L3VPN-01 / INTER-AS-L3VPN	192.168.222.1/24	-
SKNT-PE04-VMX01	GE-0/0/0.0	SKNT-AGG.SW	172.16.0.1/24	FC07::/127
SKNT-PE04-VMX01	GE-0/0/1.0	SKNT-P01-ISP	10.0.0.6/30	FC01::1/127
SKNT-PE04-VMX01	GE-0/0/2.0	SKNT-PE06-POP01	10.0.0.13/30	FC05::/127
SKNT-PE04-VMX01	GE-0/0/3	DC-EVPN-02	-	-
SKNT-PE04-VMX01	GE-0/0/3.600	DC-EVPN-02	-	-
SKNT-PE04-VMX01	GE-0/0/4	DC-EVPN-01	-	-
SKNT-PE04-VMX01	GE-0/0/4.600	DC-EVPN-01	-	-
SKNT-PE04-VMX01	GE-0/0/5.0	BGP-LINK-HE	10.205.0.1/30	FC77::1/127
SKNT-PE05-VMX02	GE-0/0/0.0	BGP-LINK-VERIZON	10.200.200.1/30	FC10::1/127
SKNT-PE05-VMX02	GE-0/0/1.0	SKNT-AGG.SW	172.16.0.2/24	FC07::1/127
SKNT-PE05-VMX02	GE-0/0/2.0	SKNT-P01-ISP	10.0.0.10/30	FC04::1/127
SKNT-PE05-VMX02	GE-0/0/3.0	SKNT-PE07-POP02	10.0.0.17/30	FC06::/127
SKNT-PE05-VMX02	GE-0/0/5	DC-EVPN-04	-	-
SKNT-PE05-VMX02	GE-0/0/5.600	DC-EVPN-04	-	-
SKNT-PE03-VMX03	GE-0/0/0	DC-EVPN-03	-	-
SKNT-PE03-VMX03	GE-0/0/0.600	DC-EVPN-03	-	-
SKNT-PE03-VMX03	GE-0/0/2.0	BGP-LINK-VERIZON	10.209.0.1/30	-
SKNT-PE03-VMX03	GE-0/0/4.0	SKNT-PE02-VIOS	10.10.11.2/30	FC79::/127
SKNT-AGG.SW	ETHERNET0/0	SKNT-PE04-VMX01	-	-
SKNT-AGG.SW	ETHERNET0/1	SKNT-PE05-VMX02	-	-
SKNT-AGG.SW	ETHERNET0/2	SKNT-PE06-POP01	-	-
SKNT-AGG.SW	ETHERNET0/3	SKNT-PE07-POP02	-	-
SKNT-AGG.SW	ETHERNET1/0	SKNT-P01-ISP	10.0.0.30/30	FC24::/127
SKNT-AGG.SW	ETHERNET1/1	SKNT-PE08-XRV	10.34.0.1/30	FC34::1/127
SKNT-PE06-POP01	TUNNEL100	VPN-TO-POP02	192.153.153.1/30	-
SKNT-PE06-POP01	GIGABITETHERNET0/1	SKNT-PE07-POP02	10.0.0.25/30	FC20::1/127
SKNT-PE06-POP01	GIGABITETHERNET0/2	SKNT-PE04-VMX01	10.0.0.14/30	FC05::1/127
SKNT-PE06-POP01	GIGABITETHERNET0/4	IPSEC-VPN01-EP01	192.151.151.2/30	-

LOCAL DEVICE	INTERFACE	PEER DESCRIPTION	IPV4	IPV6
SKNT-PE06-POP01	GIGABITETHERNET0/5	CE-01-PW01	-	-
SKNT-PE07-POP02	TUNNEL200	VPN-TO-POP01	192.153.153.2/30	-
SKNT-PE07-POP02	GIGABITETHERNET0/1	SKNT-PE06-POP01	10.0.0.26/30	FC20::/127
SKNT-PE07-POP02	GIGABITETHERNET0/3	SKNT-PE05-VMX02	10.0.0.18/30	FC06::1/127
SKNT-PE07-POP02	GIGABITETHERNET0/4	IPSEC-VPN01-EP02	192.152.152.2/30	-
SKNT-PE07-POP02	GIGABITETHERNET0/5	CE-02-PW01	-	-
ISP-VERIZON	ETHERNET0/0	PE-05-VMX-02	10.200.200.2/30	FC10::/127
ISP-VERIZON	ETHERNET0/2	SKNT-PE03-VMX03	10.209.0.2/30	-
ISP-TATA	ETHERNET0/1	PE02-VIOS-LINK-01	10.202.0.2/30	FC23::/127
ISP-TATA	ETHERNET0/2	PE02-VIOS-LINK-02	10.204.0.2/30	FC68::/127
ISP-HE	ETHERNET1/0/1	CE01-L3VPN-01	10.207.0.2/30	-
ISP-HE	ETHERNET1/0/2	SKNT-PE01-BORDER	10.206.0.2/30	FC99::/127
ISP-HE	ETHERNET1/0/5	PE-04-VMX-01	10.205.0.2/30	FC77::/127
DC-EVPN-01	ETHERNET0/0	SKNT-PE04-VMX01	-	-
DC-EVPN-01	VLAN600	SKNT-PE04-VMX01	10.35.0.1/24	2001:DB8:35:600::1/64
DC-EVPN-01	LOOPBACK0	DC-01-SERVER-01	172.31.35.1/32	2001:DB8:35:1::1/128
DC-EVPN-01	LOOPBACK1	DC-EVPN-01-SERVER-01	172.31.35.5/32	2001:DB8:35:1::5/128
DC-EVPN-01	LOOPBACK2	DC-EVPN-01-SERVER-02	172.31.35.6/32	2001:DB8:35:1::6/128
DC-EVPN-01	LOOPBACK3	DC-EVPN-01-SERVER-03	172.31.35.7/32	2001:DB8:35:1::7/128
DC-EVPN-01	LOOPBACK4	DC-EVPN-01-SERVER-04	172.31.35.8/32	2001:DB8:35:1::8/128
DC-EVPN-01	LOOPBACK5	DC-EVPN-01-SERVER-05	172.31.35.9/32	2001:DB8:35:1::9/128
DC-EVPN-02	ETHERNET0/0	SKNT-PE04-VMX01	-	-
DC-EVPN-02	VLAN600	SKNT-PE04-VMX01	10.35.0.2/24	2001:DB8:35:600::2/64
DC-EVPN-02	LOOPBACK0	DC-02-SERVER-01	172.31.35.2/32	2001:DB8:35:1::2/128
DC-EVPN-02	LOOPBACK1	DC-EVPN-02-SERVER-01	172.31.35.10/32	2001:DB8:35:1::10/128
DC-EVPN-02	LOOPBACK2	DC-EVPN-02-SERVER-02	172.31.35.11/32	2001:DB8:35:1::11/128
DC-EVPN-02	LOOPBACK3	DC-EVPN-02-SERVER-03	172.31.35.12/32	2001:DB8:35:1::12/128
DC-EVPN-02	LOOPBACK4	DC-EVPN-02-SERVER-04	172.31.35.13/32	2001:DB8:35:1::13/128
DC-EVPN-02	LOOPBACK5	DC-EVPN-02-SERVER-05	172.31.35.14/32	2001:DB8:35:1::14/128
DC-EVPN-03	ETHERNET0/0	SKNT-PE03-VMX03	-	-

LOCAL DEVICE	INTERFACE	PEER DESCRIPTION	IPV4	IPV6
DC-EVPN-03	VLAN600	SKNT-PE03-VMX03	10.35.0.3/24	2001:DB8:35:600::3/64
DC-EVPN-03	LOOPBACK0	DC-03-SERVER-01	172.31.35.3/32	2001:DB8:35:1::3/128
DC-EVPN-03	LOOPBACK1	DC-EVPN-03-SERVER-01	172.31.35.15/32	2001:DB8:35:1::15/128
DC-EVPN-03	LOOPBACK2	DC-EVPN-03-SERVER-02	172.31.35.16/32	2001:DB8:35:1::16/128
DC-EVPN-03	LOOPBACK3	DC-EVPN-03-SERVER-03	172.31.35.17/32	2001:DB8:35:1::17/128
DC-EVPN-03	LOOPBACK4	DC-EVPN-03-SERVER-04	172.31.35.18/32	2001:DB8:35:1::18/128
DC-EVPN-03	LOOPBACK5	DC-EVPN-03-SERVER-05	172.31.35.19/32	2001:DB8:35:1::19/128
DC-EVPN-04	ETHERNET0/0	SKNT-PE05-VMX02	-	-
DC-EVPN-04	VLAN600	SKNT-PE05-VMX02	10.35.0.4/24	2001:DB8:35:600::4/64
DC-EVPN-04	LOOPBACK0	DC-04-SERVER-01	172.31.35.4/32	2001:DB8:35:1::4/128
DC-EVPN-04	LOOPBACK1	DC-EVPN-04-SERVER-01	172.31.35.20/32	2001:DB8:35:1::20/128
DC-EVPN-04	LOOPBACK2	DC-EVPN-04-SERVER-02	172.31.35.21/32	2001:DB8:35:1::21/128
DC-EVPN-04	LOOPBACK3	DC-EVPN-04-SERVER-03	172.31.35.22/32	2001:DB8:35:1::22/128
DC-EVPN-04	LOOPBACK4	DC-EVPN-04-SERVER-04	172.31.35.23/32	2001:DB8:35:1::23/128
DC-EVPN-04	LOOPBACK5	DC-EVPN-04-SERVER-05	172.31.35.24/32	2001:DB8:35:1::24/128
CE-01-PW01	ETHERNET0/0	CEPC01	10.40.40.254/24	-
CE-01-PW01	ETHERNET1/1	POP-01-PW-AC	10.10.20.1/30	-
CE-02-PW01	ETHERNET0/0	CEPC02	10.50.50.254/24	-
CE-02-PW01	ETHERNET1/1	POP-02-PW-AC	10.10.20.2/30	-
CE01-L3VPN-01	ETHERNET0/0	SKNT-PE01-BORDER / INTER-AS-L3VPN	192.168.111.2/24	-
CE01-L3VPN-01	ETHERNET0/1	ISP-HE-V4	10.207.0.1/30	-
CE01-L3VPN-01	ETHERNET0/2	CE01-CUSTOMER-LAN / INTER-AS-L3VPN	192.168.245.1/30	-
CE02-L3VPN-01	ETHERNET0/0	SKNT-PE02-VIOS / INTER-AS-L3VPN	192.168.222.2/24	-
CE02-L3VPN-01	ETHERNET0/1	CE02-CUSTOMER-LAN / INTER-AS-L3VPN	192.168.246.1/30	-
SKNT-PE08-XRV	GIGABITETHERNET0/0/0/0	ISP-SPARKLE	10.201.201.1/30	FC32::1/127
SKNT-PE08-XRV	GIGABITETHERNET0/0/0/1	SKNT-AGG.SW	10.34.0.2/30	FC34::/127
ISP-SPARKLE	LOOPBACK1	SPARKLE-CE01-ADV	172.19.10.1/32	FC11::1/128
ISP-SPARKLE	LOOPBACK2	SPARKLE-CE02-ADV	172.19.20.1/32	FC12::1/128
ISP-SPARKLE	LOOPBACK3	SPARKLE-CE03-ADV	172.19.30.1/32	FC13::1/128
ISP-SPARKLE	LOOPBACK4	SPARKLE-CE04-ADV	172.19.40.1/32	FC14::1/128
ISP-SPARKLE	LOOPBACK5	SPARKLE-CE05-ADV	172.19.50.1/32	FC15::1/128

LOCAL DEVICE	INTERFACE	PEER DESCRIPTION	IPV4	IPV6
ISP-SPARKLE	ETHERNET0/0	SKNT-PE08-XRV	10.201.201.2/30	FC32::/127
SKNT-PE04-VMX01	GE-0/0/6.300	EVC-DC-01 / EVC300 VLAN300	VLAN300 ACCESS CIRCUIT	–
SKNT-PE05-VMX02	GE-0/0/6.300	EVC-DC-02 / EVC300 VLAN300	VLAN300 ACCESS CIRCUIT	–
EVC-DC-01	LOOPBACK0	EVC-DC-01-SERVER-01	10.16.16.1/32	–
EVC-DC-01	LOOPBACK1	EVC-DC-01-V4V6-01	192.168.227.1/32	FD00:16:16::1/128
EVC-DC-01	LOOPBACK2	EVC-DC-01-V4V6-02	192.168.227.2/32	FD00:16:16::2/128
EVC-DC-01	LOOPBACK3	EVC-DC-01-V4V6-03	192.168.227.3/32	FD00:16:16::3/128
EVC-DC-01	LOOPBACK4	EVC-DC-01-V4V6-04	192.168.227.4/32	FD00:16:16::4/128
EVC-DC-01	LOOPBACK5	EVC-DC-01-V4V6-05	192.168.227.5/32	FD00:16:16::5/128
EVC-DC-01	LOOPBACK300	SIMULATED-INTERNAL-LAN-DC1	172.16.172.1/32	FD00:172:16::1/128
EVC-DC-02	LOOPBACK0	EVC-DC-02-SERVER-01	10.17.17.1/32	–
EVC-DC-02	LOOPBACK1	EVC-DC-02-V4V6-01	192.168.228.1/32	FD00:17:17::1/128
EVC-DC-02	LOOPBACK2	EVC-DC-02-V4V6-02	192.168.228.2/32	FD00:17:17::2/128
EVC-DC-02	LOOPBACK3	EVC-DC-02-V4V6-03	192.168.228.3/32	FD00:17:17::3/128
EVC-DC-02	LOOPBACK4	EVC-DC-02-V4V6-04	192.168.228.4/32	FD00:17:17::4/128
EVC-DC-02	LOOPBACK5	EVC-DC-02-V4V6-05	192.168.228.5/32	FD00:17:17::5/128
EVC-DC-02	LOOPBACK300	SIMULATED-INTERNAL-LAN-DC2	172.16.173.1/32	FD00:173:16::1/128
EVC-DC-01	ETHERNET0/0	SKNT-PE04-VMX01	VLAN300 TRUNK	–
EVC-DC-01	VLAN300	EVC300 TEST SVI	192.168.131.1/24	FC91::1/64
EVC-DC-02	ETHERNET0/0	SKNT-PE05-VMX02	VLAN300 TRUNK	–
EVC-DC-02	VLAN300	EVC300 TEST SVI	192.168.131.2/24	FC91::2/64

4. PLATFORM INVENTORY



DEVICE	ROLE	MODEL / APPLIANCE	ASN	IMPLEMENTED FEATURES
SKNT-PE01-BORDER	BORDER PE / HE + CE01 L3VPN EDGE	CISCO IOSV/IOL	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, DMVPN/IPSEC
SKNT-P01-ISP	CORE P / PRIMARY ROUTE-REFLECTOR	CISCO IOSV/IOL	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, DMVPN/IPSEC
SKNT-PE02-VIOS	PE / TATA + CE02 L3VPN EDGE	CISCO IOSV/IOL	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, DMVPN/IPSEC
SKNT-PE04-VMX01	JUNIPER PE / HE PATH + EVPN/ PSEUDOWIRE + IPSEC	JUNIPER VMX	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, EVPN, EVC300
SKNT-PE05-VMX02	JUNIPER PE / VERIZON PATH + EVPN/ PSEUDOWIRE + IPSEC	JUNIPER VMX	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, EVPN, EVC300
SKNT-PE03-VMX03	JUNIPER PE / VERIZON PATH + EVPN	JUNIPER VMX	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, EVPN
SKNT-AGG.SW	AGGREGATION / PE08 EXPANSION EDGE	CISCO IOL L2/L3	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, QOS
SKNT-PE06-POP01	POP01 PE / PSEUDOWIRE + IPSEC	CISCO IOSV/IOL	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, PSEUDOWIRE
SKNT-PE07-POP02	POP02 PE / PSEUDOWIRE + IPSEC	CISCO IOSV/IOL	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, PSEUDOWIRE
ISP-VERIZON	EXTERNAL ISP AS200	CISCO IOSV	200	BGP, GRACEFUL RESTART, SSH HARDENING
ISP-TATA	EXTERNAL ISP AS500	CISCO IOSV	500	BGP, MPLS/LDP, L3VPN, GRACEFUL RESTART, SSH HARDENING
ISP-HE	EXTERNAL ISP AS400	HUAWEI NE40E	400	BGP, GRACEFUL RESTART, SSH HARDENING
DC-EVPN-01	DC EVPN CUSTOMER/SERVER SEGMENT	CISCO IOL L2/L3	-	EVPN, SSH HARDENING

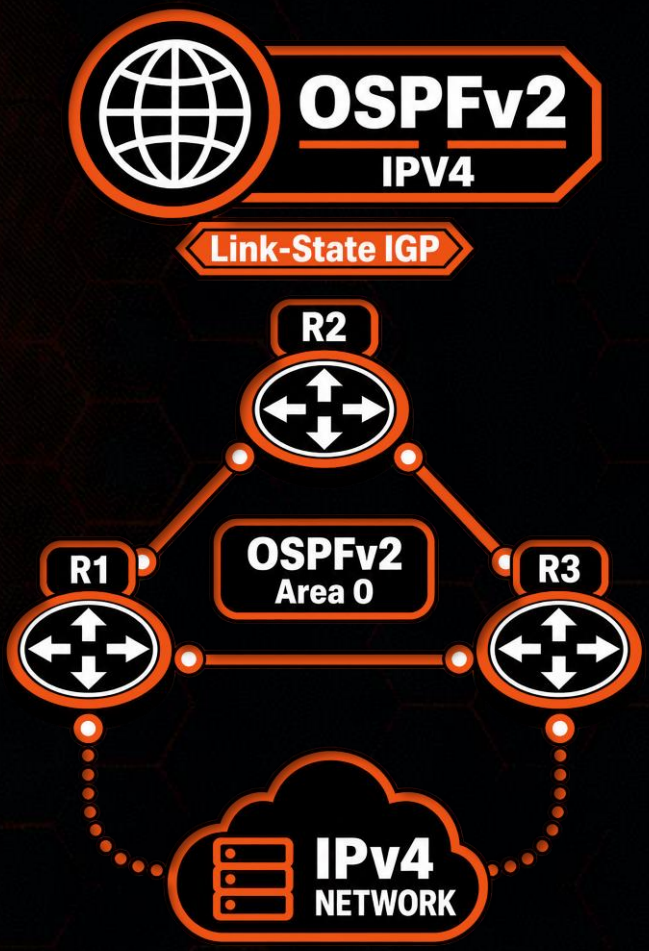
DEVICE	ROLE	MODEL / APPLIANCE	ASN	IMPLEMENTED FEATURES
DC-EVPN-02	DC EVPN CUSTOMER/SERVER SEGMENT	CISCO IOL L2/L3	-	EVPN, SSH HARDENING
DC-EVPN-03	DC EVPN CUSTOMER/SERVER SEGMENT	CISCO IOL L2/L3	-	EVPN, SSH HARDENING
DC-EVPN-04	DC EVPN CUSTOMER/SERVER SEGMENT	CISCO IOL L2/L3	-	EVPN, SSH HARDENING
CE-01-PW01	CUSTOMER EDGE - PSEUDOWIRE SERVICE	CISCO IOSV	-	MPLS/LDP, SSH HARDENING
CE-02-PW01	CUSTOMER EDGE - PSEUDOWIRE SERVICE	CISCO IOSV	-	MPLS/LDP, SSH HARDENING
CE01-L3VPN-01	CUSTOMER EDGE - L3VPN SERVICE	CISCO IOSV	301	BGP, L3VPN, QOS, GRACEFUL RESTART, SSH HARDENING
CE02-L3VPN-01	CUSTOMER EDGE - L3VPN SERVICE	CISCO IOSV	301	BGP, L3VPN, GRACEFUL RESTART, SSH HARDENING
SKNT-PE08-XRV	IOS-XR PE / SPARKLE SIMPLE EBGP EDGE	CISCO IOS XRV	300	OSPFV2, OSPFV3, BGP, MPLS/LDP, QOS, GR; VPNV4 STUB ONLY
ISP-SPARKLE	EXTERNAL ISP AS777	CISCO IOSV	777	BGP, GRACEFUL RESTART, STRICT V4/V6 PREFIX FILTERING, QOS, SSH HARDENING
EVC-DC-01	CUSTOMER EVC300 ENDPOINT A	CISCO IOL L2/L3	-	VLAN300/EVC, DUAL-STACK GRE, IKEV2/IPSEC, RIPv2, RIPng, QoS, CoPP, LLDP
EVC-DC-02	CUSTOMER EVC300 ENDPOINT B	CISCO IOL L2/L3	-	VLAN300/EVC, DUAL-STACK GRE, IKEV2/IPSEC, RIPv2, RIPng, CoPP, LLDP

5. CONTROL-PLANE REPORT

5.1. OSPFV2 - IPV4 UNDERLAY

OSPFV2 PROCESS 1 IS THE IPV4 TRANSPORT UNDERLAY FOR THE PROVIDER DOMAIN. IT GIVES DETERMINISTIC REACHABILITY TO LOOPBACKS AND TRANSIT LINKS USED BY IBGP AND LDP. THE DESIGN USES AREA 0, LOOPBACK-BASED ROUTER IDS, POINT-TO-POINT BEHAVIOR ON ROUTED LINKS, AND AN AUTO-COST REFERENCE-BANDWIDTH OF 10000 ON CORE-CAPABLE NODES.

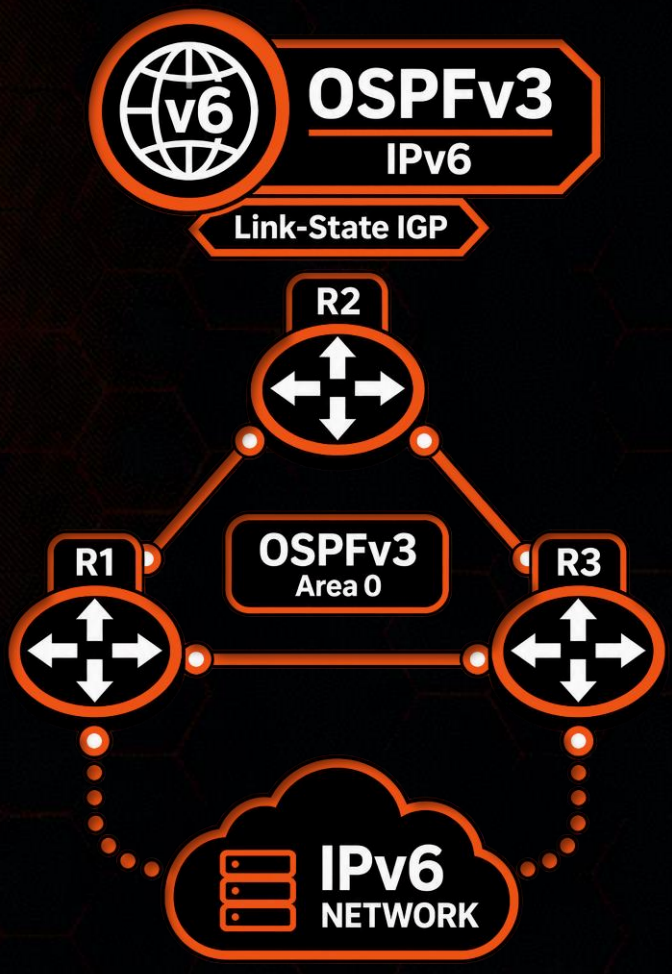
DESIGN POINT	IMPLEMENTATION DETAILS
PROCESS / AREA	ROUTER OSPF 1 OR JUNOS PROTOCOLS OSPF, AREA 0.
ROUTER ID STRATEGY	LOOPBACK0/INTERNAL-TRANSIT ADDRESSES SUCH AS 192.168.1.X ARE USED AS STABLE ROUTER IDS.
UNDERLAY PURPOSE	CARRIES REACHABILITY FOR BGP UPDATE-SOURCE LOOPBACK0 SESSIONS AND MPLS LDP ROUTER IDS.
CONVERGENCE GUARDRAIL	LDP/IGP SYNC IS CONFIGURED ON CISCO/XRV MPLS NODES TO AVOID BLACKHOLING DURING LABEL/CONTROL-PLANE TRANSITIONS.



5.2. OSPFV3 - IPV6 UNDERLAY

OSPFV3 PROCESS 1 MIRRORS THE UNDERLAY INTENT FOR IPV6. IT GIVES IPV6 CONTROL-PLANE REACHABILITY FOR IBGPV6 AND DUAL-STACK SERVICE TESTING. CISCO IOS NODES USE INTERFACE-LEVEL OSPFV3 ACTIVATION, XRV INCLUDES IPV6 ENABLE BEFORE ADDRESS ASSIGNMENT, AND JUNOS USES PROTOCOLS OSPF3.

DESIGN POINT	IMPLEMENTATION DETAILS
PROCESS / AREA	ROUTER OSPFV3 1 / JUNOS PROTOCOLS OSPFV3, AREA 0.
ADDRESS FAMILY	IPV6 UNICAST UNDERLAY USING FCXX:: ADDRESSING ACROSS INTER-ROUTER LINKS.
LOOPBACK SUPPORT	CORE LOOPBACKS CARRY /128 ADDRESSING FOR STABLE IPV6 CONTROL-PLANE IDENTITY.
PE08 INTEGRATION	PE08-XRV LOOPBACK0 USES FC93::1/128 AND AGG.SW LOOPBACK0 USES FC62::1/128 FOR LOOPBACK-TO-LOOPBACK IBGPV6.
OPERATIONAL ALIGNMENT	OSPFV2 AND OSPFV3 USE THE SAME TOPOLOGY INTENT SO IPV4 AND IPV6 FAILURES ARE EASIER TO REASON ABOUT.



5.3. EBGPV4



EBGPV4 IS USED AT ALL EXTERNAL EDGES AND IN THE CUSTOMER L3VPN HANDOFF. THE UPSTREAM EDGE DOES NOT BLINDLY ACCEPT THE INTERNET: IT FILTERS EXPECTED LAB PREFIXES, TAGS ROUTES WITH COMMUNITIES, AND EXPORTS ONLY AUTHORIZED SKYNET PREFIXES TOWARD THE ISP PEERS.

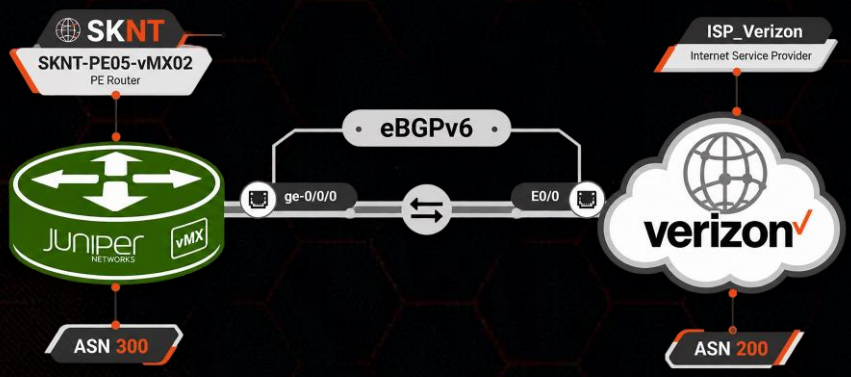
PROVIDER	AS	SKYNET EDGE	IPV4 PEER(S)	IPV6 PEER(S)	POLICY SUMMARY
TATA	500	SKNT-PE02-VIOS	10.202.0.2 / 10.204.0.2	FC23:: / FC68::	PRIMARY ISP FOR THE 172.19.X.X TEST PREFIXES. INBOUND POLICY TAGS ROUTES WITH COMMUNITY 300:500 AND TIGHT PREFIX FILTERS.
VERIZON	200	SKNT-PE05-VMX02 AND SKNT-PE03-VMX03	10.200.200.2 / 10.209.0.2	FC10::	SECONDARY ISP PATH. JUNOS POLICIES TAG/MARK ROUTES AS AS200 ORIGIN AND APPLY CONTROLLED EXPORT TOWARD VERIZON.
HE	400	SKNT-PE01-BORDER AND SKNT-PE04-VMX01	10.206.0.2 / 10.205.0.2	FC99:: / FC77::	LAST-RESORT ISP. IPV4 IMPORT POLICIES RESTRICT EXPECTED 172.19.X.X ROUTES; HE-ORIGIN ROUTES CARRY COMMUNITY 300:400 AND LOWER PREFERENCE ON PE01.
SPARKLE	777	SKNT-PE08-XRV	10.201.201.2	FC32::	FILTERED SIMPLE EBGP EDGE. PE08 IMPORTS ONLY EXPECTED 172.19.X.X AND FC11-FC15 TEST PREFIXES INTO GLOBAL UNICAST AND TAGS THEM WITH COMMUNITY 300:777; NO ACTIVE INTER-AS-L3VPN VRF IS PRESENT ON PE08.

5.4. IBGPV4

IBGPV4 INSIDE AS300 DISTRIBUTES UNDERLAY-RELEVANT ROUTES, PROVIDER LAN PREFIXES, CUSTOMER VPN REACHABILITY, EXTERNAL-ORIGIN TEST PREFIXES AND SERVICE-PLANE PREFIXES. MOST CORE IBGP SESSIONS ARE TIED TO LOOPBACKS WITH UPDATE-SOURCE LOOPBACK0, WHILE SELECTED DIRECTLY CONNECTED PEERS ARE USED WHERE THE DESIGN INTENTIONALLY MODELS EDGE ADJACENCY.

- MAXIMUM-PATHS IBGP IS CONFIGURED ON CORE NODES WHERE MULTIPLE INTERNAL PATHS ARE EXPECTED.
- SEND-COMMUNITY BOTH IS USED SO ORIGIN TAGS AND POLICY COMMUNITIES SURVIVE REFLECTION ACROSS THE CORE.
- NEXT-HOP-SELF IS APPLIED ON RR/CLIENT OR EXTERNAL-TO-INTERNAL BOUNDARIES TO PRESERVE USABLE FORWARDING NEXT HOPS.
- SOFT RECONFIGURATION/ROUTE REFRESH SUPPORT IS PRESENT ON CISCO NODES FOR SAFER POLICY VALIDATION.

5.5. EBGPV6



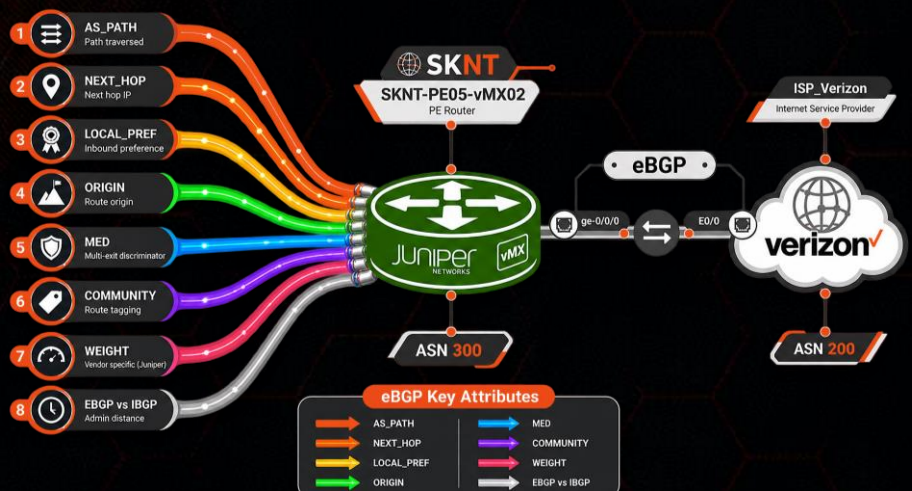
EBGPV6 MIRRORS THE V4 EXTERNAL-PROVIDER STORY FOR LAB PREFIXES FC11::1 THROUGH FC15::1 AND PE/PROVIDER LINK PREFIXES. TATA, VERIZON, HE AND SPARKLE HAVE IPV6 POLICY BOUNDARIES, WITH SPARKLE EXPLICITLY FILTERING INBOUND AND OUTBOUND IPV6 PREFIXES ON BOTH IOS AND XR SIDES.

PROVIDER	IPV6 CONTROL	CUSTOMER VALUE
TATA	FC23::/127 AND FC68::/127 TOWARD PE02-VIOS; FC11-FC15 TEST LOOPBACKS.	VALIDATES DUAL-STACK UPSTREAM BEHAVIOR ON THE PRIMARY PATH.
VERIZON	FC10::/127 TOWARD PE05-VMX02 PLUS VERIZON IPV6 LOOPBACKS.	KEEPS BACKUP PATH VALIDATION DUAL-STACK.
HE	FC99::/127 TO PE01 AND FC77::/127 TO PE04.	MAINTAINS LAST-RESORT REACHABILITY FOR IPV6 TEST PREFIXES.
SPARKLE	FC32::/127 TO PE08-XRV; FC11-FC15 ALLOWED INBOUND; FC16::/48 EXPORTED.	ADDS A STRICT IPV6 EBGP POLICY BOUNDARY WITHOUT MAKING SPARKLE PART OF THE ACTIVE L3VPN FALLBACK CHAIN.

5.6. IBGPV6

IBGPV6 CARRIES IPV6 UNICAST REACHABILITY INSIDE AS300. PE08-XRV AND AGG.SW FORM A LOOPBACK-TO-LOOPBACK IPV6 IBGP RELATIONSHIP USING FC93::1 AND FC62::1. JUNIPER VMX NODES ALSO PARTICIPATE IN IBGPV6 GROUPS USING FC07/FC60/FC61/FC63/FC64 ADDRESSING DEPENDING ON THE SEGMENT.

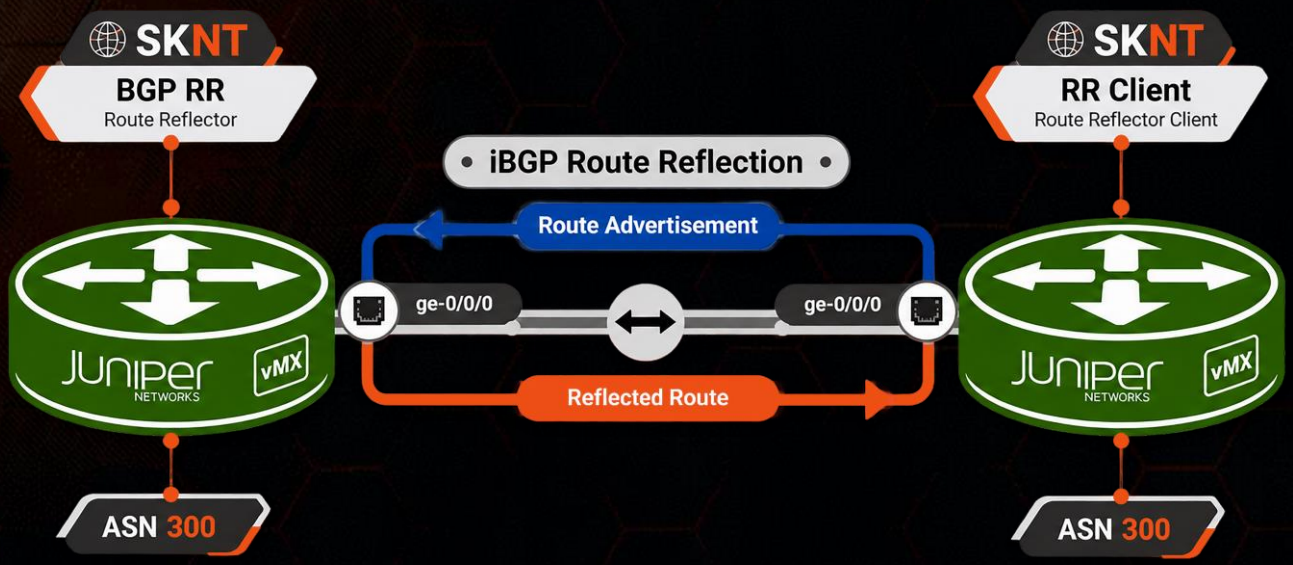
- IPV6 COMMUNITIES ARE SENT WHERE SUPPORTED, PRESERVING ORIGIN AND POLICY METADATA.
- THE DESIGN KEEPS IPV4 AND IPV6 POLICY LOGIC PARALLEL BUT NOT COPY-PASTE IDENTICAL, WHICH IS REALISTIC IN MULTIVENDOR NETWORKS.
- SPARKLE IPV6 IMPORT ON PE08 TAGS PERMITTED FC11-FC15 TEST PREFIXES WITH COMMUNITY 300:777 AND DROPS UNEXPECTED PREFIXES.



5.7. ROUTE REFLECTION SCHEME

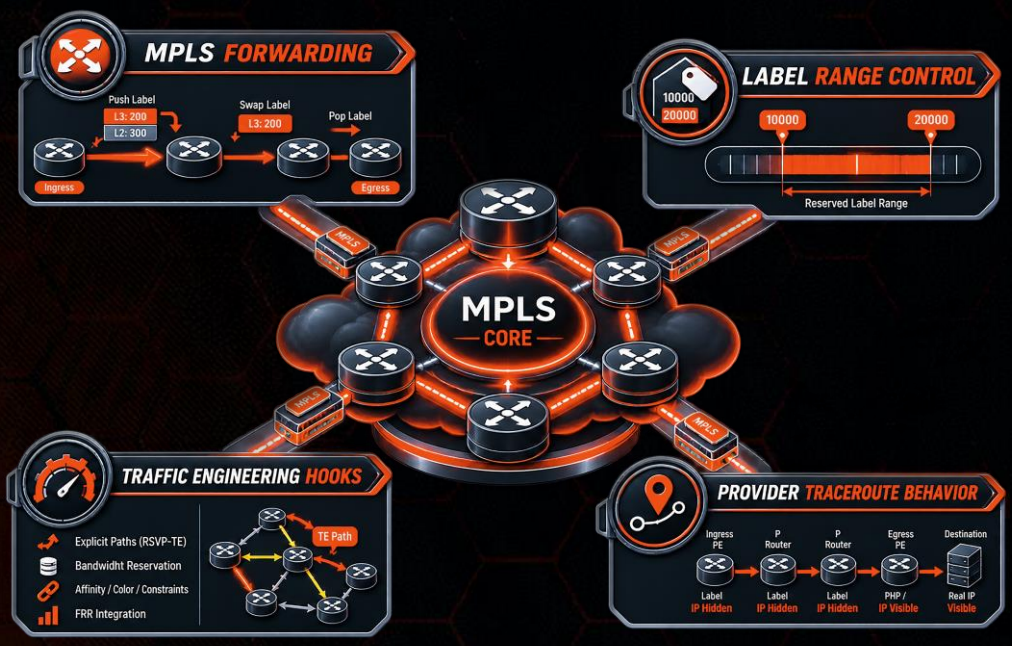
NODE	ROLE	CONTROL-PLANE IDENTITY	EXPLANATION
SKNT-P01-ISP	PRIMARY RR	192.168.1.2 / FC51::1	REFLECTS IPV4/VPNV4 AND IPV6 ROUTES FOR CORE CLIENTS INCLUDING PE01, PE02, PE05 AND AGG.SW. PE04-VMX01 IS TREATED AS A NON-CLIENT PEER IN THE RR FABRIC.
SKNT-PE04-VMX01	JUNIPER RR / EVPN CONTROL-PLANE ANCHOR	192.168.1.3 / FC60::1	USES IBGP GROUPS WITH CLUSTER IDS AND ADVERTISES INET, INET-VPN AND EVPN SIGNALING. PROVIDES ROUTE-REFLECTION BEHAVIOR INSIDE THE JUNIPER/EVPN SEGMENT WHILE MAINTAINING A NON-CLIENT TIE TO P01.
SKNT-AGG.SW	AGGREGATION RR EXTENSION	192.168.1.4 / FC62::1	ACTS AS THE LOCAL REFLECTOR TOWARD PE08-XRV FOR IPV4, VPNV4 AND IPV6 WHILE MAINTAINING CORE ADJACENCY TO P01-ISP.
SKNT-PE08-XRV	RR CLIENT / FUTURE RR-READY PE	192.168.1.101 / FC93::1	ACTIVE ROLE IS AGG-FACING PE/RR-CLIENT FOR IPV4/IPV6 UNICAST AND A SIMPLE SPARKLE EBGP EDGE. THE XRV CONFIG CARRIES A VPNV4 STUB BUT NO ACTIVE INTER-AS-L3VPN VRF.

ROUTE REFLECTION IS A BGP MECHANISM USED TO REDUCE THE NUMBER OF IBGP SESSIONS REQUIRED INSIDE AN AUTONOMOUS SYSTEM. IN A TRADITIONAL IBGP DESIGN, EVERY BGP ROUTER MUST FORM A FULL-MESH RELATIONSHIP WITH EVERY OTHER IBGP ROUTER. AS THE NETWORK GROWS, THIS FULL-MESH REQUIREMENT BECOMES DIFFICULT TO SCALE AND MANAGE. A ROUTE REFLECTOR, OR RR, SOLVES THIS PROBLEM BY REDISTRIBUTING ROUTES BETWEEN ITS IBGP CLIENTS. ROUTERS CONFIGURED AS RR CLIENTS ESTABLISH IBGP SESSIONS WITH THE ROUTE REFLECTOR INSTEAD OF WITH EVERY OTHER ROUTER. WHEN THE RR RECEIVES A ROUTE FROM ONE CLIENT, IT CAN REFLECT THAT ROUTE TO OTHER CLIENTS ACCORDING TO BGP ROUTE-REFLECTION RULES. THIS SIGNIFICANTLY REDUCES THE TOTAL NUMBER OF BGP SESSIONS AND SIMPLIFIES CONFIGURATION IN LARGE NETWORKS. ROUTE REFLECTION ALSO INTRODUCES ATTRIBUTES SUCH AS THE CLUSTER ID AND ORIGINATOR ID TO HELP PREVENT ROUTING LOOPS. MULTIPLE ROUTE REFLECTORS ARE COMMONLY DEPLOYED TO PROVIDE REDUNDANCY AND IMPROVE CONTROL-PLANE RESILIENCY. FOR SERVICE PROVIDER AND LARGE ENTERPRISE NETWORKS, ROUTE REFLECTION IS ONE OF THE MOST COMMON WAYS TO BUILD A SCALABLE IBGP ARCHITECTURE.



6. SERVICE-PLANE REPORT

6.1. MPLS FEATURES



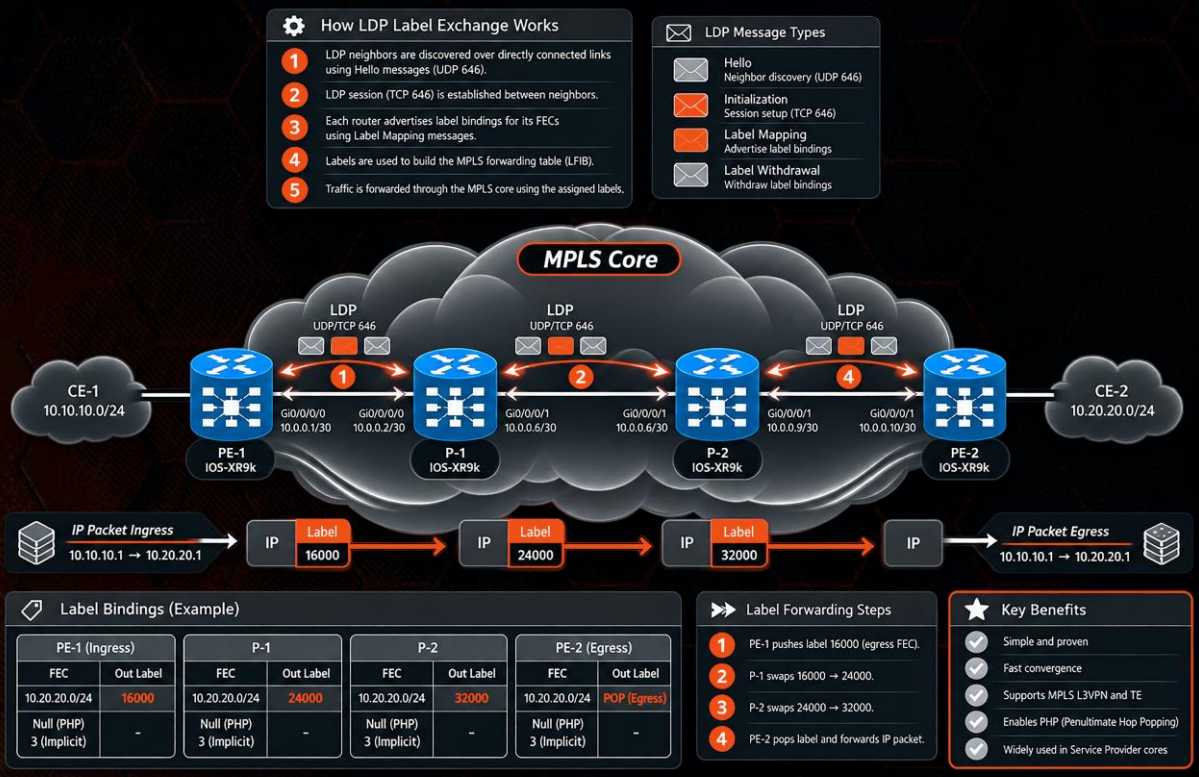
THE PROVIDER BACKBONE USES MPLS AS THE SERVICE TRANSPORT LAYER. MPLS IS ENABLED ON ROUTED CORE LINKS, WITH LDP PROVIDING LABEL DISTRIBUTION AND TRAFFIC-ENGINEERING/RSVP FEATURES PRESENT ON SELECTED TRANSIT INTERFACES. TTL PROPAGATION IS DISABLED ON CISCO NODES FOR PROVIDER-STYLE TRACEROUTE BEHAVIOR.

FEATURE	WHERE IT APPEARS	WHY IT MATTERS?
MPLS FORWARDING	CISCO IOS/IOL, IOS XRV AND JUNIPER VMX PE/CORE NODES.	CREATES THE LABEL-SWITCHED TRANSPORT NEEDED FOR VPN SERVICES.
LABEL RANGE CONTROL	AGG.SW USES 4000-4999; PE08-XRV USES 101000-101999.	MAKES LABEL OWNERSHIP PREDICTABLE DURING TROUBLESHOOTING.
TRAFFIC ENGINEERING HOOKS	MPLS TRAFFIC-ENG AND RSVP BANDWIDTH ARE PRESENT ON CORE/AGGREGATION LINKS.	KEEPS THE LAB EXTENSIBLE TOWARD RSVP-TE DEMONSTRATIONS.
PROVIDER TRACEROUTE BEHAVIOR	NO MPLS IP PROPAGATE-TTL FORWARDED ON CISCO CORE NODES.	MODELS PROVIDER OPACITY AND PREVENTS INTERNAL HOPS FROM BEING EXPOSED UNNECESSARILY.

MPLS Top of Stack and Bottom of Stack:

An MPLS packet can contain one or multiple labels in its label stack. The Top of Stack (ToS) is the label closest to the Layer 3/IP header. Routers process the top label first when forwarding an MPLS packet. The top label determines the forwarding action, such as swap, push, or pop. Additional labels below the top label provide deeper forwarding context. The Bottom of Stack (BoS) is the last label in the MPLS stack, closest to the payload. The BoS bit is set to 1 in the bottom label and 0 in all labels above it. For MPLS VPNs, the stack commonly contains a transport label on top and a VPN/service label at the bottom. The transport label gets removed or swapped as the packet traverses the MPLS core. The BoS label ultimately identifies the service/FEC context used when the packet reaches the egress PE.

6.2. MPLS LDP FEATURES



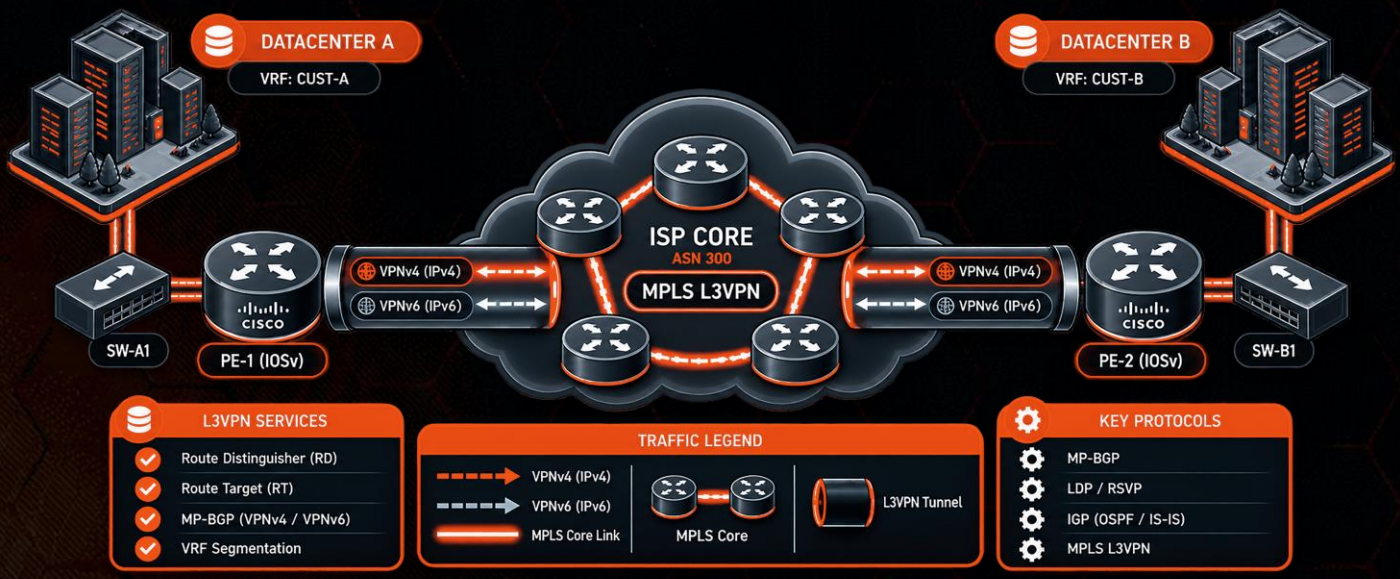
LDP IS THE LABEL DISTRIBUTION ENGINE OF THE LAB. THE DESIGN USES LOCAL LDP ADJACENCIES ON MPLS-ENABLED LINKS AND TARGETED LDP SESSIONS TO BUILD REMOTE LABEL RELATIONSHIPS BETWEEN LOOPBACK-IDENTIFIED NODES.

LDP CAPABILITY	DESIGN IMPLEMENTATION
ROUTER-ID STRATEGY	LOOPBACK0 / INTERNAL-TRANSIT LOOPBACKS ARE USED AS LDP ROUTER IDS.
TARGETED HELLOS	TARGETED LDP IS CONFIGURED ACROSS THE PE/CORE SET, INCLUDING PE08-XRV TARGETED NEIGHBORS TO LEGACY ROUTERS.
SESSION PROTECTION	CISCO/AGG/PE08 USE LDP SESSION PROTECTION AND GRACEFUL RESTART FOR CONTROL-PLANE RESILIENCE.
IGP SYNC	CISCO AND XRV MPLS NODES USE LDP/IGP SYNC SO IGP DOES NOT ADVERTISE A PATH AS FULLY USABLE BEFORE LABELS ARE READY.
PASSWORD HANDLING	PE08-XRV USES NEIGHBOR-SPECIFIC PASSWORD DISABLE SYNTAX UNDER MPLS LDP FOR COMPATIBILITY WITH THE LAB PEER SET.

FEC (Forwarding Equivalence Class)

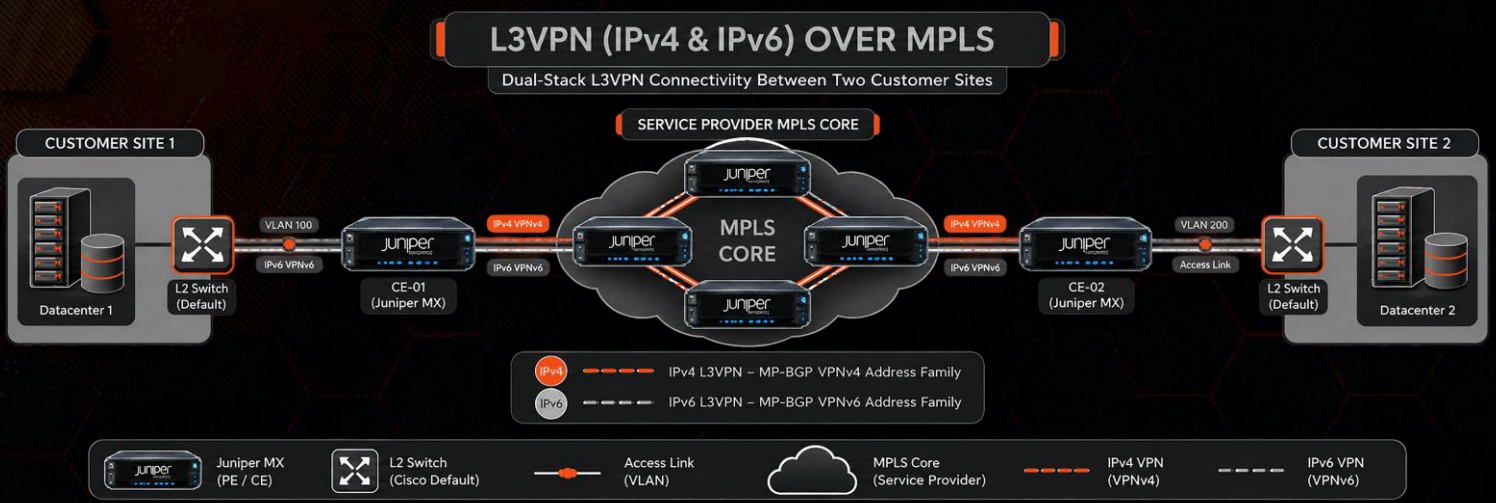
In MPLS, FEC is a group of packets that receive the same forwarding treatment through the network. Packets belonging to the same FEC are assigned the same MPLS label at the ingress LSR. The FEC can be based on criteria such as destination prefix, VPN, QoS requirements, or other forwarding information. Once labeled, intermediate LSRs forward packets based on the label rather than performing a traditional IP routing lookup. The label identifies the FEC and determines the next-hop forwarding action. This creates a Label Switched Path (LSP) through the MPLS network. For example, all packets destined for 10.10.10.0/24 may belong to the same FEC and receive the same label. FECs therefore provide the foundation for label-based forwarding, traffic engineering, QoS, and MPLS VPNs.

6.3. L3VPN FEATURES

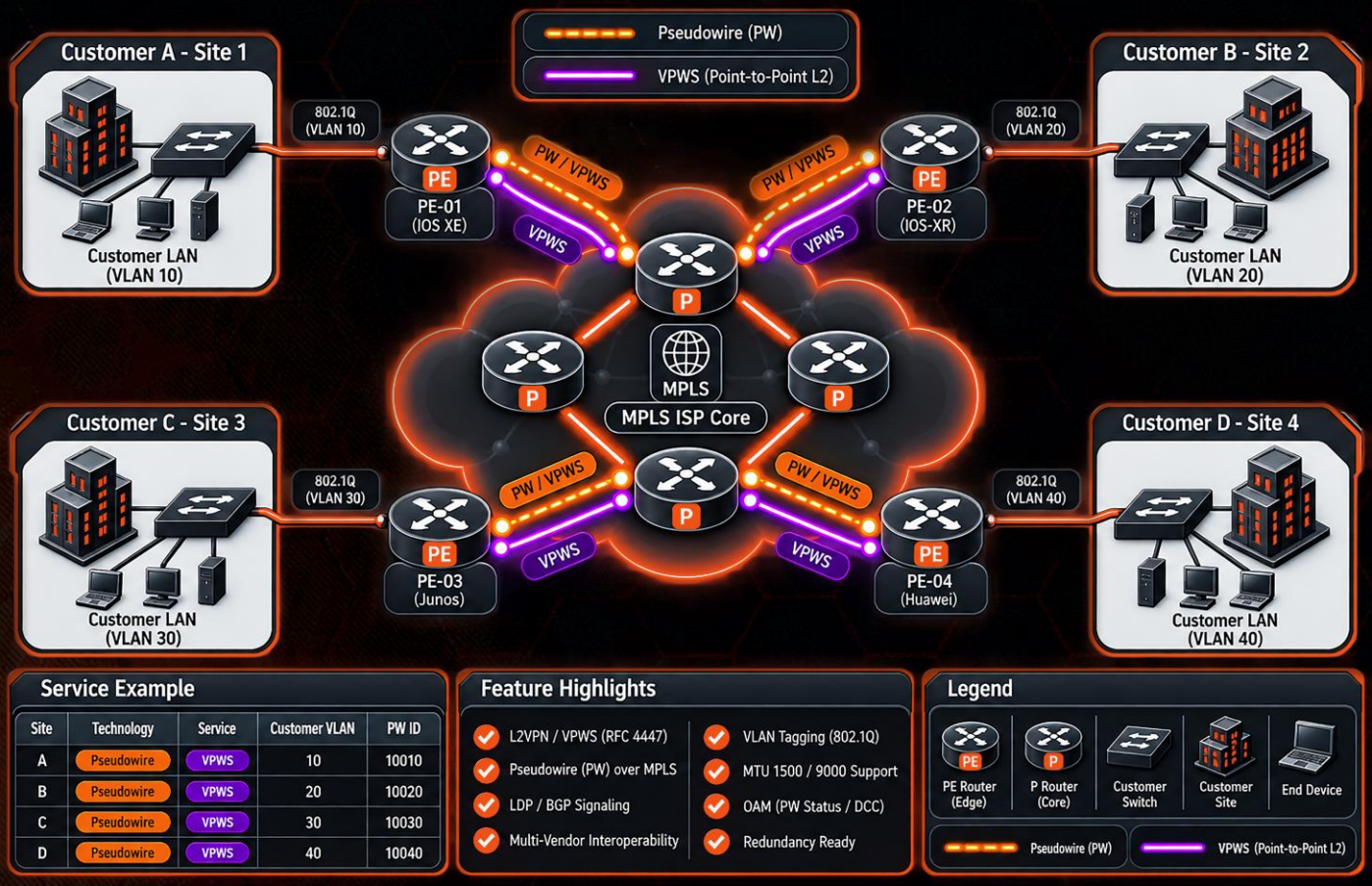


THE LAB INCLUDES MPLS L3VPN BEHAVIOR USING VRF INTER-AS-L3VPN ON THE CUSTOMER-FACING EDGES. VPNV4 IS CARRIED THROUGH THE ROUTE-REFLECTION DESIGN, AND CUSTOMER CE DEVICES USE BGP AS301 TOWARD PROVIDER PE NODES.

ELEMENT	IMPLEMENTATION
CUSTOMER VRF	INTER-AS-L3VPN ON PE/CUSTOMER-FACING NODES.
CE SITES	CE01-L3VPN-01 ON 192.168.111.0/24 AND CE02-L3VPN-01 ON 192.168.222.0/24.
CUSTOMER LANS	192.168.245.0/30 AND 192.168.246.0/30 PLUS LAB LOOPBACK/SERVICE PREFIXES.
PROVIDER CONTROL PLANE	VPNV4 AF IS ACTIVATED BETWEEN RR/PE NODES WITH EXTENDED COMMUNITIES.
POLICY OVERLAY	COMMUNITIES SUCH AS 300:101 AND 300:102 IDENTIFY CUSTOMER ORIGINS; LOCAL-PREF REQUEST COMMUNITIES SUPPORT CUSTOMER-INFLUENCED ROUTING BEHAVIOR.



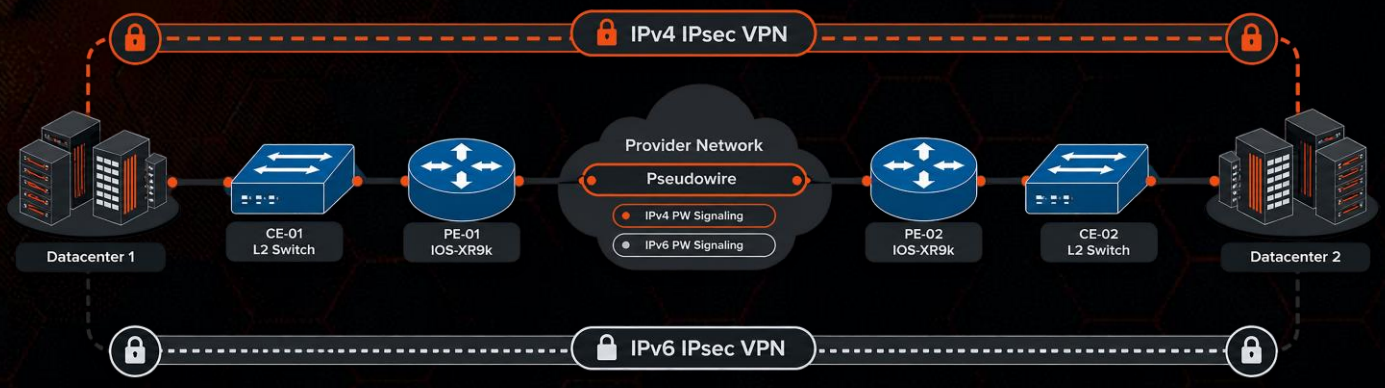
6.4. PSEUDOWIRE / VPWS FEATURES



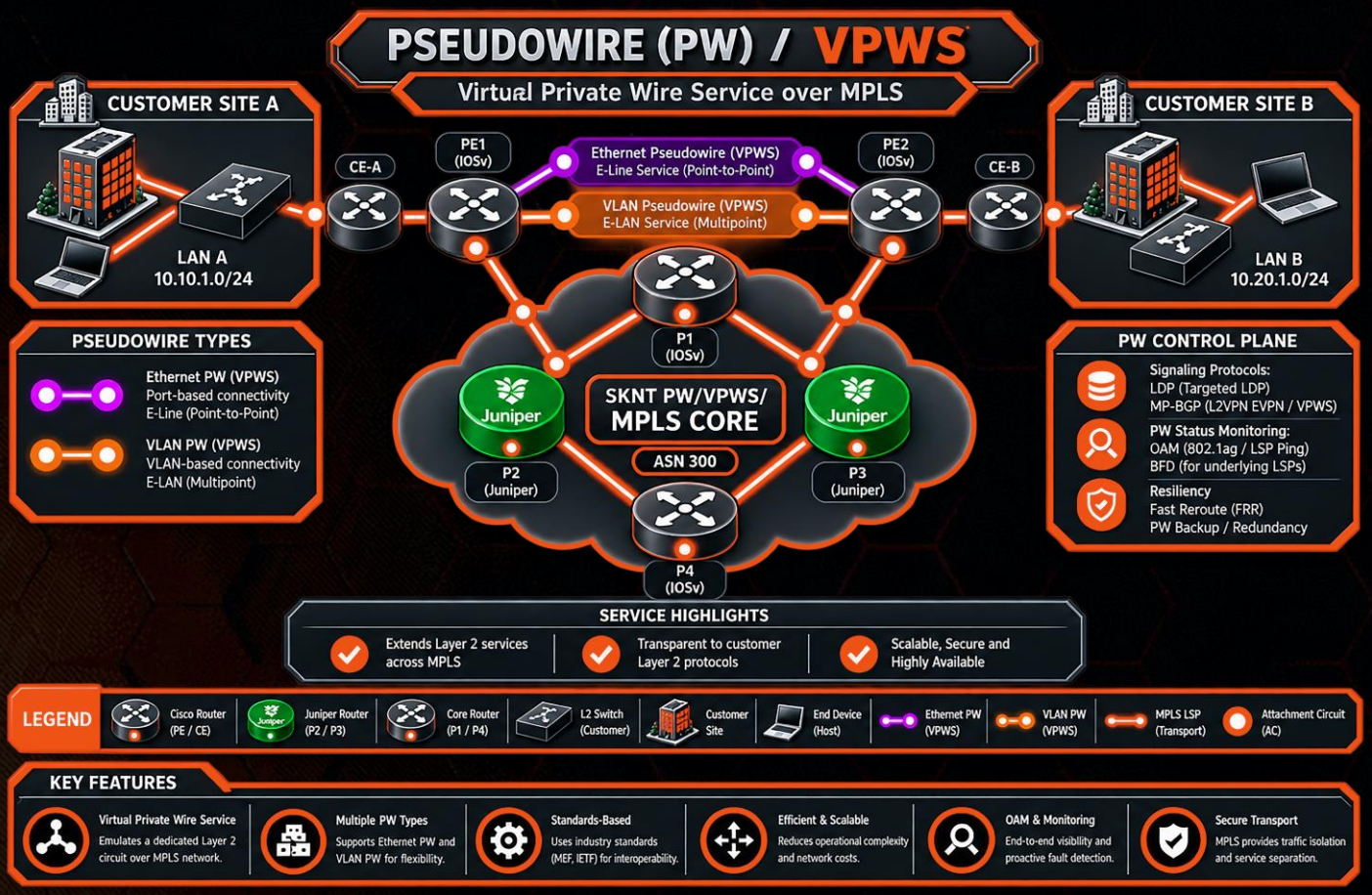
A PSEUDOWIRE IS A VIRTUAL POINT-TO-POINT CIRCUIT THAT CARRIES LAYER 2 TRAFFIC ACROSS AN IP OR MPLS NETWORK. IT MAKES TWO REMOTE CUSTOMER INTERFACES BEHAVE AS THOUGH THEY WERE CONNECTED BY A DIRECT CABLE. PSEUDOWIRES CAN TRANSPORT ETHERNET, VLAN, FRAME RELAY, ATM, AND OTHER LAYER 2 SERVICES. THEY ARE COMMONLY USED FOR VPWS, CARRIER ETHERNET, AND SITE-TO-SITE LAYER 2 EXTENSION.

THE PSEUDOWIRE SERVICE CONNECTS CE-01-PW01 AND CE-02-PW01 THROUGH POP01/POP02. ALSO THERE IS A SECOND PSEUDOWIRE SCHEME WHICH CONNECTS PC1-PW2J AND PC2-PW2J THROUGH PES VMX01 AND VMX02.

THE POP AND PE NODES CARRY XCONNECT/PSEUDOWIRE FEATURES AND APPLY A RICHER SERVICE-EDGE QOS MODEL TO PRESERVE LAN-TO-LAN TRAFFIC BEHAVIOR.

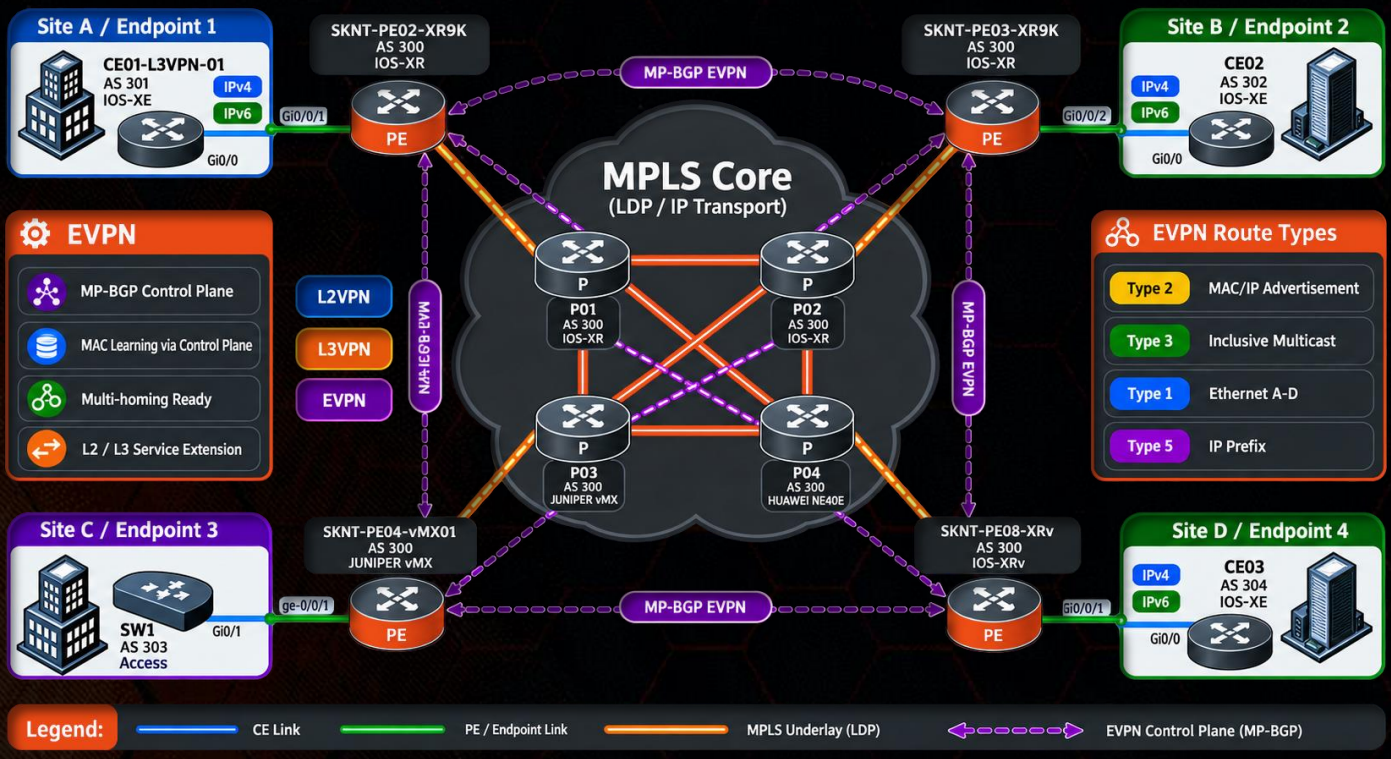


6.5. PSEUDOWIRE / VPWS ELEMENTS IMPLEMENTATION



SERVICE ELEMENT	IMPLEMENTATION
CE ENDPOINTS	CE-01-PW01, CE-02-PW01, PC1-PW2J AND PC2-PW2J.
ACCESS CIRCUITS 01	CE-01 ETHERNET1/1 TO POP01 GE0/5; CE-02 ETHERNET1/1 TO POP02 GE0/5.
ACCESS CIRCUITS 02	PC1-PW2J ETHERNET0/0 TO PE04-vMX01 GE-0/0/7.0. PC2-PW2J ETHERNET0/0 TO PE05-vMX02 GE-0/0/7.0.
CUSTOMER LANS AC01v4 (CISCO)	10.40.40.0/24 AND 10.50.50.0/24.
CUSTOMER LANS AC02v4 (JUNOS)	192.168.171.1/32 AND 192.168.167.2/32.
CUSTOMER LANS AC02v6 (JUNOS)	FD00:2026:7:10::101/128 TO FD00:2026:7:10::105/128. FD00:2026:7:10::201/128 TO FD00:2026:7:10::205/128.
PROVIDER POPs	SKNT-PE06-POP01 AND SKNT-PE07-POP02.
PROVIDER PEs	SKNT-PE04-vMX01 AND SKNT-PE05-vMX02
QOS COUPLING	POPs AND PEs CLASSIFY/MARK VOICE, SIGNALING, VIDEO, MULTIMEDIA, TRANSACTIONAL, CRITICAL-DATA, NET-CONTROL/MGMT AND SCAVENGER/DEFAULT TRAFFIC.
CUSTOMER LANS AC01v6 (CISCO)	2001:DB8:40::/64 AND 2001:DB8:50::/64.
PW TRANSIT DUAL-STACK (CISCO)	IPV4 10.10.20.0/30 AND IPV6 2001:DB8:20::/64 BETWEEN CE-01-PW01 AND CE-02-PW01 OVER VC-ID 1.
PW DYNAMIC ROUTING (CISCO)	OSPFV2 AND OSPFV3 AREA 0 RUN POINT-TO-POINT ACROSS THE PSEUDOWIRE; CUSTOMER-FACING LANS AND SERVICE LOOPBACKS ARE ADVERTISED PASSIVELY.

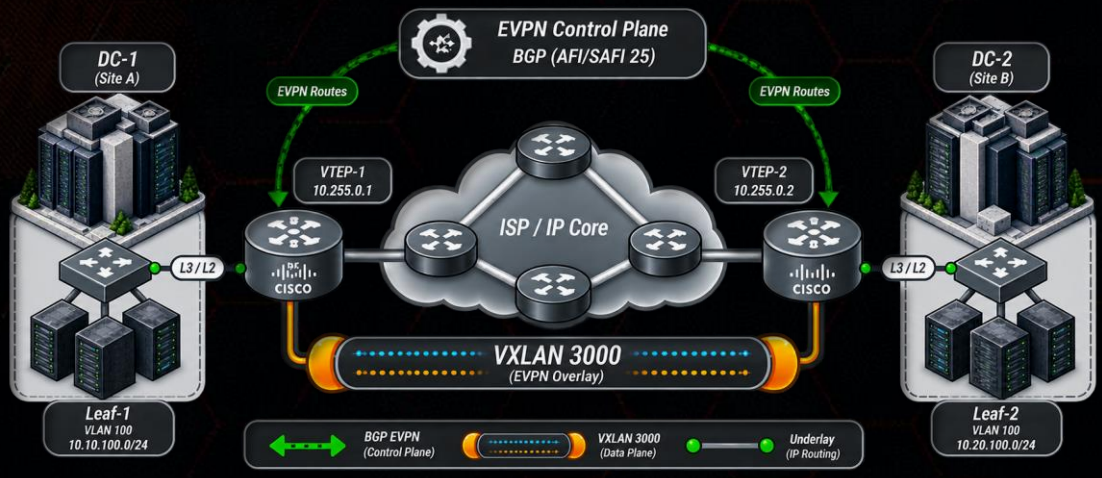
6.6. EVPN FEATURES



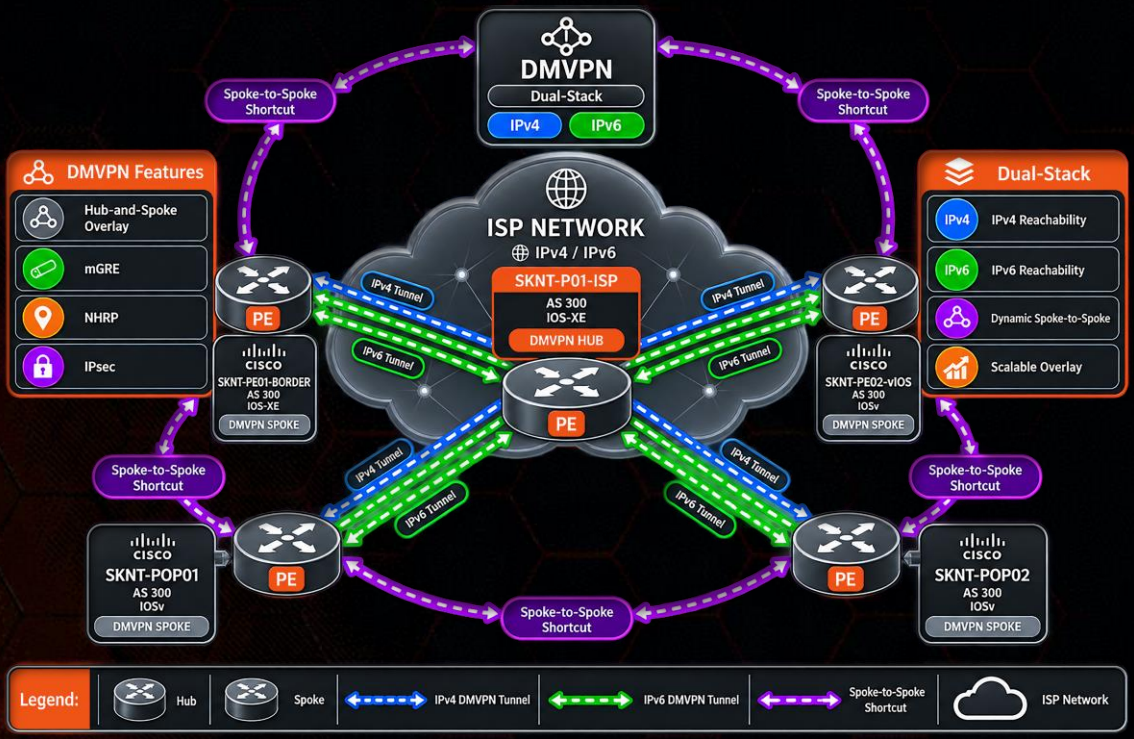
EVPN (ETHERNET VPN) IS A CONTROL-PLANE TECHNOLOGY THAT USES BGP TO ADVERTISE MAC AND IP REACHABILITY BETWEEN NETWORK SITES. IT ENABLES LAYER 2 AND LAYER 3 SERVICES ACROSS AN MPLS OR VXLAN INFRASTRUCTURE. UNLIKE TRADITIONAL FLOODING-BASED DESIGNS, EVPN LEARNS ENDPOINTS THROUGH BGP, IMPROVING SCALABILITY AND STABILITY. IT ALSO SUPPORTS MULTIHOMING, REDUNDANCY, WORKLOAD MOBILITY, AND EFFICIENT TRAFFIC FORWARDING.

THE EVPN SEGMENT DEMONSTRATES DATA-CENTER ATTACHMENT AND VLAN EXTENSION BEHAVIOR AROUND VLAN 600. THE JUNIPER VMX NODES CARRY EVPN SIGNALING, WHILE THE DC-EVPN NODES PROVIDE SERVER-FACING VLAN600 INTERFACES AND LOOPBACK IDENTITIES.

EVPN COMPONENTS	IMPLEMENTATION DETAILS
DC VLAN	VLAN600 / 10.35.0.0/24 ACROSS DC-EVPN-01 THROUGH DC-EVPN-04.
DC NODES	DC-EVPN-01/02 ATTACH TO PE04-VMX01, DC-EVPN-03 TO PE03-VMX03, DC-EVPN-04 TO PE05-VMX02.
CONTROL PLANE	JUNIPER IBGP FAMILIES INCLUDE EVPN SIGNALING WHERE CONFIGURED.
CUSTOMER PROOF POINT	THE LAB CAN DEMONSTRATE DC SEGMENT REACHABILITY INDEPENDENT OF TRADITIONAL ROUTED L3VPN SERVICES.



6.7. DMVPN AND IPSEC FEATURES



THE LAB INCLUDES ENCRYPTED DUAL-STACK OVERLAY FEATURES TO MODEL BRANCH/POP SECURITY SERVICES OVER THE PROVIDER NETWORK. P01-ISP ACTS AS THE DMVPN HUB FOR BOTH IPV4 AND IPV6 DMVPN, WHILE PE01, PE02, PE06 AND PE07 CARRY THE SAME HUB-AND-SPOKE DESIGN USING TUNNEL10 (IPV4) AND TUNNEL20 (IPV6) OVERLAY FUNCTIONS. POP01 AND POP02 ALSO HAVE POINT-TO-POINT IPSEC VPN TUNNEL ADDRESSING USING OSPFV2 FOR ROUTE ADVERTISEMENT AND CONVERGENCE.

OVERLAY FEATURE	IMPLEMENTATION
DMVPN HUB	SKNT-P01-ISP TUNNEL10 10.15.150.1/24, DMVPN ENDPOINT LOOPBACK300 10.254.252.1/32.
DMVPN SPOKES	PE01, PE02, PE06 AND PE07 TUNNEL10 ADDRESSES IN 10.15.150.0/24 SUBNET.
IPSEC POP OVERLAY	PE06 TUNNEL100 192.153.153.1/30 AND PE07 TUNNEL200 192.153.153.2/30.
VALUE	DEMONSTRATES THAT THE PROVIDER LAB CAN CARRY SECURE OVERLAYS WHILE STILL VALIDATING MPLS AND BGP CONTROL-PLANE BEHAVIOR.

DMVPN, OR DYNAMIC MULTIPOINT VPN, IS A CISCO VPN TECHNOLOGY DESIGNED TO CONNECT MULTIPLE REMOTE SITES DYNAMICALLY OVER AN IP NETWORK. IT COMBINES MGRE (MULTIPOINT GRE), NHRP, AND IPSEC TO CREATE SCALABLE ENCRYPTED TUNNELS BETWEEN ROUTERS. INSTEAD OF CONFIGURING A SEPARATE TUNNEL FOR EVERY BRANCH, A CENTRAL HUB CAN SUPPORT MANY SPOKE ROUTERS THROUGH ONE MULTIPOINT INTERFACE. SPOKES INITIALLY REGISTER WITH THE HUB USING NHRP, WHICH KEEPS TRACK OF THEIR PUBLIC IP ADDRESSES. WHEN TWO SPOKES NEED TO COMMUNICATE, DMVPN CAN DYNAMICALLY CREATE A DIRECT SPOKE-TO-SPOKE TUNNEL WITHOUT SENDING ALL TRAFFIC THROUGH THE HUB. THIS REDUCES LATENCY, BANDWIDTH USAGE, AND CONFIGURATION COMPLEXITY IN LARGE WAN ENVIRONMENTS. DMVPN IS COMMONLY USED TO SECURELY CONNECT BRANCH OFFICES, DATACENTERS, AND REMOTE LOCATIONS OVER THE INTERNET OR OTHER IP TRANSPORT NETWORKS. ITS MAIN ADVANTAGE IS PROVIDING A SCALABLE, FLEXIBLE, AND ENCRYPTED WAN ARCHITECTURE WITHOUT REQUIRING A FULL MESH OF STATICALLY CONFIGURED TUNNELS.

7. DETAILED 8 CLASS QOS MODEL ADOPTION



AN 8-CLASS QOS MODEL DIVIDES TRAFFIC INTO EIGHT PRIORITY CATEGORIES BASED ON BUSINESS IMPORTANCE AND PERFORMANCE NEEDS. TYPICAL CLASSES INCLUDE NETWORK CONTROL, REAL-TIME VOICE, VIDEO, CRITICAL APPLICATIONS, SIGNALING, STANDARD DATA, BULK TRAFFIC, AND SCAVENGER TRAFFIC. EACH CLASS RECEIVES SPECIFIC BANDWIDTH, QUEUEING, LATENCY, POLICING, AND DROP TREATMENT. THIS ENSURES CRITICAL TRAFFIC IS PROTECTED DURING CONGESTION WHILE LOWER-PRIORITY TRAFFIC USES REMAINING CAPACITY.

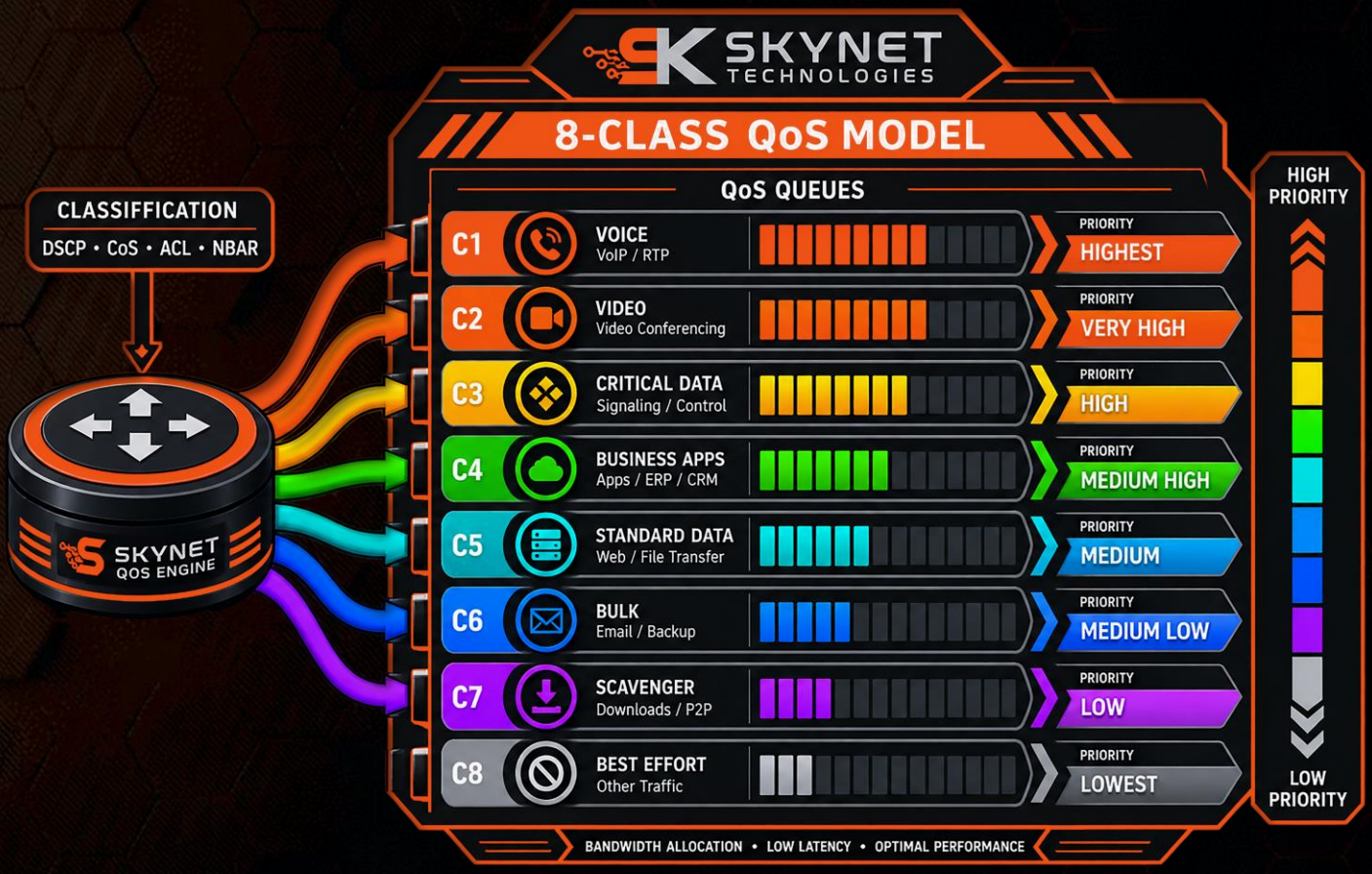
THE QOS ARCHITECTURE FOLLOWS THE PPDIOO MINDSET: DETAILED CLASSIFICATION WHERE THE TRAFFIC ENTERS A SERVICE, SCALABLE TREATMENT IN THE TRANSPORT CORE, AND VENDOR-APPROPRIATE SCHEDULER TRANSLATION ON CISCO, JUNIPER AND IOS-XR. THE INTENT IS NOT JUST TO MARK PACKETS; IT IS TO PROVE CONGESTION BEHAVIOR BEFORE THE DESIGN TOUCHES A CUSTOMER NETWORK.

CLASS / MODEL	TRAFFIC MATCHED	TREATMENT AND DESIGN VALUE
NETWORK CONTROL / ROUTING	OSPF, BGP, LDP, RSVP, BFD/NTP/SSH DEPENDING ON NODE ACL	PRIORITY/STRICT-HIGH TREATMENT OR PROTECTED QUEUE. KEEPS THE CONTROL-PLANE STABLE DURING CONGESTION.
VOICE	DSCP EF	PRIORITY QUEUE/POLICER ON CISCO AND HIGH-PRIORITY SCHEDULER ON JUNOS.
CRITICAL DATA	DSCP AF31 / CRITICAL FORWARDING CLASS	RESERVED BANDWIDTH/MEDIUM-HIGH SCHEDULER, USED FOR BUSINESS-CRITICAL TRAFFIC.
SCAVENGER	DSCP CS1	POLICED OR LOW-PRIORITY QUEUE WITH MINIMAL BANDWIDTH TO PREVENT BULK TRAFFIC FROM HURTING SERVICES.
DEFAULT / BEST-EFFORT	ALL REMAINING TRAFFIC	FAIR-QUEUE OR REMAINDER BANDWIDTH.
PSEUDOWIRE 8-CLASS EDGE MODEL	VOICE, SIGNALING, INTERACTIVE VIDEO, MULTIMEDIA, CRITICAL-DATA, TRANSACTIONAL, NET-CONTROL/MGMT, SCAVENGER/DEFAULT	POP PE NODES PERFORM DETAILED SERVICE-EDGE MARKING AND SCHEDULING FOR LAN-TO-LAN PSEUDOWIRE TRANSPORT.

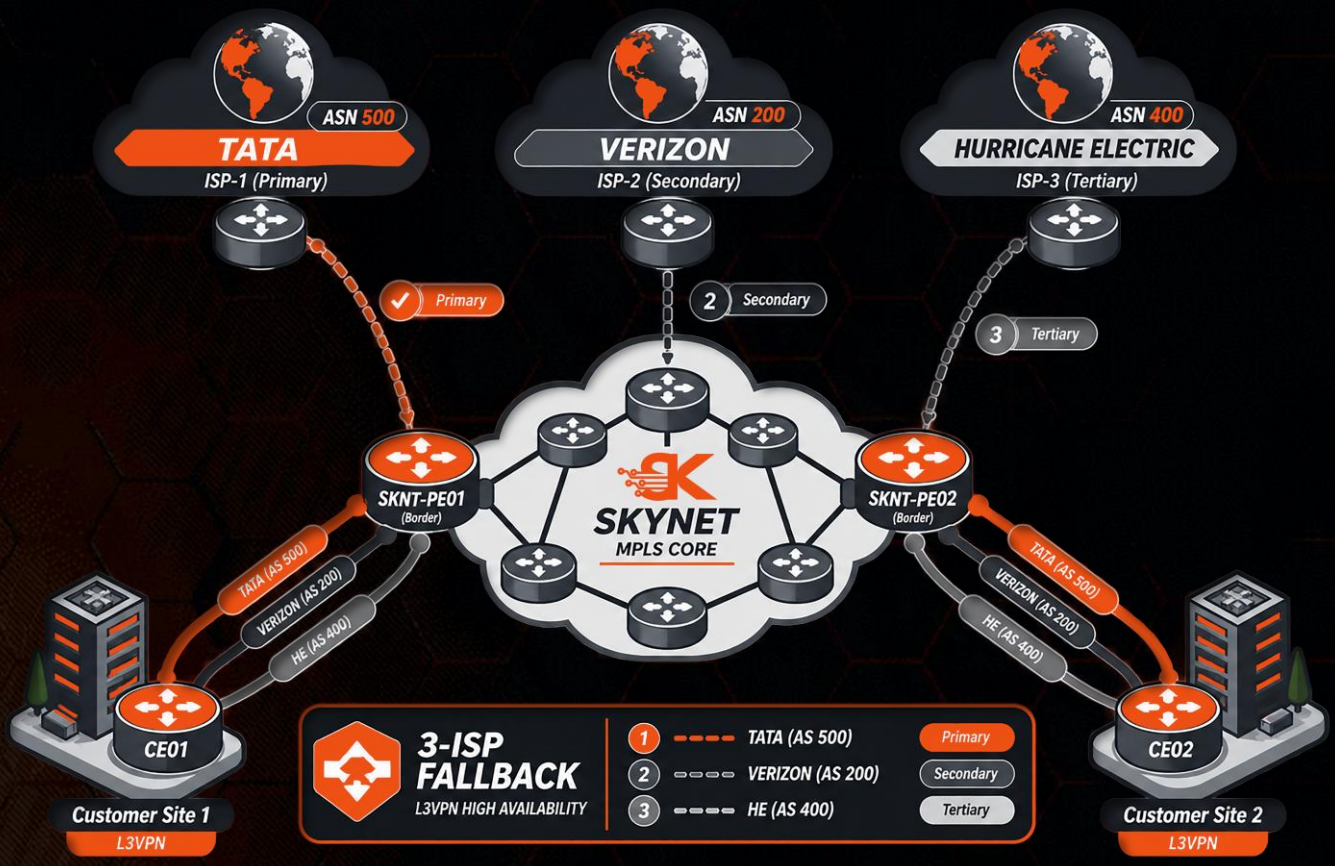


7.1. QOS BY DEVICE FAMILY

FAMILY	CONFIGURATION STYLE	OBSERVED BEHAVIOR
CISCO IOS/IOL	CLASS-MAP, POLICY-MAP, PARENT/CHILD SHAPING, PRIORITY, BANDWIDTH PERCENT, POLICE AND FAIR-QUEUE.	APPLIED ON PE/CORE/CUSTOMER/POP INTERFACES; CORE MODEL PROTECTS CONTROL, VOICE, CRITICAL AND SCAVENGER.
CISCO IOS-XRV PE08	CLASS-MAP/POLICY-MAP USING PRIORITY LEVELS, POLICE RATES AND PARENT SHAPE AVERAGE 20000 KBPS.	MIRRORS THE AGG.SW 20 MBPS SHAPED MODEL ON THE PE08-FACING AGGREGATION LINK.
JUNIPER VMX	CLASS-OF-SERVICE DSCP CLASSIFIERS, FORWARDING-CLASSES, SCHEDULER MAPS AND SCHEDULERS.	MAPS DSCP TO BEST-EFFORT, SCAVENGER, CRITICAL, VOICE AND NETWORK-CONTROL FORWARDING CLASSES.
PSEUDOWIRE SERVICE EDGE	DEDICATED PW CLASS-MAPS AND MARKING POLICIES ON PE06/PE07.	THE MOST DETAILED TRAFFIC CLASSIFICATION LIVES CLOSEST TO THE CUSTOMER SERVICE EDGE.



8. L3VPN CUSTOMER THREE-ISP FALLBACK

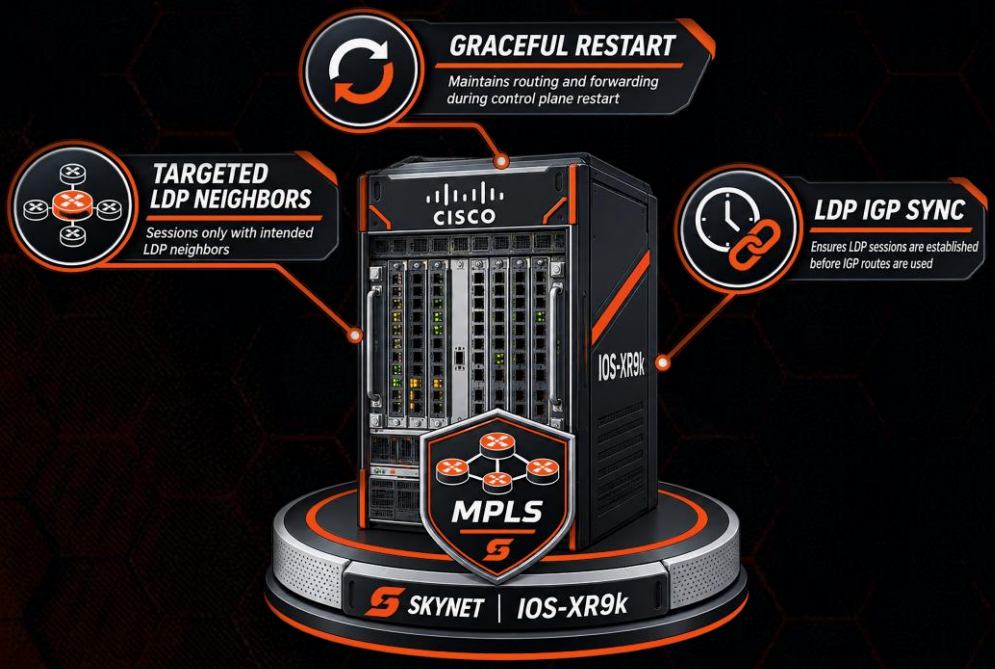


AN L3VPN IS A PROVIDER-MANAGED LAYER 3 SERVICE THAT CONNECTS MULTIPLE CUSTOMER SITES USING ISOLATED VRFS. CUSTOMER ROUTES ARE EXCHANGED THROUGH ROUTING PROTOCOLS SUCH AS BGP, OSPF, OR STATIC ROUTING. ACROSS THE PROVIDER BACKBONE, VPN ROUTES ARE CARRIED USING MP-BGP AND TYPICALLY FORWARDED WITH MPLS LABELS. EACH CUSTOMER REMAINS LOGICALLY SEPARATED EVEN WHILE SHARING THE SAME PHYSICAL INFRASTRUCTURE.

THE CUSTOMER-FACING L3VPN FALLBACK STORY REMAINS TATA FIRST, VERIZON SECOND AND HE LAST RESORT FOR THE 172.19.X.X SERVICE PREFIXES. SPARKLE ISP DOES NOT PARTICIPATE IN THE FALLBACK SCHEME.

ORDER	PROVIDER	ROLE	POLICY CONTROL
1	**TATA AS500**	PREFERRED UPSTREAM FOR 172.19.10.1 THROUGH 172.19.50.1	COMMUNITY 300:500; STRICT INBOUND/OUTBOUND PREFIX LISTS; DUAL IPV4 LINKS FROM PE02.
2	**VERIZON AS200**	BACKUP WHEN TATA PATH IS REMOVED	COMMUNITY 300:200; JUNOS IMPORT/EXPORT POLICIES; VMX03/PE05 PARTICIPATION.
3	**HE AS400**	LAST RESORT IN THE ORIGINAL THREE-ISP DESIGN	COMMUNITY 300:400; HE-SPECIFIC INBOUND PREFIX FILTERS; PE01 POLICY LOWERS LOCAL PREFERENCE FOR HE-LEARNED IPV4 TEST PREFIXES.

9. MPLS RESILIENCE, SECURITY AND OPERATIONS



GRACEFUL RESTART

GRACEFUL RESTART ALLOWS A ROUTER TO RESTART ITS ROUTING PROCESS WITHOUT IMMEDIATELY DISRUPTING PACKET FORWARDING. NEIGHBORING ROUTERS TEMPORARILY PRESERVE THE AFFECTED ROUTES WHILE THE CONTROL PLANE RECOVERS. THIS REDUCES TRAFFIC LOSS DURING SOFTWARE UPGRADES, PROCESS RESTARTS, OR BRIEF CONTROL-PLANE FAILURES. IT REQUIRES COMPATIBLE PROTOCOL SUPPORT AND CORRECT CONFIGURATION ON CONNECTED PEERS.

BGP GRACEFUL RESTART APPEARS ACROSS THE MAJOR BGP NODES, INCLUDING CISCO IOS, JUNIPER AND IOS-XRV COMPONENTS. THIS GIVES THE CONTROL PLANE A REALISTIC RESTART BEHAVIOR AND LETS THE LAB DEMONSTRATE SESSION RESILIENCY DURING PLANNED PROCESS RESETS OR DEVICE MAINTENANCE SIMULATIONS.

TARGETED LDP NEIGHBORS

TARGETED LDP EXTENDS LABEL BINDING BEYOND DIRECTLY CONNECTED ADJACENCIES. THE DESIGN INCLUDES TARGETED NEIGHBOR DEFINITIONS ACROSS CORE AND PE LOOPBACKS, INCLUDING THE PE08-XRV INTEGRATION TO THE LDP ROUTER-ID SET.

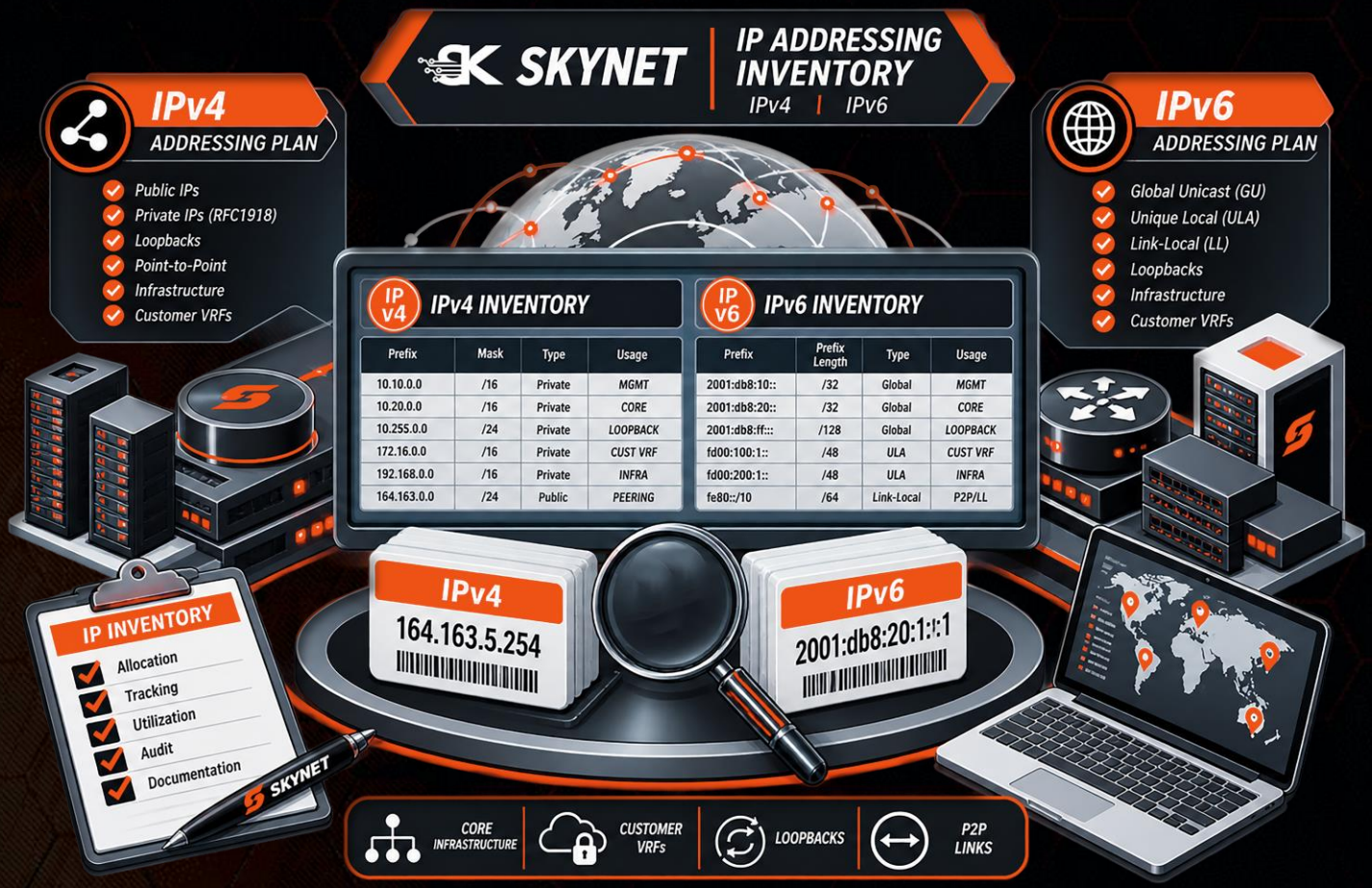
LDP IGP SYNC

LDP/IGP SYNC PREVENTS IGP FROM ATTRACTING TRAFFIC OVER AN MPLS PATH BEFORE LABEL DISTRIBUTION IS READY. THIS IS ESPECIALLY IMPORTANT IN SERVICE-PROVIDER NETWORKS CARRYING L3VPN OR PSEUDOWIRE TRAFFIC, BECAUSE UNLABELED TRANSIT CAN BECOME A SILENT SERVICE KILLER.

10. FEATURE MATRIX

DEVICE	EIGRP V4 V6	RIP V2	RIP-NG	OSPFV2	OSPFV3	BGP	MPLS/LDP	L3VPN	PW	EVPN	QOS	DMVPN/ NHRP V6	GR
SKNT-PE01-BORDER	-	-	-	YES	YES	YES	YES	YES	-	-	YES	YES	YES
SKNT-P01-ISP	-	-	-	YES	YES	YES	YES	YES	-	-	YES	YES	YES
SKNT-PE02-VIOS	-	-	-	YES	YES	YES	YES	YES	-	-	YES	YES	YES
SKNT-PE04-VMX01	-	-	-	YES	YES	YES	YES	YES	-	YES	YES	-	YES
SKNT-PE05-VMX02	-	-	-	YES	YES	YES	YES	YES	-	YES	YES	-	YES
SKNT-PE03-VMX03	-	-	-	YES	YES	YES	YES	YES	-	YES	YES	-	YES
SKNT-AGG.SW	-	-	-	YES	YES	YES	YES	YES	-	-	YES	-	YES
SKNT-PE06-POP01	-	-	-	YES	YES	YES	YES	YES	YES	-	YES	YES	YES
SKNT-PE07-POP02	-	-	-	YES	YES	YES	YES	YES	YES	-	YES	YES	YES
ISP-VERIZON	-	-	-	-	-	YES	-	-	-	-	-	-	YES
ISP-TATA	-	-	-	-	-	YES	YES	YES	-	-	-	-	YES
ISP-HE	-	-	-	-	-	YES	-	-	-	-	-	-	YES
DC-EVPN-01	YES	-	-	-	-	-	-	-	-	YES	-	-	-
DC-EVPN-02	YES	-	-	-	-	-	-	-	-	YES	-	-	-
DC-EVPN-03	YES	-	-	-	-	-	-	-	-	YES	-	-	-
DC-EVPN-04	YES	-	-	-	-	-	-	-	-	YES	-	-	-
CE-01-PW01	-	-	-	-	-	-	YES	-	-	-	-	-	-
CE-02-PW01	-	-	-	-	-	-	YES	-	-	-	-	-	-
CE01-L3VPN-01	-	-	-	-	-	YES	-	YES	-	-	YES	-	YES
CE02-L3VPN-01	-	-	-	-	-	YES	-	YES	-	-	-	-	YES
SKNT-PE08-XRV	-	-	-	YES	YES	YES	YES	STUB	-	-	YES	-	YES
ISP-SPARKLE	-	-	-	-	-	YES	-	-	-	-	-	-	-
EVC-DC-01	-	YES	YES	-	-	-	-	-	-	-	-	-	-
EVC-DC-02	-	YES	YES	-	-	-	-		-	-		-	-

11. DEVICE ADDRESSING INVENTORY



THE FOLLOWING INVENTORY LISTS THE CONFIGURED IPV4 AND IPV6 INTERFACES PER DEVICE. L2-ONLY ATTACHMENT PORTS ARE INCLUDED WHEN THE INTERFACE DESCRIPTION IS RELEVANT TO TOPOLOGY OR SERVICE MAPPING. THIS APPENDIX IS INTENTIONALLY COMPACT SO THE DOCUMENT REMAINS USEFUL AS AN EDITABLE OPERATIONAL REFERENCE.



SKNT-PE01-BORDER

BORDER PE / HE + CE01 L3VPN EDGE | CISCO IOSV/IOL 15.6 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, DMVPN/IPSEC, QOS, GRACEFUL RESTART, TARGETED LDP, LDP IGP SYNC, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	INTERNAL-TRANSIT	192.168.1.1/32	FC50::1/128	UP
LOOPBACK1	PROVIDER-LAN	172.16.0.254/32	-	UP
LOOPBACK10	CE01-L3VPN-DC01	192.168.98.1/32	-	UP
LOOPBACK20	CE01-L3VPN-DC02	192.168.98.2/32	-	UP
LOOPBACK30	CE01-L3VPN-DC03	192.168.98.3/32	-	UP
LOOPBACK40	CE01-L3VPN-DC04	192.168.98.4/32	-	UP
LOOPBACK50	CE01-L3VPN-DC05	192.168.98.5/32	-	UP
LOOPBACK300	DMVPN-ENDPOINT-PE01	10.254.250.1/32	-	UP
TUNNEL10	DMVPN-SPOKE-TO-P01	10.15.150.8/24	-	UP
GIGABITETHERNET0/0	TO-SKNT-P01-ISP	10.0.0.1/30	FC02::/127	UP
GIGABITETHERNET0/1	TO-HOME-WAN	192.168.100.100/24	-	UP
GIGABITETHERNET0/2	BGP-LINK-HE-SECONDARY	10.206.0.1/30	FC99::1/127	UP
GIGABITETHERNET0/3	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/4	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/5	TO-CE01-L3VPN-01 / INTER-AS-L3VPN	192.168.111.1/24	-	UP
GIGABITETHERNET0/6	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/7	NOT-USED	-	-	SHUTDOWN

SKNT-P01-ISP

CORE P / PRIMARY ROUTE-REFLECTOR | CISCO IOSV/IOL 15.6 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, DMVPN/IPSEC, QOS, GRACEFUL RESTART, TARGETED LDP, LDP IGP SYNC, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	INTERNAL-TRANSIT	192.168.1.2/32	FC51::1/128	UP
LOOPBACK1	PROVIDER-LAN	172.16.0.3/32	-	UP
LOOPBACK300	DMVPN-ENDPOINT-HUB	10.254.252.1/32	-	UP
TUNNEL10	DMVPN-HUB	10.15.150.1/24	-	UP
GIGABITETHERNET0/0	TO-SKNT-PE01-BORDER	10.0.0.2/30	FC02::1/127	UP
GIGABITETHERNET0/1	TO-SKNT-AGG.SW	10.0.0.29/30	FC24::1/127	UP
GIGABITETHERNET0/2	TO-SKNT-PE04-VMX01	10.0.0.5/30	FC01::/127	UP
GIGABITETHERNET0/3	TO-SKNT-PE02-VIOS	10.0.0.21/30	FC03::/127	UP
GIGABITETHERNET0/4	TO-SKNT-PE05-VMX02	10.0.0.9/30	FC04::/127	UP
GIGABITETHERNET0/5	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/6	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/7	NOT-USED	-	-	SHUTDOWN

SKNT-PE02-VIOS

PE / TATA + CE02 L3VPN EDGE | CISCO IOSV/IOL 15.6 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, DMVPN/IPSEC, QOS, GRACEFUL RESTART, TARGETED LDP, LDP IGP SYNC, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	INTERNAL-TRANSIT	192.168.1.250/32	FC65::1/128	UP
LOOPBACK1	PROVIDER-LAN	172.16.0.250/32	-	UP
LOOPBACK10	CE02-L3VPN-DC01	192.168.99.1/32	-	UP
LOOPBACK20	CE02-L3VPN-DC02	192.168.99.2/32	-	UP
LOOPBACK30	CE02-L3VPN-DC03	192.168.99.3/32	-	UP
LOOPBACK40	CE02-L3VPN-DC04	192.168.99.4/32	-	UP
LOOPBACK50	CE02-L3VPN-DC05	192.168.99.5/32	-	UP
LOOPBACK300	DMVPN-ENDPOINT-PE02	10.254.251.1/32	-	UP
TUNNEL10	DMVPN-SPOKE-TO-P01	10.15.150.9/24	-	UP
GIGABITETHERNET0/0	TO-SKNT-P01-ISP	10.0.0.22/30	FC03::1/127	UP
GIGABITETHERNET0/1	TO-TATA-LINK1-V4V6	10.202.0.1/30	FC23::1/127	UP
GIGABITETHERNET0/2	TO-TATA-LINK2-V4V6	10.204.0.1/30	FC68::1/127	UP
GIGABITETHERNET0/3	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/4	TO-SKNT-PE03-VMX03	10.10.11.1/30	FC79::1/127	UP
GIGABITETHERNET0/5	TO-CE02-L3VPN-01 / INTER-AS-L3VPN	192.168.222.1/24	-	UP
GIGABITETHERNET0/6	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/7	NOT-USED	-	-	SHUTDOWN

SKNT-PE04-VMX01

JUNIPER PE / HE PATH + EVPN | JUNIPER VMX 14.1R1.10 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, EVPN, QOS, GRACEFUL RESTART, TARGETED LDP, VRRP.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
GE-0/0/0.0	TO-SKNT-AGG.SW	172.16.0.1/24	FC07::/127	UP
GE-0/0/1.0	TO-SKNT-P01-ISP	10.0.0.6/30	FC01::1/127	UP
GE-0/0/2.0	TO-SKNT-PE06-POP01	10.0.0.13/30	FC05::/127	UP
GE-0/0/3	TO-DC-EVPN-02	-	-	UP
GE-0/0/3.600	TO-DC-EVPN-02	-	-	L2/UP
GE-0/0/4	TO-DC-EVPN-01	-	-	UP
GE-0/0/4.600	TO-DC-EVPN-01	-	-	L2/UP
GE-0/0/5.0	BGP-LINK-HE	10.205.0.1/30	FC77::1/127	UP
GE-0/0/6.300	To-EVC300-CUSTOMER	-	-	L2/UP
GE-0/0/7	-	-	-	SHUTDOWN
GE-0/0/8	-	-	-	SHUTDOWN
GE-0/0/9	-	-	-	SHUTDOWN
LO0.1	INTERNAL-TRANSIT	192.168.1.3/32	FC60::1/128	UP

SKNT-PE05-VMX02

JUNIPER PE / VERIZON PATH + EVPN | JUNIPER VMX 14.1R1.10 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, EVPN, QOS, GRACEFUL RESTART, TARGETED LDP, VRRP.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
GE-0/0/0.0	BGP-LINK-VERIZON	10.200.200.1/30	FC10::1/127	UP
GE-0/0/1.0	TO-SKNT-AGG.SW	172.16.0.2/24	FC07::1/127	UP
GE-0/0/2.0	TO-SKNT-P01-ISP	10.0.0.10/30	FC04::1/127	UP
GE-0/0/3.0	TO-SKNT-PE07-POP02	10.0.0.17/30	FC06::/127	UP
GE-0/0/4	-	-	-	SHUTDOWN
GE-0/0/5	TO-DC-EVPN-04	-	-	UP
GE-0/0/5.600	TO-DC-EVPN-04	-	-	L2/UP
GE-0/0/6.300	To-EVC300-CUSTOMER	-	-	L2/UP
GE-0/0/7	-	-	-	SHUTDOWN
GE-0/0/8	-	-	-	SHUTDOWN
GE-0/0/9	-	-	-	SHUTDOWN
LO0.1	INTERNAL-TRANSIT	192.168.1.5/32	FC61::1/128	UP

SKNT-PE03-VMX03

JUNIPER PE / VERIZON PATH + EVPN | JUNIPER VMX 14.1R1.10 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, EVPN, QOS, GRACEFUL RESTART, TARGETED LDP.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
GE-0/0/0	TO-DC-EVPN-03	-	-	UP
GE-0/0/0.600	TO-DC-EVPN-03	-	-	L2/UP
GE-0/0/1	-	-	-	SHUTDOWN
GE-0/0/2.0	BGP-LINK-VERIZON	10.209.0.1/30	-	UP
GE-0/0/3	-	-	-	SHUTDOWN
GE-0/0/4.0	TO-SKNT-PE02-VIOS	10.10.11.2/30	FC79::/127	UP
GE-0/0/5	-	-	-	SHUTDOWN
GE-0/0/6	-	-	-	SHUTDOWN
GE-0/0/7	-	-	-	SHUTDOWN
GE-0/0/8	-	-	-	SHUTDOWN
GE-0/0/9	-	-	-	SHUTDOWN
LO0.0	INTERNAL-TRANSIT	192.168.1.251/32, 172.16.0.251/32	FC66::1/128	UP

SKNT-AGG.SW

AGGREGATION / PE08 EXPANSION EDGE | CISCO IOL L2/L3 15.2 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, QOS, GRACEFUL RESTART, TARGETED LDP, LDP IGP SYNC, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	INTERNAL-TRANSIT	192.168.1.4/32	FC62::1/128	UP
LOOPBACK1	PROVIDER-LAN	172.16.0.4/32	-	UP
ETHERNET0/0	TO-SKNT-PE04-VMX01	-	-	L2/UP
ETHERNET0/1	TO-SKNT-PE05-VMX02	-	-	L2/UP
ETHERNET0/2	TO-SKNT-PE06-POP01	-	-	L2/UP
ETHERNET0/3	TO-SKNT-PE07-POP02	-	-	L2/UP
ETHERNET1/0	TO-SKNT-P01-ISP	10.0.0.30/30	FC24::/127	UP
ETHERNET1/1	TO-SKNT-PE08-XRV	10.34.0.1/30	FC34::1/127	UP
ETHERNET1/2	NOT-USED	-	-	SHUTDOWN
ETHERNET1/3	NOT-USED	-	-	SHUTDOWN

SKNT-PE06-POP01

POP01 PE / PSEUDOWIRE + IPSEC | CISCO IOSV/IOL 15.6 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, PSEUDOWIRE, DMVPN/IPSEC, QOS, GRACEFUL RESTART, TARGETED LDP, LDP IGP SYNC, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	INTERNAL-TRANSIT	192.168.1.100/32	FC63::1/128	UP
LOOPBACK1	PROVIDER-LAN	172.16.0.100/32	-	UP
LOOPBACK10	PRIVATE-SUBNET-POP01	10.151.151.1/30	-	UP
LOOPBACK300	DMVPN-ENDPOINT-PE06	10.254.253.1/32	-	UP
TUNNEL10	DMVPN-SPOKE-TO-P01	10.15.150.6/24	-	UP
TUNNEL100	VPN-TO-POP02	192.153.153.1/30	-	UP
GIGABITETHERNET0/0	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/1	TO-SKNT-PE07-POP02	10.0.0.25/30	FC20::1/127	UP
GIGABITETHERNET0/2	TO-SKNT-PE04-VMX01	10.0.0.14/30	FC05::1/127	UP
GIGABITETHERNET0/3	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/4	TO-IPSEC-VPN01-EP01	192.151.151.2/30	-	UP
GIGABITETHERNET0/5	TO-CE-01-PW01	-	-	UP
GIGABITETHERNET0/6	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/7	NOT-USED	-	-	SHUTDOWN

SKNT-PE07-POP02

POP02 PE / PSEUDOWIRE + IPSEC | CISCO IOSV/IOL 15.6 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, PSEUDOWIRE, DMVPN/IPSEC, QOS, GRACEFUL RESTART, TARGETED LDP, LDP IGP SYNC, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	INTERNAL-TRANSIT	192.168.1.200/32	FC64::1/128	UP
LOOPBACK1	PROVIDER-LAN	172.16.0.200/32	-	UP
LOOPBACK20	PRIVATE-SUBNET-POP02	10.152.152.1/30	-	UP
LOOPBACK300	DMVPN-ENDPOINT-PE07	10.254.254.1/32	-	UP
TUNNEL10	DMVPN-SPOKE-TO-P01	10.15.150.7/24	-	UP
TUNNEL200	VPN-TO-POP01	192.153.153.2/30	-	UP
GIGABITETHERNET0/0	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/1	TO-SKNT-PE06-POP01	10.0.0.26/30	FC20::/127	UP
GIGABITETHERNET0/2	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/3	TO-SKNT-PE05-VMX02	10.0.0.18/30	FC06::1/127	UP
GIGABITETHERNET0/4	TO-IPSEC-VPN01-EP02	192.152.152.2/30	-	UP
GIGABITETHERNET0/5	TO-CE-02-PW01	-	-	UP
GIGABITETHERNET0/6	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/7	NOT-USED	-	-	SHUTDOWN

ISP-VERIZON

EXTERNAL ISP AS200 | CISCO IOSV 15.4 | AS 200 | FEATURES: BGP, GRACEFUL RESTART, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK1	ANUNCIO-EBGP1	172.19.10.1/32	FC11::1/128	UP
LOOPBACK2	ANUNCIO-EBGP2	172.19.20.1/32	FC12::1/128	UP
LOOPBACK3	ANUNCIO-EBGP3	172.19.30.1/32	FC13::1/128	UP
LOOPBACK4	ANUNCIO-EBGP4	172.19.40.1/32	FC14::1/128	UP
LOOPBACK5	ANUNCIO-EBGP5	172.19.50.1/32	FC15::1/128	UP
ETHERNET0/0	TO-PE-05-VMX-02	10.200.200.2/30	FC10::/127	UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	TO-SKNT-PE03-VMX03	10.209.0.2/30	-	UP
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN

ISP-TATA

EXTERNAL ISP AS500 | CISCO IOSV 15.4 | AS 500 | FEATURES: BGP, MPLS/LDP, L3VPN, GRACEFUL RESTART, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK1	ANUNCIO-EBGP1	172.19.10.1/32	FC11::1/128	UP
LOOPBACK2	ANUNCIO-EBGP2	172.19.20.1/32	FC12::1/128	UP
LOOPBACK3	ANUNCIO-EBGP3	172.19.30.1/32	FC13::1/128	UP
LOOPBACK4	ANUNCIO-EBGP4	172.19.40.1/32	FC14::1/128	UP
LOOPBACK5	ANUNCIO-EBGP5	172.19.50.1/32	FC15::1/128	UP
ETHERNET0/0	NOT-USED	-	-	SHUTDOWN
ETHERNET0/1	TO-PE02-VIOS-LINK-01	10.202.0.2/30	FC23::/127	UP
ETHERNET0/2	TO-PE02-VIOS-LINK-02	10.204.0.2/30	FC68::/127	UP
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN

ISP-HE

EXTERNAL ISP AS400 | HUAWEI NE40E V800R011C00SPC607B607 | AS 400 | FEATURES: BGP, GRACEFUL RESTART, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
ETHERNET1/0/0	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0/1	TO-CE01-L3VPN-01	10.207.0.2/30	-	UP
ETHERNET1/0/2	TO-SKNT-PE01-BORDER	10.206.0.2/30	FC99::/127	UP
ETHERNET1/0/3	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0/4	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0/5	TO-PE-04-VMX-01	10.205.0.2/30	FC77::/127	UP
ETHERNET1/0/6	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0/7	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0/8	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0/9	NOT-USED	-	-	SHUTDOWN
LOOPBACK1	ANUNCIO-EBGP1	172.19.10.1/32	FC11::1/128	UP
LOOPBACK2	ANUNCIO-EBGP2	172.19.20.1/32	FC12::1/128	UP
LOOPBACK3	ANUNCIO-EBGP3	172.19.30.1/32	FC13::1/128	UP
LOOPBACK4	ANUNCIO-EBGP4	172.19.40.1/32	FC14::1/128	UP
LOOPBACK5	ANUNCIO-EBGP5	172.19.50.1/32	FC15::1/128	UP
NULL0	BLACKHOLE-ROUTES	-	-	UP

ISP-SPARKLE

EXTERNAL ISP AS777 | CISCO IOSV 15.4 | AS 777 | FEATURES: BGP, MPLS/LDP, L3VPN, GRACEFUL RESTART, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK1	SPARKLE-CE01-ADV	172.19.10.1/32	FC11::1/128	UP
LOOPBACK2	SPARKLE-CE02-ADV	172.19.20.1/32	FC12::1/128	UP
LOOPBACK3	SPARKLE-CE03-ADV	172.19.30.1/32	FC13::1/128	UP
LOOPBACK4	SPARKLE-CE04-ADV	172.19.40.1/32	FC14::1/128	UP
LOOPBACK5	SPARKLE-CE05-ADV	172.19.50.1/32	FC15::1/128	UP
ETHERNET0/0	TO-SKNT-PE08-XRV	10.201.201.2/30	FC32::/127	UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN

DC-EVPN-01 (VLAN 600)

DC EVPN CUSTOMER/SERVER SEGMENT | CISCO IOL L2/L3 15.2 | EIGRP v4 AND v6 AT ASN 300 | FEATURES: EVPN, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	DC-01-SERVER-01	172.31.35.1/32	2001:DB8:35:1::1/128	UP
LOOPBACK1	DC-EVPN-01-SERVER-01	172.31.35.5/32	2001:DB8:35:1::5/128	UP
LOOPBACK2	DC-EVPN-01-SERVER-02	172.31.35.6/32	2001:DB8:35:1::6/128	UP
LOOPBACK3	DC-EVPN-01-SERVER-03	172.31.35.7/32	2001:DB8:35:1::7/128	UP
LOOPBACK4	DC-EVPN-01-SERVER-04	172.31.35.8/32	2001:DB8:35:1::8/128	UP
LOOPBACK5	DC-EVPN-01-SERVER-05	172.31.35.9/32	2001:DB8:35:1::9/128	UP
ETHERNET0/0	TO-SKNT-PE04-VMX01	-	-	L2/UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN
VLAN600	TO-SKNT-PE04-VMX01	10.35.0.1/24	-	UP

DC-EVPN-02 (VLAN 600)

DC EVPN CUSTOMER/SERVER SEGMENT | CISCO IOL L2/L3 15.2 | EIGRP v4 AND v6 AT ASN 300 | FEATURES: EVPN, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	DC-02-SERVER-01	172.31.35.2/32	2001:DB8:35:1::2/128	UP
LOOPBACK1	DC-EVPN-02-SERVER-01	172.31.35.10/32	2001:DB8:35:1::10/128	UP
LOOPBACK2	DC-EVPN-02-SERVER-02	172.31.35.11/32	2001:DB8:35:1::11/128	UP
LOOPBACK3	DC-EVPN-02-SERVER-03	172.31.35.12/32	2001:DB8:35:1::12/128	UP
LOOPBACK4	DC-EVPN-02-SERVER-04	172.31.35.13/32	2001:DB8:35:1::13/128	UP
LOOPBACK5	DC-EVPN-02-SERVER-05	172.31.35.14/32	2001:DB8:35:1::14/128	UP
ETHERNET0/0	TO-SKNT-PE04-VMX01	-	-	L2/UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN
VLAN600	TO-SKNT-PE04-VMX01	10.35.0.2/24	-	UP

DC-EVPN-03 (VLAN 600)

DC EVPN CUSTOMER/SERVER SEGMENT | CISCO IOL L2/L3 15.2 | EIGRP v4 AND v6 AT ASN 300 | FEATURES: EVPN, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	DC-03-SERVER-01	172.31.35.3/32	2001:DB8:35:1::3/128	UP
LOOPBACK1	DC-EVPN-03-SERVER-01	172.31.35.15/32	2001:DB8:35:1::15/128	UP
LOOPBACK2	DC-EVPN-03-SERVER-02	172.31.35.16/32	2001:DB8:35:1::16/128	UP
LOOPBACK3	DC-EVPN-03-SERVER-03	172.31.35.17/32	2001:DB8:35:1::17/128	UP
LOOPBACK4	DC-EVPN-03-SERVER-04	172.31.35.18/32	2001:DB8:35:1::18/128	UP
LOOPBACK5	DC-EVPN-03-SERVER-05	172.31.35.19/32	2001:DB8:35:1::19/128	UP
ETHERNET0/0	TO-SKNT-PE03-VMX03	-	-	L2/UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN
VLAN600	TO-SKNT-PE03-VMX03	10.35.0.3/24	-	UP

DC-EVPN-04 (VLAN 600)

DC EVPN CUSTOMER/SERVER SEGMENT | CISCO IOL L2/L3 15.2 | EIGRP v4 AND v6 AT ASN 300 | FEATURES: EVPN, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	DC-04-SERVER-01	172.31.35.4/32	2001:DB8:35:1::4/128	UP
LOOPBACK1	DC-EVPN-04-SERVER-01	172.31.35.20/32	2001:DB8:35:1::20/128	UP
LOOPBACK2	DC-EVPN-04-SERVER-02	172.31.35.21/32	2001:DB8:35:1::21/128	UP
LOOPBACK3	DC-EVPN-04-SERVER-03	172.31.35.22/32	2001:DB8:35:1::22/128	UP
LOOPBACK4	DC-EVPN-04-SERVER-04	172.31.35.23/32	2001:DB8:35:1::23/128	UP
LOOPBACK5	DC-EVPN-04-SERVER-05	172.31.35.24/32	2001:DB8:35:1::24/128	UP
ETHERNET0/0	TO-SKNT-PE05-VMX02	-	-	L2/UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN
VLAN600	TO-SKNT-PE05-VMX02	10.35.0.4/24	-	UP

CE-01-PW01

CUSTOMER EDGE - PSEUDOWIRE SERVICE | CISCO IOSV 15.4 | AS N/A | FEATURES: OSPFV2, OSPFV3, MPLS/LDP, DUAL-STACK PW ROUTING, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	CE-01-LO0-PW	192.168.1.6/32	-	UP
ETHERNET0/0	TO-CEPC01	10.40.40.254/24	2001:DB8:40::254/64	UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0	NOT-USED	-	-	SHUTDOWN
ETHERNET1/1	TO-POP-01-PW-AC	10.10.20.1/30	2001:DB8:20::1/64	UP
ETHERNET1/2	NOT-USED	-	-	SHUTDOWN
ETHERNET1/3	NOT-USED	-	-	SHUTDOWN
LOOPBACK10-50	PW-SERVICE-01..05	10.201.1.1/32 ... 10.201.1.5/32	2001:DB8:201:1::1/128 ... ::5/128	UP

CE-02-PW01

CUSTOMER EDGE - PSEUDOWIRE SERVICE | CISCO IOSV 15.4 | AS N/A | FEATURES: OSPFV2, OSPFV3, MPLS/LDP, DUAL-STACK PW ROUTING, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	CE-02-LO0-PW	192.168.1.7/32	-	UP
ETHERNET0/0	TO-CEPC02	10.50.50.254/24	2001:DB8:50::254/64	UP
ETHERNET0/1	NOT-USED	-	-	SHUTDOWN
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN
ETHERNET1/0	NOT-USED	-	-	SHUTDOWN
ETHERNET1/1	TO-POP-02-PW-AC	10.10.20.2/30	2001:DB8:20::2/64	UP
ETHERNET1/2	NOT-USED	-	-	SHUTDOWN
ETHERNET1/3	NOT-USED	-	-	SHUTDOWN
LOOPBACK10-50	PW-SERVICE-01..05	10.202.1.1/32 ... 10.202.1.5/32	2001:DB8:202:1::1/128 ... ::5/128	UP

CE01-L3VPN-01

CUSTOMER EDGE - L3VPN SERVICE | CISCO IOSV 15.4 | AS 301 | FEATURES: BGP, L3VPN, QOS, GRACEFUL RESTART, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
ETHERNET0/0	TO-SKNT-PE01-BORDER / INTER-AS-L3VPN	192.168.111.2/24	-	UP
ETHERNET0/1	TO-ISP-HE-V4	10.207.0.1/30	-	UP
ETHERNET0/2	CE01-CUSTOMER-LAN / INTER-AS-L3VPN	192.168.245.1/30	-	UP
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN

CE02-L3VPN-01

CUSTOMER EDGE - L3VPN SERVICE | CISCO IOSV 15.4 | AS 301 | FEATURES: BGP, L3VPN, GRACEFUL RESTART, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
ETHERNET0/0	TO-SKNT-PE02-VIOS / INTER-AS-L3VPN	192.168.222.2/24	-	UP
ETHERNET0/1	CE02-CUSTOMER-LAN / INTER-AS-L3VPN	192.168.246.1/30	-	UP
ETHERNET0/2	NOT-USED	-	-	SHUTDOWN
ETHERNET0/3	NOT-USED	-	-	SHUTDOWN

SKNT-PE08-XRV

IOS-XR PE / SPARKLE SIMPLE EBGP EDGE | CISCO IOS XRV 6.1.3 | AS 300 | FEATURES: OSPFV2, OSPFV3, BGP, MPLS/LDP, L3VPN, QOS, GRACEFUL RESTART, TARGETED LDP, LDP IGP SYNC, SSH HARDENING.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	INTERNAL-TRANSIT	192.168.1.101/32	FC93::1/128	UP
LOOPBACK1	PROVIDER-LAN	172.16.0.101/32	-	UP
LOOPBACK16	SKNT-IPV6-ISP-AGGREGATE-ANCHOR	-	FC16::101/128	UP
MGMTETH0/0/CPU0/0	NOT-USED	-	-	SHUTDOWN
GIGABITETHERNET0/0/0/0	TO-ISP-SPARKLE	10.201.201.1/30	FC32::1/127	UP
GIGABITETHERNET0/0/0/1	TO-SKNT-AGG.SW	10.34.0.2/30	FC34::/127	UP
GIGABITETHERNET0/0/0/2	NOT-USED	-	-	SHUTDOWN

EVC-DC-01 (EVC300 VLAN 300)

CUSTOMER EDGE - EVC SERVICE | CISCO IOSV 15.4 | VLAN 300.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	EVC-DC-01-SERVER-01	10.16.16.1/32	-	UP
LOOPBACK1	EVC-DC-01-V4V6-01	192.168.227.1/32	FD00:16:16::1/128	UP
LOOPBACK2	EVC-DC-01-V4V6-02	192.168.227.2/32	FD00:16:16::2/128	UP
LOOPBACK3	EVC-DC-01-V4V6-03	192.168.227.3/32	FD00:16:16::3/128	UP
LOOPBACK4	EVC-DC-01-V4V6-04	192.168.227.4/32	FD00:16:16::4/128	UP
LOOPBACK5	EVC-DC-01-V4V6-05	192.168.227.5/32	FD00:16:16::5/128	UP
LOOPBACK300	SIMULATED-INTERNAL-LAN-DC1	172.16.172.1/32	FD00:172:16::1/128	UP
ETHERNET0/0	To-SKNT-PE04-vMX01	VLAN300 TRUNK	-	UP
VLAN300	EVC300-Test-SVI	192.168.131.1/24	-	UP

EVC-DC-02 (EVC300 VLAN 300)

CUSTOMER EDGE - EVC SERVICE | CISCO IOSV 15.4 | VLAN 300.

INTERFACE	DESCRIPTION	IPV4	IPV6	STATE
LOOPBACK0	EVC-DC-02-SERVER-01	10.17.17.1/32	-	UP
LOOPBACK1	EVC-DC-02-V4V6-01	192.168.228.1/32	FD00:17:17::1/128	UP
LOOPBACK2	EVC-DC-02-V4V6-02	192.168.228.2/32	FD00:17:17::2/128	UP
LOOPBACK3	EVC-DC-02-V4V6-03	192.168.228.3/32	FD00:17:17::3/128	UP
LOOPBACK4	EVC-DC-02-V4V6-04	192.168.228.4/32	FD00:17:17::4/128	UP
LOOPBACK5	EVC-DC-02-V4V6-05	192.168.228.5/32	FD00:17:17::5/128	UP
LOOPBACK300	SIMULATED-INTERNAL-LAN-DC2	172.16.173.1/32	FD00:173:16::1/128	UP
ETHERNET0/0	To-SKNT-PE05-vMX02	VLAN300 TRUNK	-	UP
VLAN300	EVC300-Test-SVI	192.168.131.2/24	-	UP

12. CUSTOMER VALUE STORYLINE

THIS LAB CAN BE POSITIONED AS A PRE-PRODUCTION CARRIER BLUEPRINT. IT GIVES THE CUSTOMER A SAFE PLACE TO VALIDATE FAILURE BEHAVIOR, VENDOR INTEROPERABILITY, SERVICE PROVISIONING AND OPERATIONAL PROCEDURES BEFORE PRODUCTION CHANGE WINDOWS.

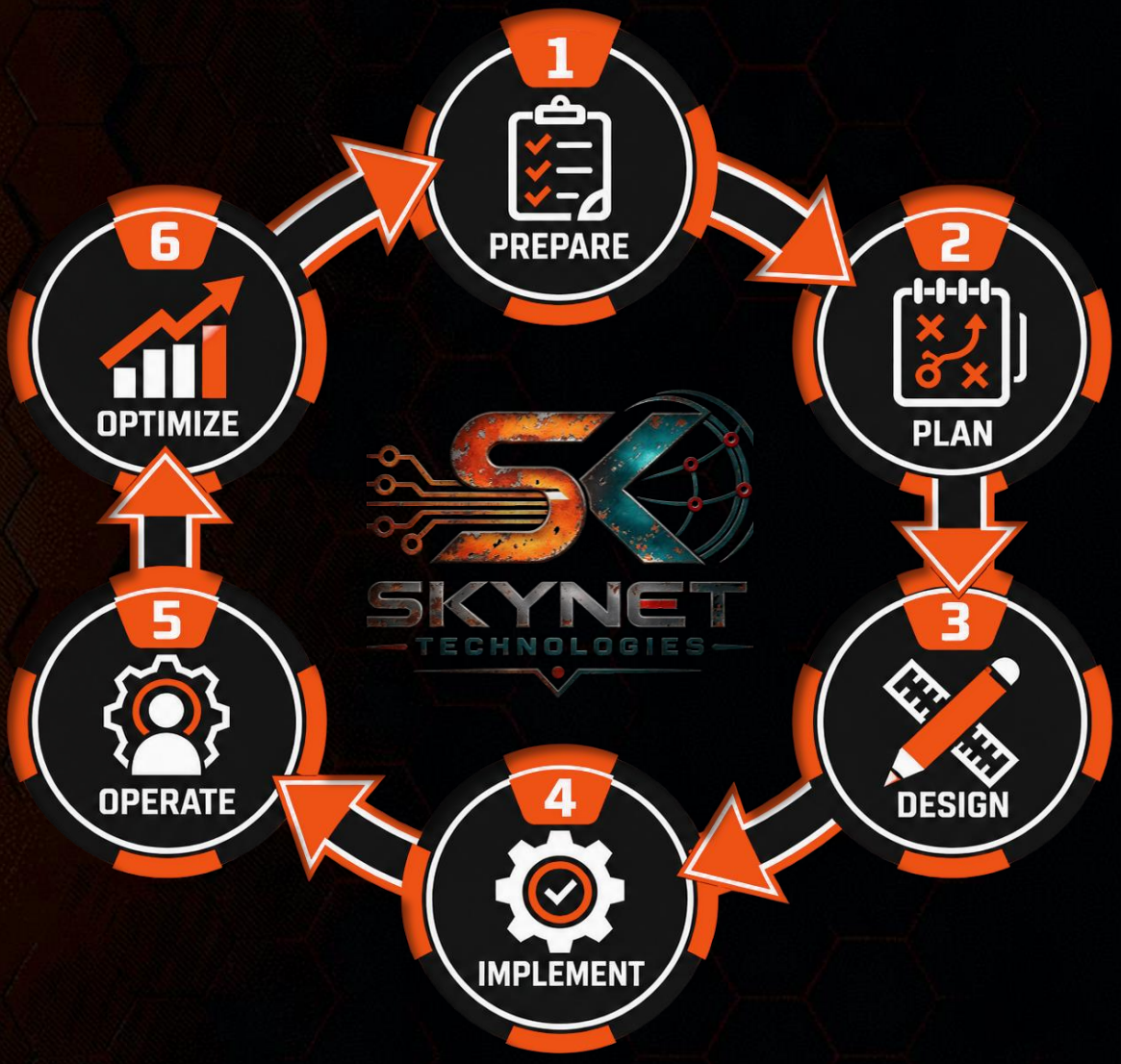
CUSTOMER QUESTION	SKYNET'S DEVELOPER / ARCHITECT ANSWER
CAN THE NETWORK SURVIVE PROVIDER FAILURE?	YES. THE ISP FALLBACK POLICY IS DEMONSTRABLE AND ORDERED: TATA, VERIZON AND HE FOR THE ACTIVE L3VPN STORY. SPARKLE REMAINS AVAILABLE AS A SEPARATE FILTERED EBGP LAB EDGE, NOT AS AN ACTIVE VRF FALLBACK PATH.
CAN THIS SCALE BEYOND A SMALL FULL-MESH?	YES. IBGP ROUTE-REFLECTION IS BUILT INTO THE DESIGN WITH P01, PE04 AND AGG/PE08 EXPANSION ROLES.
CAN WE CARRY REAL CUSTOMER SERVICES?	YES. L3VPN, PSEUDOWIRE, EVPN DC SERVICE AND ENCRYPTED OVERLAYS ARE ALL REPRESENTED.
CAN WE PROVE QOS BEFORE PRODUCTION?	YES. THE LAB HAS CISCO, XR AND JUNIPER QOS MODELS WITH SERVICE-EDGE AND CORE TREATMENT.
CAN OPERATIONS TROUBLESHOOT IT?	YES. INTERFACE DESCRIPTIONS, LLDP, DETERMINISTIC ADDRESSING, COMMUNITIES, ROUTE TAGS AND DEVICE ROLES MAKE THE DESIGN TRACEABLE.



13. PPDIOO FRAMEWORK REFERENCE FOR THE SKYNET ISP

PPDIOO IS USED HERE AS THE LIFECYCLE REFERENCE FOR THE SKYNET ISP LAB: **PREPARE**, **PLAN**, **DESIGN**, **IMPLEMENT**, **OPERATE AND OPTIMIZE**. THE LAB IS NOT ONLY A COLLECTION OF PROTOCOLS; IT IS A CONTROLLED ENGINEERING WORKFLOW WHERE REQUIREMENTS, RISKS, TECHNICAL DESIGN, IMPLEMENTATION STEPS, VALIDATION AND FUTURE OPTIMIZATION ARE ALL DOCUMENTED AS PART OF THE SAME SERVICE-PROVIDER BLUEPRINT.

THE BGP COMMUNITY CONTROL DESIGN FOLLOWS THIS LIFECYCLE DIRECTLY. THE GOAL IS TO PROVIDE CUSTOMER-CONTROLLED ROUTE INTENT AND PROVIDER-SIDE ENFORCEMENT WITHOUT CHANGING THE CORE PREFERENCE ORDER.



PPDIOO PHASE	HOW IT MAPS TO SKYNET	EVIDENCES IN THIS LAB
PREPARE	THE AS300 SERVICE-PROVIDER BASELINE WAS REVIEWED BEFORE FINAL RELEASE. THE LAB CONSTRAINTS ARE RESPECTED: TATA REMAINS PREFERRED, VERIZON REMAINS SECONDARY, HE REMAINS LAST RESORT, SPARKLE REMAINS A SIMPLE FILTERED EBGP/GLOBAL-UNICAST EDGE, SEGMENT ROUTING IS NOT USED, AND BROAD BFD IS AVOIDED.	EXISTING TOPOLOGY, ROUTE-REFLECTOR DESIGN, L3VPN CUSTOMER EDGES, ISP LINKS, COMMUNITY CONVENTIONS AND DEVICE LIMITATIONS WERE PRESERVED.
PLAN	THE NEW CONTROL MODEL WAS PLANNED AROUND ROUTE INTENT. CUSTOMERS CAN REQUEST ROUTING TREATMENT THROUGH BGP COMMUNITIES, WHILE PROVIDER EDGES ENFORCE WHAT MAY LEAVE THE NETWORK.	COMMUNITY NAMESPACE WAS STANDARDIZED WITH 300:777 FOR SPARKLE ORIGIN AND 300:778 FOR INTERNAL-ONLY ROUTES.
DESIGN	THE FEATURE WAS SPLIT INTO TWO LAYERS: CUSTOMER CONTROL CENTER AND EDGE ENFORCEMENT CENTER. CUSTOMER CES SIGNAL INTENT; PROVIDER PES VALIDATE AND TAG; ISP-FACING EDGES BLOCK DANGEROUS OR UNAUTHORIZED EXPORTS.	CE ROUTE-MAPS, PE INBOUND ROUTE-MAPS, JUNOS POLICY-STATEMENTS AND IOS-XR ROUTE-POLICIES WERE ALIGNED TO THE SAME CATALOG.
IMPLEMENT	THE IMPLEMENTATION USES ROUTE-MAPS, COMMUNITY-LISTS, JUNOS COMMUNITIES, JUNOS POLICY-STATEMENTS, IOS-XR COMMUNITY-SETS, SPARKLE ROUTE-POLICIES, AND THE JUNOS EVC300 L2CIRCUIT SERVICE.	CE01/CE02, PE01/PE02, PE03/PE04/PE05, PE08-XRV, ISP-SPARKLE, EVC-DC-01 AND EVC-DC-02 ARE REPRESENTED IN THE RELEASE CONFIGS.
OPERATE	POST-CHANGE VALIDATION VERIFIED BGP ROUTE STATE, COMMUNITIES, LOCAL-PREFERENCE BEHAVIOR, ADVERTISED ROUTES, IMPORT/EXPORT POLICIES AND END-TO-END PINGS.	CE01 TO CE02 AND CE02 TO CE01 REACHED 100 PERCENT SUCCESS, INCLUDING SOURCED PINGS.
OPTIMIZE	THE DESIGN HAS A CLEAN FOUNDATION FOR AUTOMATION, NMS MONITORING, ROUTE-LEAK ALARMS, CONTROLLED RTBH PROCEDURES, EVC SERVICE TURN-UP TESTS AND CUSTOMER-FACING POLICY DOCUMENTATION.	FUTURE WORK CAN INCLUDE MONITORING DASHBOARDS AND CONTROLLED TEST CASES FOR NO-EXPORT AND RTBH BEHAVIOR.

13.1. WHY PPDIOO MATTERS IN THIS ENVIRONMENT?

THE PPDIOO REFERENCE GIVES THE CUSTOMER A WAY TO UNDERSTAND THE LAB AS AN ENGINEERING LIFECYCLE, NOT AS ISOLATED CLI WORK. EVERY FEATURE IN THE LAB IS TRACEABLE FROM A BUSINESS OR TECHNICAL NEED, THROUGH DESIGN, INTO IMPLEMENTATION, OPERATION AND OPTIMIZATION.

- PREPARE: PROTECT THE L3VPN FALLBACK ORDER AND IDENTIFY THE ACTIVE POLICY BOUNDARIES.
- PLAN: DEFINE THE CUSTOMER-FACING AND PROVIDER-FACING COMMUNITY CATALOG BEFORE TOUCHING DEVICES.
- DESIGN: SEPARATE CUSTOMER ROUTE INTENT FROM PROVIDER EXPORT ENFORCEMENT.
- IMPLEMENT: APPLY THE CONFIGURATION USING THE RELEASE DEVICE NAMING AND POLICY NAMING PRESENT IN THE LAB.
- OPERATE: VALIDATE CONTROL-PLANE STATE, COMMUNITIES, ROUTE ADVERTISEMENTS AND END-TO-END REACHABILITY.
- OPTIMIZE: KEEP THE POLICY CATALOG EXTENSIBLE FOR FUTURE CUSTOMER SERVICES AND SECURITY AUTOMATION.

14. BGP COMMUNITY CONTROL AND EDGE ENFORCEMENT CENTER

THE POLICY DESIGN USES AN ISP-GRADE BGP COMMUNITY CONTROL SYSTEM. THE GOAL IS TO MAKE ROUTES CARRY POLICY MEANING, NOT ONLY REACHABILITY. A ROUTE CAN IDENTIFY ITS ORIGIN, REQUEST A LOCAL-PREFERENCE TREATMENT, REQUEST INTERNAL-ONLY HANDLING, OR TRIGGER CONTROLLED RTBH BEHAVIOR.

THE IMPLEMENTATION WAS DIVIDED INTO TWO COORDINATED COMPONENTS: THE CUSTOMER BGP COMMUNITY CONTROL CENTER AND THE EDGE COMMUNITY ENFORCEMENT CENTER. THE FIRST COMPONENT HANDLES CUSTOMER INTENT ON CE01 AND CE02. THE SECOND COMPONENT ENFORCES ROUTE-LEAK PROTECTION AND ISP-SPECIFIC EXPORT POLICY ON CISCO, JUNOS AND IOS-XR EDGE ROUTERS.



14.1. FINAL BGP COMMUNITY NAMESPACE

COMMUNITY	MEANING	OPERATIONAL USE
300:101	CE01 ORIGIN	APPLIED BY THE PROVIDER WHEN ROUTES ARE LEARNED FROM CE01-L3VPN-01.
300:102	CE02 ORIGIN	APPLIED BY THE PROVIDER WHEN ROUTES ARE LEARNED FROM CE02-L3VPN-01.
300:150	NORMAL LOCAL-PREFERENCE REQUEST	DEFAULT CUSTOMER EXPORT COMMUNITY; PROVIDER SETS NORMAL LOCAL PREFERENCE.
300:250	HIGH LOCAL-PREFERENCE REQUEST	OPTIONAL CUSTOMER ROUTE-INTENT COMMUNITY FOR PREFERRED TREATMENT.
300:90	BACKUP LOCAL-PREFERENCE REQUEST	OPTIONAL CUSTOMER ROUTE-INTENT COMMUNITY FOR BACKUP TREATMENT.
300:50	LAST-RESORT LOCAL-PREFERENCE REQUEST	OPTIONAL CUSTOMER ROUTE-INTENT COMMUNITY FOR LOWEST PREFERENCE TREATMENT.
300:200	VERIZON ORIGIN	APPLIED TO ROUTES LEARNED FROM VERIZON AS200.
300:300	SKYNET ORIGIN	APPLIED TO AUTHORIZED SKYNET-ORIGINATED EXPORTS.
300:400	HE ORIGIN	APPLIED TO ROUTES LEARNED FROM HE AS400.
300:500	TATA ORIGIN	APPLIED TO ROUTES LEARNED FROM TATA AS500.
300:777	SPARKLE ORIGIN	RESERVED FOR ROUTES LEARNED FROM ISP-SPARKLE AS777.
300:778	INTERNAL-ONLY	BLOCKS EXPORT OUTSIDE AS300 AT PROVIDER AND ISP-FACING EDGE POLICIES.
300:666	RTBH / BLACKHOLE	USED ONLY FOR AUTHORIZED /32 BLACKHOLE SIGNALING; BLOCKED TOWARD EXTERNAL ISPS.
300:701	DO NOT EXPORT TO HE	CUSTOMER/PROVIDER ROUTE-LEAK PREVENTION TOWARD HE.
300:702	DO NOT EXPORT TO TATA	CUSTOMER/PROVIDER ROUTE-LEAK PREVENTION TOWARD TATA.
300:703	DO NOT EXPORT TO VERIZON	CUSTOMER/PROVIDER ROUTE-LEAK PREVENTION TOWARD VERIZON.

14.2. CUSTOMER CONTROL CENTER - CE01 AND CE02

THE CUSTOMER EDGE ROUTERS ADVERTISE CUSTOMER PREFIXES WITH AN EXPLICIT POLICY COMMUNITY. THE DEFAULT FINAL STATE IS NORMAL TREATMENT, USING COMMUNITY 300:150. THIS KEEPS ROUTING BEHAVIOR DETERMINISTIC WHILE ALLOWING CONTROLLED TESTS FOR HIGH PREFERENCE, BACKUP PREFERENCE, LAST-RESORT PREFERENCE, INTERNAL-ONLY AND RTBH REQUESTS.

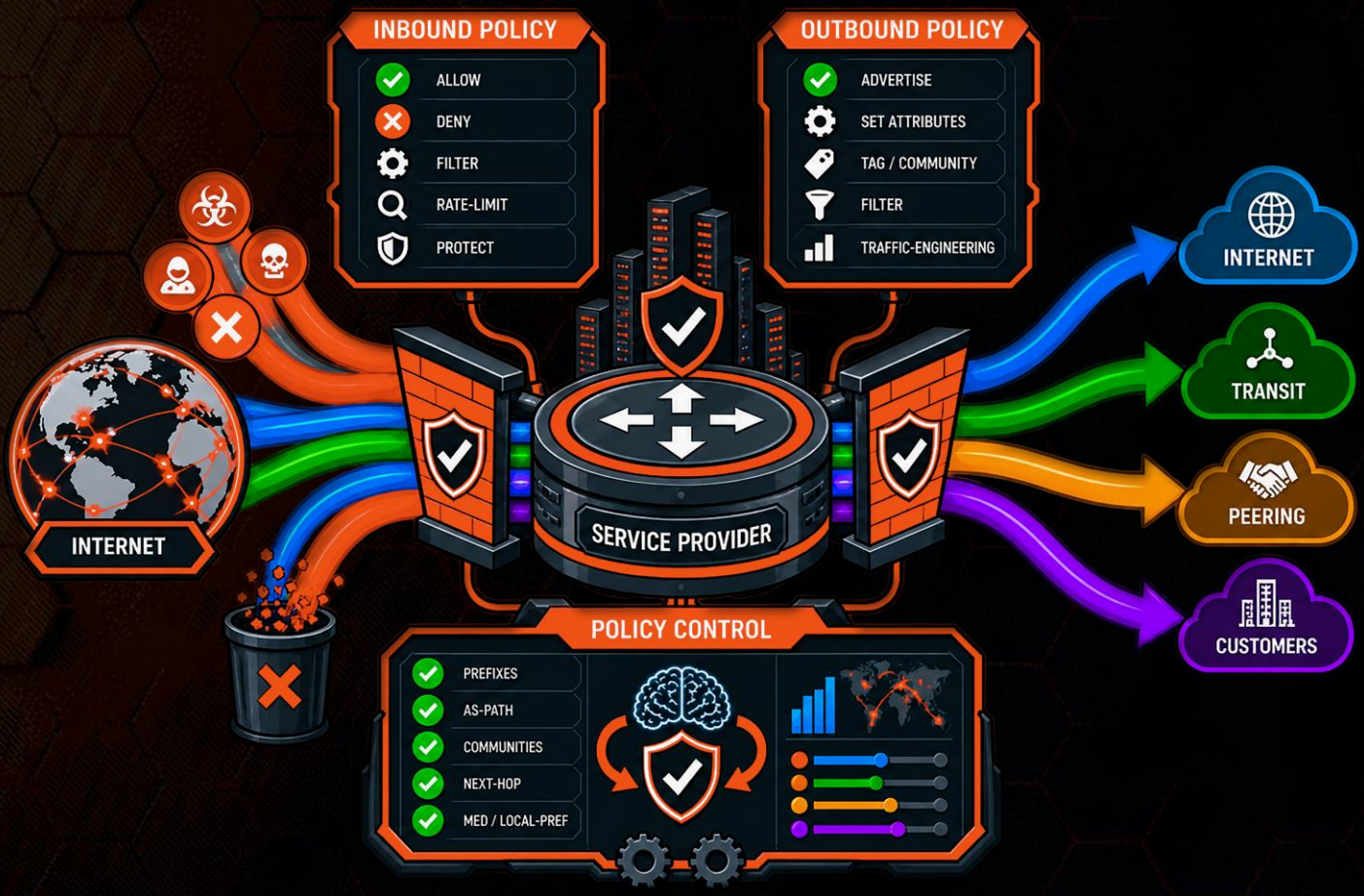
DEVICE	POLICY OBJECT	FINAL DEFAULT BEHAVIOR	PROTECTED PREFIXES
CE01-L3VPN-01	RM-CE01-TO-PE	EXPORTS CE01 CUSTOMER PREFIX WITH COMMUNITY 300:150 ADDITIVE.	PL-CE01-TO-PE PERMITS 192.168.245.0/30.
CE02-L3VPN-01	RM-CE02-TO-PE	EXPORTS CE02 CUSTOMER PREFIX WITH COMMUNITY 300:150 ADDITIVE.	PL-CE02-TO-PE PERMITS 192.168.246.0/30.

THE CE POLICIES REMAIN INTENTIONALLY SIMPLE. THEY DO NOT DECIDE THE PROVIDER-WIDE RESULT BY THEMSELVES; THEY SIGNAL INTENT. THE PROVIDER PE ROUTERS ARE RESPONSIBLE FOR VALIDATING THE ROUTE, APPLYING ORIGIN COMMUNITIES AND ENFORCING SAFE BEHAVIOR.

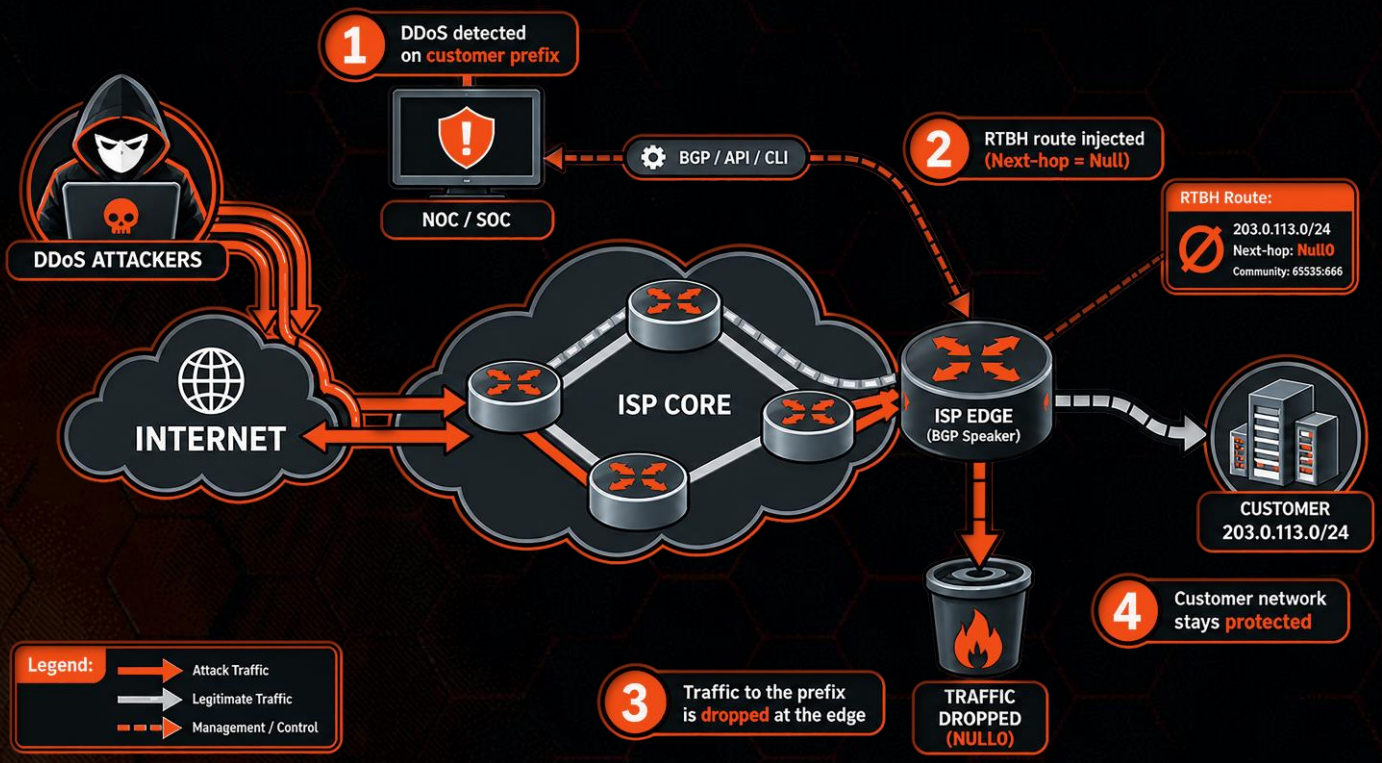
14.3. PROVIDER PE INBOUND CONTROL - PE01 AND PE02

THE PROVIDER PE INBOUND POLICIES ARE THE TRUST BOUNDARY BETWEEN CUSTOMER INTENT AND PROVIDER ROUTING CONTROL. THEY ACCEPT ONLY AUTHORIZED CUSTOMER PREFIXES, THEN TRANSLATE CUSTOMER COMMUNITIES INTO PROVIDER ACTIONS. UNAUTHORIZED PREFIXES AND UNSAFE BLACKHOLE REQUESTS ARE DENIED.

PROVIDER PE	CUSTOMER PEER	INBOUND ROUTE-MAP	ORIGIN TAG	DEFAULT RESULT
SKNT-PE01-BORDER	CE01-L3VPN-01 / 192.168.111.2	RM-SKNT-CE01-IN	300:101	CE01 ROUTES RECEIVE NORMAL LOCAL PREFERENCE AND COMMUNITY 300:101 300:150.
SKNT-PE02-VIOS	CE02-L3VPN-01 / 192.168.222.2	RM-SKNT-CE02-IN	300:102	CE02 ROUTES RECEIVE NORMAL LOCAL PREFERENCE AND COMMUNITY 300:102 300:150.



14.4. RTBH EXPLANATION AND GUARDRAILS



RTBH (REMOTELY TRIGGERED BLACK HOLE) IS A DDOS MITIGATION TECHNIQUE. IT USES BGP TO ANNOUNCE A TARGETED IP ADDRESS—USUALLY A /32—WITH A SPECIAL COMMUNITY. ROUTERS MATCH THAT COMMUNITY AND REDIRECT TRAFFIC TO A DISCARD/NULL ROUTE. THIS QUICKLY PROTECTS THE WIDER NETWORK, BUT THE TARGETED IP BECOMES UNREACHABLE. RTBH CAN BLOCK TRAFFIC BY DESTINATION OR, WITH URPF (UNICAST REVERSE PATH FORWARDING), BY ATTACK SOURCE.

RTBH WAS IMPLEMENTED AS A CONTROLLED SERVICE CAPABILITY, NOT AS AN OPEN CUSTOMER PRIVILEGE. THE PROVIDER ACCEPTS RTBH ONLY WHEN THE CUSTOMER ADVERTISES AN AUTHORIZED /32 HOST ROUTE AND ATTACHES COMMUNITY 300:666. THE ACCEPTED BLACKHOLE ROUTE IS POINTED TO THE PROVIDER NULL0 NEXT-HOP 192.0.2.1 INSIDE VRF INTER-AS-L3VPN. BROAD PREFIXES OR UNAUTHORIZED RTBH REQUESTS ARE EXPLICITLY DENIED.

CONTROL	IMPLEMENTATION DETAIL	REASON
RTBH COMMUNITY	SKNT-RTBH = 300:666.	IDENTIFIES A CUSTOMER BLACKHOLE REQUEST.
AUTHORIZED CE01 HOST	SKNT-CE01-RTBH-HOSTS PERMITS ONLY 192.168.245.2/32.	PREVENTS CE01 FROM BLACKHOLING BROAD OR UNRELATED PREFIXES.
AUTHORIZED CE02 HOST	SKNT-CE02-RTBH-HOSTS PERMITS ONLY 192.168.246.2/32.	PREVENTS CE02 FROM BLACKHOLING BROAD OR UNRELATED PREFIXES.
PROVIDER DISCARD NEXT-HOP	192.0.2.1/32 IS INSTALLED TO NULL0 IN VRF INTER-AS-L3VPN.	GIVES A DETERMINISTIC DISCARD TARGET FOR ACCEPTED BLACKHOLE ROUTES.
EXTERNAL LEAKAGE PROTECTION	RTBH COMMUNITY IS BLOCKED BY CISCO ROUTE-MAPS, JUNOS POLICIES AND XRV ROUTE-POLICIES TOWARD ISPS.	BLACKHOLE INTENT REMAINS INTERNAL TO THE PROVIDER DOMAIN.

14.5. INTERNAL-ONLY AND NO-EXPORT ENFORCEMENT

THE COMMUNITY NAMESPACE RESERVES 300:777 FOR SPARKLE ORIGIN AND 300:778 FOR INTERNAL-ONLY ROUTES. THIS AVOIDS CONFUSING SPARKLE-ORIGIN ROUTES WITH INTERNAL-ONLY POLICY ROUTES.

ROUTES MARKED AS INTERNAL-ONLY MUST REMAIN INSIDE AS300. THE SAME LOGIC APPLIES TO PROVIDER-SPECIFIC NO-EXPORT REQUESTS: 300:701 BLOCKS EXPORT TO HE, 300:702 BLOCKS EXPORT TO TATA AND 300:703 BLOCKS EXPORT TO VERIZON.

POLICY INTENT	COMMUNITY	ENFORCEMENT LOCATION
INTERNAL-ONLY	300:778	BLOCKED BY CISCO ISP ROUTE-MAPS, JUNOS EBGP-OUT POLICIES AND XRV SPARKLE OUTBOUND ROUTE-POLICIES.
NO EXPORT TO HE	300:701	BLOCKED BY RM-OUT-TO-HE-AS400, RM-OUT-TO-HE-AS400-V6 AND JUNOS HE EBGP POLICIES.
NO EXPORT TO TATA	300:702	BLOCKED BY RM-OUT-TO-TATA-AS500 AND RM-OUT-TO-TATA-AS500-V6.
NO EXPORT TO VERIZON	300:703	BLOCKED BY JUNOS VERIZON EXPORT POLICIES.
RTBH	300:666	BLOCKED TOWARD ALL EXTERNAL ISP EDGES.

14.6. EDGE ENFORCEMENT CENTER BY DEVICE FAMILY

THE EDGE ENFORCEMENT LAYER PREVENTS ACCIDENTAL OR INTENTIONAL ROUTE LEAKAGE. IT ALSO KEEPS ISP ORIGIN TAGGING CONSISTENT SO THAT OPERATORS CAN IDENTIFY WHERE ROUTES CAME FROM DURING TROUBLESHOOTING.

DEVICE	VENDOR/ROLE	ACTUAL POLICY AND GROUP NAMES PRESERVED	IMPLEMENTED ENFORCEMENT
SKNT-PE01-BORDER	CISCO IOS / HE EDGE	RM-OUT-TO-HE-AS400, RM-OUT-TO-HE-AS400-V6, SKNT-RTBH, SKNT-INTERNAL-ONLY, SKNT-NO-EXPORT-HE	BLOCKS RTBH, INTERNAL-ONLY AND NO-EXPORT-TO-HE ROUTES BEFORE ALLOWING ONLY AUTHORIZED SKYNET PUBLIC PREFIXES.
SKNT-PE02-VIOS	CISCO IOS / TATA EDGE	RM-OUT-TO-TATA-AS500, RM-OUT-TO-TATA-AS500-V6, SKNT-RTBH, SKNT-INTERNAL-ONLY, SKNT-NO-EXPORT-TATA	BLOCKS RTBH, INTERNAL-ONLY AND NO-EXPORT-TO-TATA ROUTES BEFORE ALLOWING ONLY AUTHORIZED SKYNET PUBLIC PREFIXES.
SKNT-PE03-VMX03	JUNOS / VERIZON VRF EDGE	BGP GROUP VERIZON, IMPORT-FROM-VERIZON, EXPORT-TO-VERIZON, COMM-FROM-VERIZON-AS200, SKNT-ORIGIN-VERIZON	IMPORTS VERIZON ROUTES WITH 300:200 TAGGING AND BLOCKS RTBH/INTERNAL-ONLY/NO-EXPORT-VERIZON ROUTES OUTBOUND.
SKNT-PE04-VMX01	JUNOS / HE EDGE	BGP-V4-HURRICANE, BGP-V6-HURRICANE, IMPORT-FROM-HE-V4, IMPORT-FROM-HE-V6, EBGP-OUT-V4, EBGP-OUT-V6, COMM-FROM-HE-AS400	IMPORTS HE ROUTES WITH 300:400 TAGGING AND BLOCKS RTBH/INTERNAL-ONLY/NO-EXPORT-HE ROUTES OUTBOUND.
SKNT-PE05-VMX02	JUNOS / VERIZON GLOBAL EDGE	BGP-V4-VERIZON, BGP-V6-VERIZON, TAG-FROM-VERIZON-AS200, EBGP-OUT-V4, EBGP-OUT-V6, COMM-FROM-VERIZON-AS200	TAGS VERIZON ROUTES WITH 300:200 AND BLOCKS RTBH/INTERNAL-ONLY/NO-EXPORT-VERIZON ROUTES OUTBOUND.
SKNT-PE08-XRV	CISCO IOS-XRV / SPARKLE EDGE	CS-SKNT-RTBH, CS-SKNT-INTERNAL-ONLY, CS-SKNT-ORIGIN-SPARKLE, RP-SPARKLE-IN-V4, RP-SPARKLE-IN-V6, RP-SPARKLE-OUT-V4, RP-SPARKLE-OUT-V6	TAGS SPARKLE ROUTES WITH 300:777 AND BLOCKS RTBH/INTERNAL-ONLY TOWARD SPARKLE.

DEVICE	IMPORTANT FINAL COMMUNITY OBJECTS	FINAL MEANING
SKNT-PE03-VMX03	COMM-FROM-VERIZON-AS200 = 300:200; SKNT-ORIGIN-VERIZON = 300:200	VERIZON VRF-EDGE ROUTES ARE CLEARLY TAGGED AS VERIZON-ORIGINATED.
SKNT-PE04-VMX01	COMM-FROM-HE-AS400 = 300:400; SKNT-INTERNAL-ONLY = 300:778	HE ROUTES ARE TAGGED AS HE-ORIGINATED AND INTERNAL-ONLY LEAKAGE IS BLOCKED.
SKNT-PE05-VMX02	COMM-FROM-VERIZON-AS200 = 300:200; SKNT-INTERNAL-ONLY = 300:778	VERIZON GLOBAL EDGE ROUTES ARE TAGGED AND INTERNAL-ONLY LEAKAGE IS BLOCKED.

14.7. IOS-XR X SPARKLE EDGE POLICY



SPARKLE REMAINS OUTSIDE THE ACTIVE INTER-AS-L3VPN FALLBACK CHAIN AND OPERATES AS A FILTERED SIMPLE EBGP/GLOBAL-UNICAST EDGE. THE XRV POLICIES TAG INBOUND SPARKLE ROUTES AND TIGHTLY CONTROL WHAT IS ADVERTISED BACK TO ISP-SPARKLE.

XRV OBJECT	FINAL FUNCTION
CS-SKNT-RTBH	COMMUNITY-SET MATCHING 300:666. ANY ROUTE CARRYING THIS COMMUNITY IS BLOCKED OUTBOUND TO SPARKLE.
CS-SKNT-INTERNAL-ONLY	COMMUNITY-SET MATCHING 300:778. INTERNAL-ONLY ROUTES ARE BLOCKED OUTBOUND TO SPARKLE.
CS-SKNT-ORIGIN-SPARKLE	COMMUNITY-SET MATCHING 300:777, THE SPARKLE-ORIGIN IDENTITY.
RP-SPARKLE-IN-V4 / RP-SPARKLE-IN-V6	PERMIT ONLY EXPECTED SPARKLE TEST PREFIXES AND TAG ACCEPTED ROUTES WITH 300:777.
RP-SPARKLE-OUT-V4 / RP-SPARKLE-OUT-V6	BLOCK RTBH AND INTERNAL-ONLY ROUTES, THEN PERMIT ONLY APPROVED OUTBOUND PREFIXES.

14.8. RETURN-PATH CORRECTION BETWEEN CE01 AND CE02

DURING VALIDATION, CE01-TO-CE02 REACHABILITY WORKED, BUT CE02-TO-CE01 INITIALLY FAILED. THE ISSUE WAS NOT THE BGP COMMUNITY DESIGN. IT WAS A RETURN-PATH FILTERING ISSUE CAUSED BY PE-CE TRANSPORT-SOURCE ADDRESSING. WHEN CE02 PINGED CE01 WITHOUT AN EXPLICIT SOURCE, IOS COULD USE 192.168.222.2 AS THE SOURCE. CE01 NEEDED TO ACCEPT THE CE02 PE-CE TRANSPORT PREFIX FROM THE PROVIDER SO THE RETURN PATH COULD BE RESOLVED.

DEVICE	PREFIX-LIST SCOPE	PREFIX CONFIRMED	REASON
CE01-L3VPN-01	PL-CE01-FROM-PE	192.168.222.0/24	ALLOWS CE01 TO RETURN TRAFFIC TO CE02 PE-CE TRANSPORT-SOURCED PACKETS.
CE02-L3VPN-01	PL-CE02-FROM-PE	192.168.111.0/24	MAINTAINS SYMMETRIC HANDLING FOR CE01 PE-CE TRANSPORT-SOURCED PACKETS.

14.9. VALIDATION SUMMARY

VALIDATION CONFIRMS THAT THE FEATURE IS OPERATIONAL AND THAT CUSTOMER REACHABILITY REMAINS INTACT. THE COMMUNITY CONTROL CENTER AND EDGE ENFORCEMENT CENTER WORK TOGETHER WITHOUT CHANGING THE INTENDED PROVIDER PREFERENCE ORDER.

VALIDATION AREA	RESULT
CE01 CUSTOMER ROUTE	PE01 SEES 192.168.245.0/30 WITH COMMUNITY 300:101 300:150 AND LOCAL-PREFERENCE 150.
CE02 CUSTOMER ROUTE	PE02 SEES 192.168.246.0/30 WITH COMMUNITY 300:102 300:150 AND LOCAL-PREFERENCE 150.
CE01 TO CE02 REACHABILITY	PING TO 192.168.246.1 SUCCEEDS WITH 100 PERCENT SUCCESS, INCLUDING SOURCE 192.168.245.1.
CE02 TO CE01 REACHABILITY	PING TO 192.168.245.1 SUCCEEDS WITH 100 PERCENT SUCCESS, INCLUDING SOURCE 192.168.246.1.
HE EDGE	HE IPV4 AND IPV6 ROUTES ARE RECEIVED AND TAGGED WITH COMMUNITY 300:400.
VERIZON EDGE	VERIZON ROUTES ARE TAGGED WITH 300:200; INCORRECT SKNT-ORIGIN-VERIZON MEMBER 300 WAS REMOVED.
SPARKLE EDGE	SPARKLE ROUTES ARE TAGGED WITH 300:777; ONLY 172.16.0.0/24 AND FC16::/48 ARE ADVERTISED TO SPARKLE.
ROUTE-LEAK PROTECTION	RTBH 300:666 AND INTERNAL-ONLY 300:778 ARE BLOCKED TOWARD EXTERNAL ISP EDGES.

14.10. CUSTOMER-FACING OUTCOME

THE CUSTOMER-FACING VALUE IS THAT SKYNET HAS A REALISTIC ISP ROUTING-POLICY SERVICE MODEL. THE NETWORK DOES NOT MERELY EXCHANGE PREFIXES; IT EXCHANGES PREFIXES WITH IDENTITY, INTENT AND PROTECTION. OPERATORS CAN IDENTIFY ROUTE ORIGIN, HONOR CUSTOMER INTENT, PROTECT AGAINST ROUTE LEAKS AND KEEP THE PROVIDER EDGE CLEAN.

- CUSTOMER ROUTES ARE TAGGED WITH CLEAR ORIGIN COMMUNITIES.
- EXTERNAL ISP ROUTES ARE TAGGED BY SOURCE PROVIDER: TATA, VERIZON, HE OR SPARKLE.
- INTERNAL-ONLY ROUTES USE 300:778 AND ARE BLOCKED AT EVERY EXTERNAL EDGE.
- SPARKLE ORIGIN USES 300:777 AND NO LONGER CONFLICTS WITH INTERNAL-ONLY BEHAVIOR.
- RTBH IS AVAILABLE AS A CONTROLLED SERVICE CAPABILITY WITH STRICT /32 GUARDRAILS.
- THE LAB REPRESENTS A PRODUCTION-FLAVORED SERVICE-PROVIDER POLICY ARCHITECTURE.

14.11. OPERATIONAL GUARDRAILS

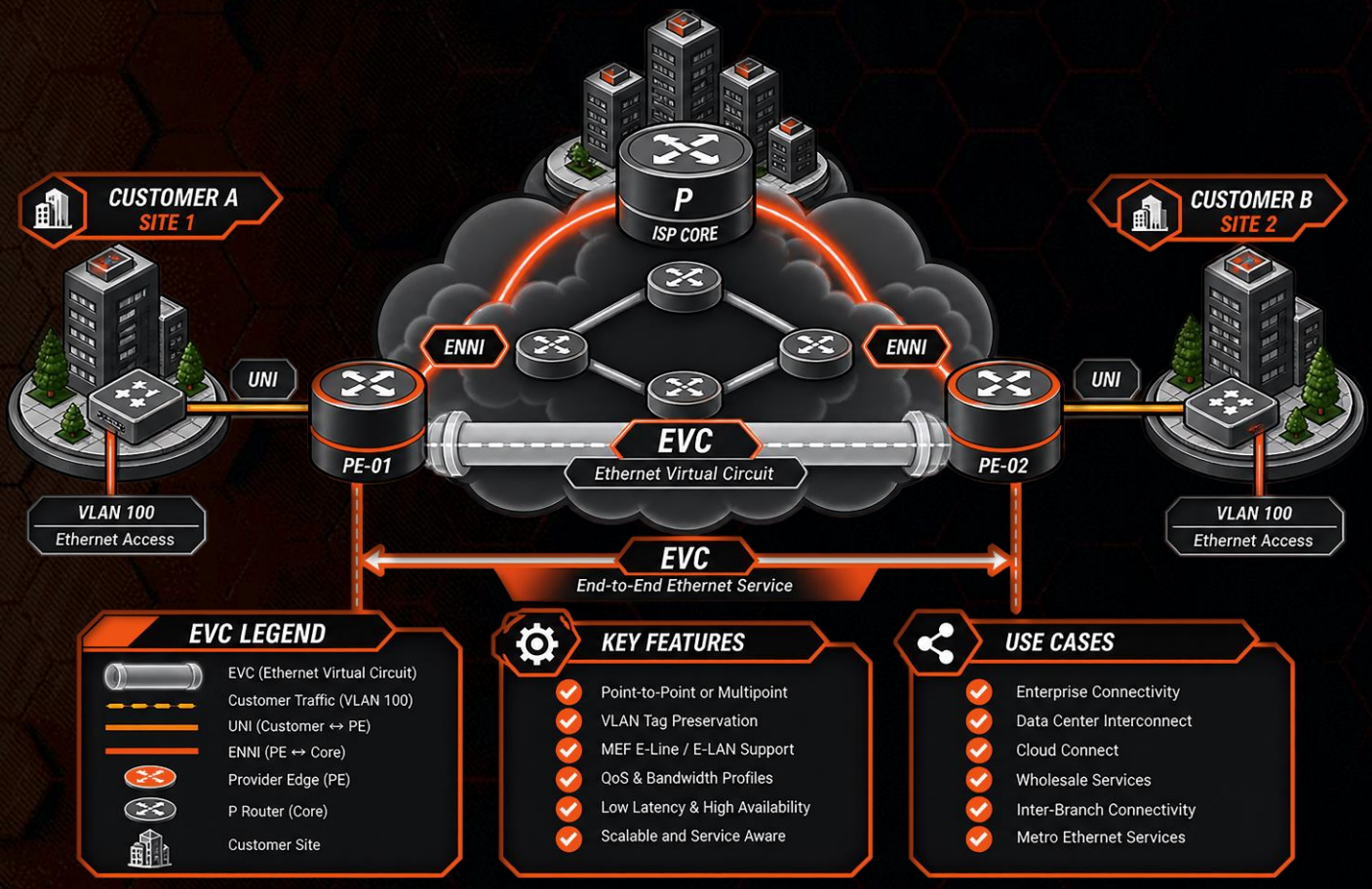
GUARDRAIL	OPERATIONAL RULE
FALLBACK ORDER	THE CUSTOMER L3VPN FALLBACK ORDER REMAINS TATA FIRST, VERIZON SECOND AND HE LAST RESORT. SPARKLE REMAINS OUTSIDE THE ACTIVE L3VPN FALLBACK SCHEME.
COMMUNITY NAMESPACE	DO NOT REUSE 300:777 FOR INTERNAL-ONLY ROUTES. 300:777 IS SPARKLE ORIGIN. 300:778 IS INTERNAL-ONLY.
RTBH AFETY	DO NOT ACCEPT BROAD RTBH PREFIXES. RTBH IS RESTRICTED TO AUTHORIZED /32 HOSTS ONLY.
ISP EXPORTS	DO NOT EXPORT ROUTES CARRYING RTBH, INTERNAL-ONLY OR PROVIDER-SPECIFIC NO-EXPORT COMMUNITIES TO EXTERNAL ISPS.
POLICY NAMING	PRESERVE EXISTING BGP GROUP, ROUTE-MAP, POLICY-STATEMENT, COMMUNITY-LIST AND ROUTE-POLICY NAMES TO KEEP THE DOCUMENTATION ALIGNED WITH LIVE DEVICE CONFIGURATION.

15. EVC300 ETHERNET VIRTUAL CIRCUIT SERVICE

15.1. WHAT IS AN EVC?

AN ETHERNET VIRTUAL CIRCUIT, OR EVC, IS A CUSTOMER-FACING CARRIER ETHERNET SERVICE CONTAINER. IT DEFINES HOW A CUSTOMER ETHERNET FLOW ENTERS THE PROVIDER EDGE, HOW IT IS IDENTIFIED BY VLAN OR SERVICE INSTANCE, AND HOW IT IS DELIVERED TO ANOTHER CUSTOMER-FACING PORT WITHOUT THE CUSTOMER NEEDING TO SEE THE PROVIDER MPLS CORE.

IN THIS LAB, EVC300 IS A POINT-TO-POINT E-LINE STYLE SERVICE BETWEEN TWO JUNIPER VMX PES. THE CUSTOMER SEES A SIMPLE VLAN300 ETHERNET SEGMENT BETWEEN EVC-DC-01 AND EVC-DC-02, WHILE THE PROVIDER CORE TRANSPORTS THE SERVICE USING JUNOS L2CIRCUIT SIGNALING OVER MPLS. THE SAME ENDPOINTS ALSO BUILD SEPARATE IPV4 AND IPV6 GRE TUNNELS PROTECTED BY IKEV2/IPSEC, PROVIDING AN ENCRYPTED DUAL-STACK OVERLAY ACROSS THE EVC300 TRANSPORT.



ANGLE	EVC VIEW	PSEUDOWIRE VIEW
SERVICE ABSTRACTION	EVC DESCRIBES THE CUSTOMER-FACING ETHERNET SERVICE: VLAN ID, ACCESS PORT, DEMARCATION, AND END-TO-END SERVICE INTENT.	A PSEUDOWIRE DESCRIBES THE TRANSPORT CIRCUIT USED TO CARRY LAYER-2 FRAMES ACROSS MPLS.
CUSTOMER BENEFIT	EVC MAKES THE SERVICE EASY TO DOCUMENT, SELL, OPERATE AND TROUBLESHOOT AS A CARRIER ETHERNET PRODUCT.	PSEUDOWIRE IS POWERFUL, BUT BY ITSELF IT IS MORE OF A TRANSPORT MECHANISM THAN A CUSTOMER SERVICE MODEL.
VLAN HANDLING	EVC CLEANLY MODELS CUSTOMER VLAN ATTACHMENT, INCLUDING POP/PUSH BEHAVIOR AT THE PE.	A BASIC XCONNECT MAY BRIDGE OR TRANSPORT FRAMES BUT MAY NOT EXPRESS THE SERVICE DEMARCATION AS CLEARLY.
LAB RELATIONSHIP	EVC300 IS THE CUSTOMER-FACING SERVICE NAME AND VLAN300 SERVICE CONTEXT.	JUNOS L2CIRCUIT IS THE UNDERLAY MECHANISM USED TO TRANSPORT EVC300 BETWEEN VMX PES.



15.2. EVC300 IMPLEMENTATION DETAILS

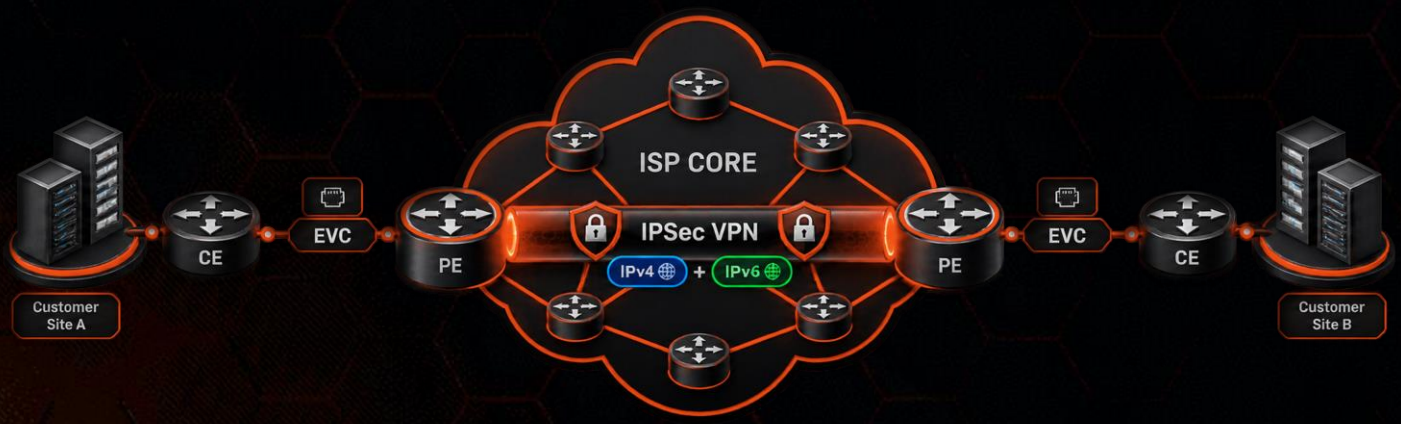
ITEM	DETAIL
SERVICE NAME	EVC300
SERVICE TYPE	POINT-TO-POINT ETHERNET VIRTUAL CIRCUIT / E-LINE STYLE SERVICE
CUSTOMER VLAN	300
VIRTUAL CIRCUIT ID	300
CUSTOMER SUBNET	192.168.131.0/24
ENDPOINT A	EVC-DC-01 VLAN300 SVI 192.168.131.1/24 CONNECTED TO SKNT-PE04-VMX01 GE-0/0/6
ENDPOINT B	EVC-DC-02 VLAN300 SVI 192.168.131.2/24 CONNECTED TO SKNT-PE05-VMX02 GE-0/0/6
PROVIDER PE A	SKNT-PE04-VMX01 LOOPBACK 192.168.1.3
PROVIDER PE B	SKNT-PE05-VMX02 LOOPBACK 192.168.1.5
TRANSPORT	MPLS/LDP WITH JUNOS L2CIRCUIT BETWEEN PE LOOPBACKS

JUNOS PE	SERVICE INTERFACE	ACCESS ENCAPSULATION	L2CIRCUIT CONTROL PLANE
SKNT-PE04-VMX01	GE-0/0/6.300	VLAN-CCC, VLAN-ID 300, INPUT-VLAN-MAP POP, OUTPUT-VLAN-MAP PUSH	NEIGHBOR 192.168.1.5, VC-ID 300
SKNT-PE05-VMX02	GE-0/0/6.300	VLAN-CCC, VLAN-ID 300, INPUT-VLAN-MAP POP, OUTPUT-VLAN-MAP PUSH	NEIGHBOR 192.168.1.3, VC-ID 300
ENDPOINT	ACCESS PORT	VLAN TREATMENT	TEST SVI
EVC-DC-01	ETHERNET0/0	TRUNK ALLOWING VLAN 300	VLAN300 192.168.131.1/24
EVC-DC-02	ETHERNET0/0	TRUNK ALLOWING VLAN 300	VLAN300 192.168.131.2/24

15.3. OPERATIONAL VALIDATION

- EVC-DC-01 LLDP SEES SKNT-PE04-VMX01 ON ETHERNET0/0 TOWARD GE-0/0/6.
- EVC-DC-02 LLDP SEES SKNT-PE05-VMX02 ON ETHERNET0/0 TOWARD GE-0/0/6.
- A SUCCESSFUL PING BETWEEN 192.168.131.1 AND 192.168.131.2 VALIDATES THE END-TO-END VLAN300 EVC DATA PLANE.
- EVC300 IS SEPARATE FROM THE EVPN VLAN600 SERVICE. EVPN VLAN600 USES 10.35.0.0/24; EVC300 USES 192.168.131.0/24.

15.4. VPN V4 AND V6 BETWEEN EVC ENDPOINTS



EVC-DC-01 AND EVC-DC-02 ARE CONFIGURED WITH TWO PARALLEL ENCRYPTED OVERLAYS ACROSS THE VLAN300 EVC: AN IPV4 GRE TUNNEL AND AN IPV6 GRE TUNNEL. BOTH USE IKEV2/IPSEC WITH AES-256 ENCRYPTION, SHA-512 INTEGRITY, DIFFIE-HELLMAN/PFS GROUP 20, PERIODIC DEAD-PEER DETECTION, AND 1,800-SECOND IPSEC SECURITY-ASSOCIATION LIFETIMES.

- VPN V4: EVC-DC-01 TUNNEL0 (10.177.177.1/30) AND EVC-DC-02 TUNNEL0 (10.177.177.2/30) FORM A GRE VPN SOURCED FROM VLAN300. THE UNDERLAY ENDPOINTS ARE 192.168.131.1/24 AND 192.168.131.2/24, AND THE TUNNEL IS PROTECTED BY THE IPSEC-IKEV2 PROFILE.
- VPN V6: EVC-DC-01 TUNNEL1 (2001:177:177::1/64) AND EVC-DC-02 TUNNEL1 (2001:177:177::2/64) FORM A GRE-OVER-IPV6 VPN SOURCED FROM VLAN300. THE UNDERLAY ENDPOINTS ARE FC91::1/64 AND FC91::2/64, PROTECTED BY THE DC1-IPV6-IPSEC AND DC2-IPV6-IPSEC PROFILES.
- ROUTE EXCHANGE: RIPv2 OPERATES ACROSS TUNNEL0 FOR IPV4 REACHABILITY, WHILE RIP-NG-SKYNET OPERATES ACROSS TUNNEL1 FOR IPV6 REACHABILITY. THE ENDPOINT LOOPBACK PREFIXES REPRESENT THE DUAL-STACK CUSTOMER/SERVER NETWORKS ADVERTISED THROUGH THE ENCRYPTED OVERLAYS.
- IPV4 PREFIXES INCLUDE 192.168.227.1-5/32 AND 172.16.172.1/32 AT DC1, AND 192.168.228.1-5/32 AND 172.16.173.1/32 AT DC2. IPV6 PREFIXES INCLUDE FD00:16:16::1-5/128 AND FD00:172:16::1/128 AT DC1, AND FD00:17:17::1-5/128 AND FD00:173:16::1/128 AT DC2.
- OPERATIONAL PURPOSE: THE EVC PROVIDES THE CARRIER ETHERNET TRANSPORT, WHILE THE GRE/IKEV2/IPSEC LAYERS PROVIDE ROUTED IPV4/IPV6 CONNECTIVITY, CONFIDENTIALITY, INTEGRITY, PEER AUTHENTICATION, AND INDEPENDENT DUAL-STACK ROUTING BETWEEN THE TWO EVC CUSTOMER ENDPOINTS.

16. BGP MD5 AUTHENTICATION



16.1. OVERVIEW

BGP MD5 AUTHENTICATION IS A SECURITY MECHANISM THAT PROTECTS TCP CONNECTIONS BETWEEN BGP PEERS USING A CRYPTOGRAPHIC CHECKSUM. IT ENSURES THAT EVERY BGP MESSAGE EXCHANGED BETWEEN TWO ROUTERS IS VERIFIED WITH A SHARED SECRET KEY, PREVENTING UNAUTHORIZED DEVICES FROM INJECTING FALSE ROUTING INFORMATION OR HIJACKING BGP SESSIONS.

IN THE SKYNET ISP LAB, MD5 AUTHENTICATION WAS DEPLOYED ON THE PRE-EXISTING EXTERNAL BGP (EBGP) SESSIONS DOCUMENTED IN SECTION 16.4—BETWEEN THE PROVIDER EDGE ROUTERS (PES) AND THE UPSTREAM ISP ROUTERS, AS WELL AS BETWEEN THE PES AND THE CUSTOMER EDGE (CE) ROUTERS. THE DIRECT CE01-L3VPN-01 ↔ ISP-HE AS301/AS400 SESSION ADDED ON 10-AUG-2026 IS DOCUMENTED IN SECTION 30 AND WAS NOT PART OF THE ORIGINAL MD5 DEPLOYMENT.

16.2. WHY BGP MD5 AUTHENTICATION MATTERS?

BGP IS THE FOUNDATION OF INTERNET ROUTING. WITHOUT AUTHENTICATION, AN ATTACKER WHO GAINS ACCESS TO A NETWORK SEGMENT CAN:

- ESTABLISH A ROGUE BGP SESSION AND INJECT FALSE PREFIXES.
- HIJACK LEGITIMATE BGP SESSIONS BY SENDING SPOOFED TCP PACKETS.
- DISRUPT THE ROUTING TABLE BY RESETTING SESSIONS WITH FORGED RST PACKETS.
- MD5 AUTHENTICATION ADDS A LIGHTWEIGHT BUT EFFECTIVE LAYER OF PROTECTION. THE SHARED SECRET KEY IS NEVER TRANSMITTED OVER THE WIRE; ONLY A CRYPTOGRAPHIC HASH OF THE PACKET AND THE KEY IS APPENDED. IF AN ATTACKER DOES NOT KNOW THE KEY, THEIR PACKETS ARE REJECTED AT THE TCP LAYER, PREVENTING THE BGP SESSION FROM EVEN BEING ESTABLISHED.

16.3. HOW IT WORKS?

THE MECHANISM OPERATES AT THE TCP LEVEL, NOT AT THE BGP APPLICATION LEVEL. WHEN A TCP CONNECTION IS ESTABLISHED BETWEEN TWO BGP SPEAKERS THAT ARE CONFIGURED WITH A MATCHING PASSWORD, EVERY TCP SEGMENT (SYN, ACK, DATA, RST) INCLUDES AN MD5 DIGEST IN THE TCP OPTIONS FIELD. THE DIGEST IS CALCULATED OVER:

THE TCP PSEUDO-HEADER (SOURCE AND DESTINATION IP ADDRESSES, PROTOCOL NUMBER, SEGMENT LENGTH). THE TCP SEGMENT ITSELF (HEADER AND PAYLOAD). A SHARED SECRET KEY (THE CONFIGURED PASSWORD).

THE RECEIVING ROUTER COMPUTES THE EXPECTED DIGEST USING THE SAME KEY AND COMPARES IT WITH THE RECEIVED DIGEST. IF THEY MATCH, THE SEGMENT IS ACCEPTED; IF THEY DO NOT, THE SEGMENT IS SILENTLY DISCARDED. THIS EFFECTIVELY PREVENTS TCP INJECTION ATTACKS AND ENSURES THAT ONLY ROUTERS POSSESSING THE CORRECT KEY CAN PARTICIPATE IN THE BGP SESSION.

16.4. IMPLEMENTATION SCOPE

IN THE SKYNET ISP LAB, MD5 AUTHENTICATION WAS CONFIGURED ON THE FOLLOWING EBGP SESSIONS:

PEER A	PEER B	AS NUMBERS	ADDRESS FAMILY
SKNT-PE01-BORDER	ISP-HE (HUAWEI)	300 ↔ 400	IPV4, IPV6
SKNT-PE04-VMX01	ISP-HE (HUAWEI)	300 ↔ 400	IPV4, IPV6
SKNT-PE02-VIOS	ISP-TATA (CISCO)	300 ↔ 500	IPV4, IPV6 (DUAL LINK)
SKNT-PE05-VMX02	ISP-VERIZON (CISCO)	300 ↔ 200	IPV4, IPV6
SKNT-PE03-VMX03	ISP-VERIZON (CISCO)	300 ↔ 200 (VRF)	IPV4
SKNT-PE08-XRV	ISP-SPARKLE (CISCO)	300 ↔ 777	IPV4, IPV6
CE01-L3VPN-01	SKNT-PE01-BORDER	301 ↔ 300 (VRF)	IPV4
CE02-L3VPN-01	SKNT-PE02-VIOS	301 ↔ 300 (VRF)	IPV4

FOR LAB ONLY PURPOSES, THE SAME PASSWORD **Evelab@2025** WAS USED ACROSS ALL SESSIONS TO SIMPLIFY KEY MANAGEMENT WHILE MAINTAINING CRYPTOGRAPHIC ISOLATION—ONLY DEVICES WITHIN THE LAB DOMAIN POSSESS THE SHARED SECRET.

16.5. CONCLUSION:

THE IMPLEMENTATION OF BGP MD5 AUTHENTICATION ACROSS THE PRE-EXISTING EXTERNAL BGP SESSIONS DOCUMENTED IN THIS SECTION PROVIDES A ROBUST AND VERIFIABLE SECURITY LAYER. THE DIRECT CE01-L3VPN-01 ↔ ISP-HE AS301/AS400 SESSION ADDED IN SECTION 30 WAS INTRODUCED WITHOUT MD5 AUTHENTICATION TO KEEP THE MIGRATION MINIMAL AND REVERSIBLE; AUTHENTICATION FOR THAT NEW PEER IS RESERVED FOR A SEPARATE HARDENING WINDOW. THE EXISTING AUTHENTICATED PEERINGS REMAIN UNCHANGED.

17. COPP (CONTROL PLANE POLICING)



CONTROL PLANE POLICING PROTECTS A ROUTER'S CPU FROM BEING OVERWHELMED BY EXCESSIVE TRAFFIC DESTINED TO THE DEVICE ITSELF—SUCH AS BGP, OSPF, SSH, OR ICMP—BY FILTERING AND RATE-LIMITING PACKETS BEFORE THEY REACH THE CONTROL PLANE. WITHOUT COPP, A FLOOD OF MANAGEMENT OR ATTACK TRAFFIC CAN EXHAUST CPU RESOURCES, CAUSING ROUTING PROTOCOL FLAPS, MANAGEMENT INACCESSIBILITY, AND SERVICE OUTAGES.

IN THE SKYNET ISP LAB, COPP WAS DEPLOYED ON ALL 10 CORE DEVICES (CISCO IOSV, JUNIPER VMX, AND IOS-XRV). TRAFFIC IS CLASSIFIED INTO THREE CATEGORIES:

- **CRITICAL** - OSPF, BGP, LDP (POLICED AT 1 MBPS, GUARANTEED DELIVERY).
- **MANAGEMENT** - SSH FROM THE PROVIDER LAN (512 KBPS).
- **MONITORING** - ICMP DIAGNOSTICS AND NTP (256 KBPS).

ALL OTHER UNCLASSIFIED TRAFFIC IS RATE-LIMITED TO 64 KBPS. EXCESS TRAFFIC IN ANY CLASS IS DROPPED, ENSURING THAT EVEN DURING A DDOS ATTACK OR MISCONFIGURATION, THE ROUTER'S CONTROL PLANE REMAINS RESPONSIVE.

18. VRF MIGRATION (PE01/PE02)

18.1. BACKGROUND

THE ORIGINAL INTER-AS-L3VPN CUSTOMER VRF WAS CREATED USING THE LEGACY IP VRF COMMAND, WHICH SUPPORTS ONLY IPV4. TO ENABLE IPV6 L3VPN (6VPE) SERVICES, THE VRF HAD TO BE MIGRATED TO THE MODERN VRF DEFINITION FORMAT, WHICH CAN ACCOMMODATE BOTH IPV4 AND IPV6 ADDRESS FAMILIES UNDER A SINGLE ROUTING INSTANCE.

18.2. SCOPE

- THE MIGRATION WAS PERFORMED ON THE TWO PROVIDER EDGE ROUTERS THAT TERMINATE THE CUSTOMER L3VPN:
- SKNT-PE01-BORDER (SERVING CE01-L3VPN-01)
- SKNT-PE02-VIOS (SERVING CE02-L3VPN-01)
- THE CORE ROUTE REFLECTOR (P01-ISP) AND THE CUSTOMER EDGE DEVICES WERE NOT CHANGED, AS THE CES REMAINED IPV4-ONLY BY DESIGN.

18.3. PROCEDURE

THE MIGRATION FOLLOWED THESE STEPS ON EACH PE:

1. REMOVE THE LEGACY VRF - NO IP VRF INTER-AS-L3VPN. THIS DELETES THE OLD IPV4-ONLY VRF AND TEMPORARILY UNBINDS ALL ASSOCIATED INTERFACES.
2. CREATE THE NEW DUAL-STACK VRF - VRF DEFINITION INTER-AS-L3VPN WITH THE SAME ROUTE DISTINGUISHER (300:100). INSIDE IT, TWO ADDRESS FAMILIES WERE DEFINED:
3. ADDRESS-FAMILY IPV4 WITH THE ORIGINAL ROUTE-TARGETS (IMPORT/EXPORT 300:100).
4. ADDRESS-FAMILY IPV6 WITH THE IDENTICAL ROUTE-TARGETS, PRESERVING THE EXISTING VPN TOPOLOGY.
5. RE-ASSIGN INTERFACES - ALL CUSTOMER-FACING AND LOOPBACK INTERFACES WERE PLACED BACK INTO THE VRF USING VRF FORWARDING INTER-AS-L3VPN, AND THEIR IPV4 ADDRESSES WERE RE-APPLIED. IPV6 ADDRESSES WERE ADDED TO THE LOOPBACKS AT THE SAME TIME.
6. RESTORE BGP CONFIGURATION - THE IPV4 VRF ADDRESS FAMILY WAS RECONFIGURED WITH ITS ORIGINAL NEIGHBORS, ROUTE-MAPS, AND PREFIX-LISTS. A NEW IPV6 VRF ADDRESS FAMILY WAS ADDED, INCLUDING REDISTRIBUTE CONNECTED TO ADVERTISE THE NEW IPV6 LOOPBACK PREFIXES.
7. ACTIVATE VPNV6 IBGP - THE VPNV6 ADDRESS FAMILY WAS ACTIVATED ON THE IBGP SESSIONS TO THE ROUTE REFLECTOR, ALLOWING IPV6 VRF ROUTES TO BE EXCHANGED BETWEEN PE01 AND PE02.

18.4. SERVICE RESTORATION

DURING THE MIGRATION, TWO CRITICAL SETS OF STATIC ROUTES WERE LOST:

TATA PRIMARY/BACKUP STATIC ROUTES (ON PE02) FOR THE ISP FALLBACK CHAIN.

HE LAST-RESORT STATIC ROUTES (ON PE01) FOR THE ISP FALLBACK CHAIN.

BOTH WERE MANUALLY RE-ENTERED AFTER THE VRF WAS RECREATED. NAT ACLS COVERING THE L3VPN SOURCE PREFIXES WERE ALSO VERIFIED AND RESTORED.

18.5. OUTCOME

THE MIGRATION SUCCESSFULLY ENABLED DUAL-STACK SUPPORT IN THE CUSTOMER VRF WITHOUT ALTERING THE EXISTING IPV4 L3VPN SERVICE. THE THREE-ISP FALLBACK ORDER (TATA → VERIZON → HE) REMAINED INTACT. IPV6 CONNECTIVITY BETWEEN THE PE LOOPBACKS WAS VALIDATED VIA PING, CONFIRMING END-TO-END 6VPE OPERATION ACROSS THE MPLS CORE. THE CES CONTINUED TO OPERATE AS IPV4-ONLY, IN LINE WITH THE LAB'S DESIGN CHOICE.

19. 6PVE/6PE (IPv6 L3VPN)



19.1. OVERVIEW

6VPE EXTENDS MPLS L3VPN TO SUPPORT IPV6 TRAFFIC OVER AN EXISTING IPV4 MPLS CORE. IT USES THE SAME VRF, ROUTE TARGETS, AND ROUTE REFLECTORS AS IPV4 VPNS, BUT ADDS THE VPNV6 ADDRESS FAMILY TO CARRY IPV6 PREFIXES ACROSS THE PROVIDER BACKBONE. THIS ENABLES CUSTOMER SITES TO RUN DUAL-STACK WITHOUT DEPLOYING A SEPARATE IPV6 TRANSPORT NETWORK.

IN THE SKYNET ISP LAB, 6VPE WAS ACTIVATED ON THE EXISTING INTER-AS-L3VPN VRF, ALLOWING PE01-BORDER AND PE02-VIOS TO EXCHANGE IPV6 ROUTES WHILE THE CUSTOMER EDGE DEVICES REMAINED IPV4-ONLY. THE IMPLEMENTATION PRESERVED ALL EXISTING IPV4 L3VPN SERVICES AND THE THREE-ISP FALLBACK CHAIN.

19.2. IMPLEMENTATION STEPS

1. VRF MIGRATION - THE EXISTING IP VRF WAS REPLACED WITH VRF DEFINITION, ADDING AN ADDRESS-FAMILY IPV6 BLOCK WITH THE SAME ROUTE-TARGETS (300:100). INTERFACE ADDRESSING - IPV6 ADDRESSES WERE ASSIGNED TO THE VRF LOOPBACKS ON BOTH PES (E.G., 2001:DB8:98::1/128 ON PE01, 2001:DB8:99::1/128 ON PE02).

2. VPNV6 IBGP - THE VPNV6 ADDRESS FAMILY WAS ACTIVATED ON IBGP SESSIONS BETWEEN PE01/PE02 AND THE ROUTE REFLECTOR (P01-ISP). P01 WAS CONFIGURED TO REFLECT VPNV6 ROUTES JUST AS IT DOES FOR VPNV4.

3. VRF ROUTE ADVERTISEMENT - UNDER THE IPV6 VRF ADDRESS FAMILY, REDISTRIBUTE CONNECTED WAS USED TO INJECT THE NEW LOOPBACK PREFIXES INTO BGP.

19.3. VERIFICATION

- BIDIRECTIONAL IPV6 CONNECTIVITY BETWEEN THE PE VRF LOOPBACKS WAS CONFIRMED:

- PE01# PING VRF INTER-AS-L3VPN 2001:DB8:99::1 SOURCE 2001:DB8:98::1
  SUCCESS RATE IS 100 PERCENT

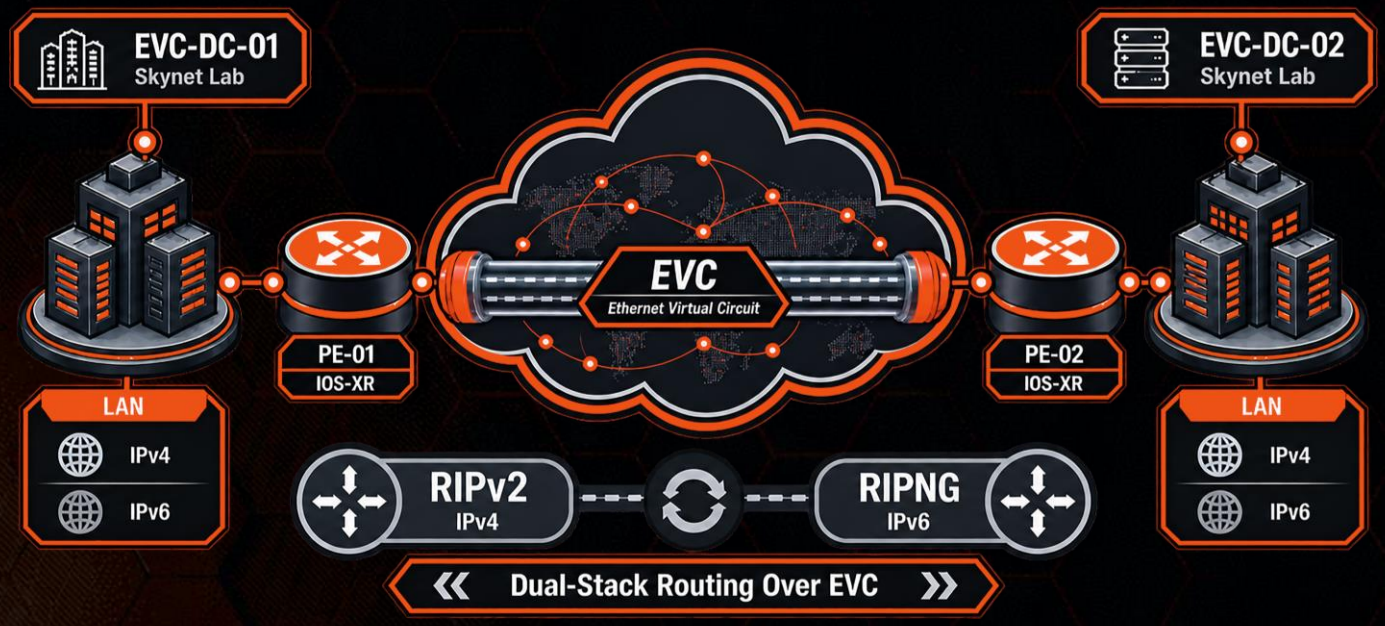
- PE02# PING VRF INTER-AS-L3VPN 2001:DB8:98::1 SOURCE 2001:DB8:99::1
  SUCCESS RATE IS 100 PERCENT

- MPLS LABEL FORWARDING WAS VERIFIED END-TO-END, WITH CORRECT TRANSPORT AND VPN LABELS IMPOSED AT THE INGRESS PE.

19.4. OUTCOME

THE CUSTOMER VRF NOW SUPPORTS BOTH IPV4 AND IPV6 ADDRESSING, DEMONSTRATING A REALISTIC DUAL-STACK L3VPN SERVICE OVER A SINGLE MPLS CORE. ALL EXISTING IPV4 ROUTES, ISP FALLBACK POLICIES, AND SECURITY HARDENING REMAINED UNCHANGED. THE CES WERE INTENTIONALLY LEFT IPV4-ONLY, MATCHING THE LAB’S ORIGINAL DESIGN SCOPE.

20. RIPv2 & RIP-NG IMPLEMENTATION



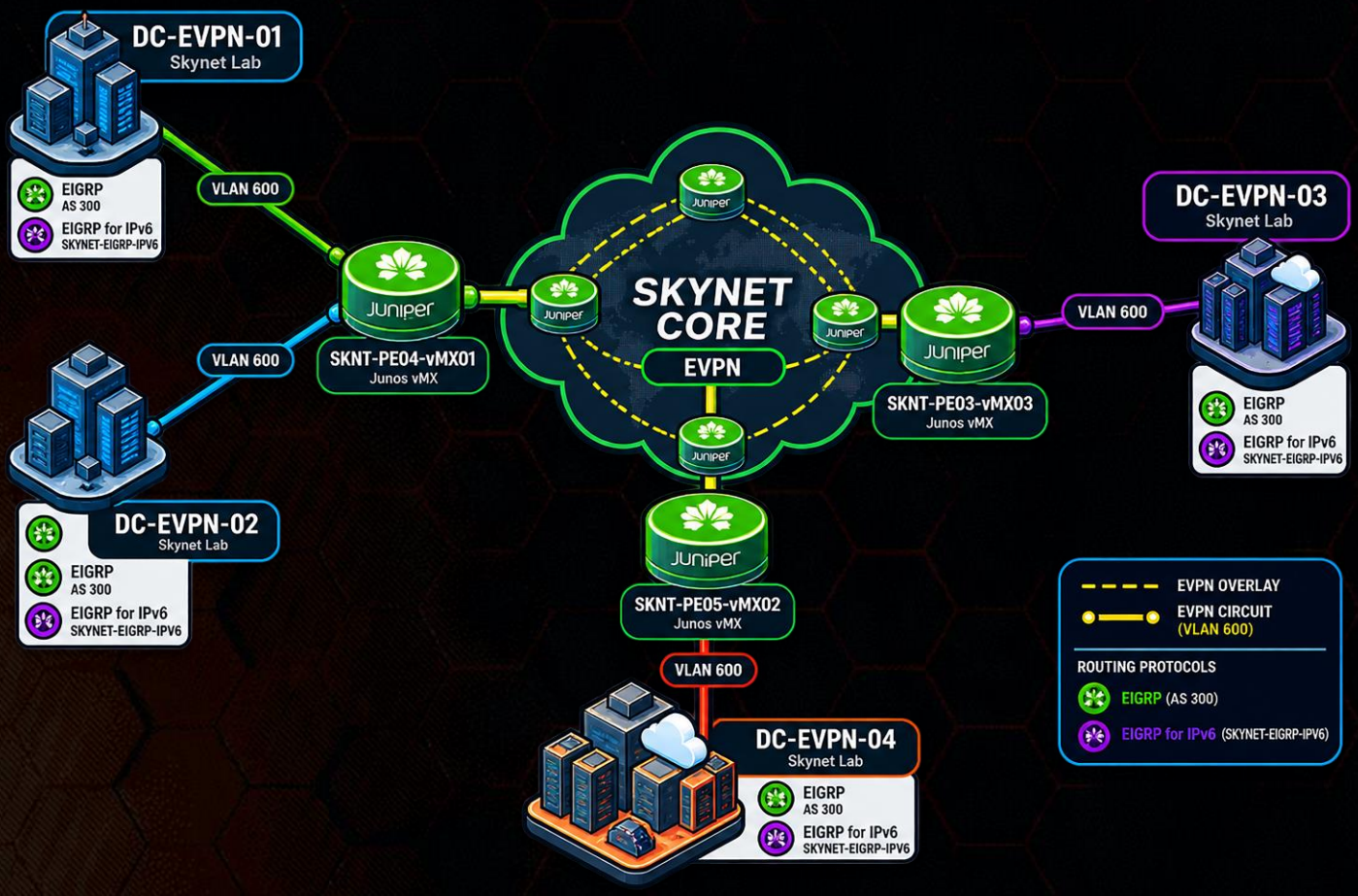
20.1. INTRODUCTION

IN MODERN NETWORK ARCHITECTURES, THE COEXISTENCE OF IPV4 AND IPV6 REQUIRES ROBUST DYNAMIC ROUTING PROTOCOLS. FOR THE EVC (EDGE VIRTUAL CONNECTIVITY) ENDPOINTS, A STRAIGHTFORWARD YET EFFECTIVE DISTANCE-VECTOR APPROACH WAS CHOSEN TO FACILITATE ROUTING BETWEEN THE TWO DATA CENTER LOCATIONS. RIPV2 HANDLES THE LEGACY IPV4 TRAFFIC, WHILE RIPNG MANAGES THE IPV6 PREFIX ADVERTISEMENTS. THE PRIMARY GATEWAY INTERFACES RESIDE ON **VLAN300**, WHICH ACTS AS THE TRANSPORT LINK BETWEEN THE TWO ROUTERS.

20.2. CONCLUSION

THE IMPLEMENTATION OF RIPV2 AND RIPNG ON THE EVC ENDPOINTS SUCCESSFULLY ESTABLISHED A RESILIENT, DUAL-STACK ROUTING ENVIRONMENT. THE MIGRATION FROM OSPFV3 TO RIPNG DEMONSTRATED THE FLEXIBILITY OF DISTANCE-VECTOR PROTOCOLS IN SMALL-SCALE DATA CENTER INTERCONNECTS. THE TROUBLESHOOTING PHASE HIGHLIGHTED THE IMPORTANCE OF STRICT ADHERENCE TO IPV6 SYNTAX—SPECIFICALLY, THE 4-DIGIT HEXTET LIMITATION—AND THE NECESSITY OF EXPLICITLY ENABLING ROUTING PROTOCOLS ON ALL INTENDED INTERFACES. THE FINAL TOPOLOGY IS NOW FULLY CONVERGED, WITH THE RIP-NG-SKYNET PROCESS EFFICIENTLY DISTRIBUTING REACHABILITY INFORMATION ACROSS THE NETWORK. REST IN PEACE, DEAR USELESS OLD FRIEND.

21. EIGRP AND EIGRP FOR IPv6



21.1. OVERVIEW

ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL (EIGRP) IS AN ADVANCED DISTANCE-VECTOR ROUTING PROTOCOL ORIGINALLY DEVELOPED BY CISCO. IT USES THE DIFFUSING UPDATE ALGORITHM (DUAL) TO ACHIEVE RAPID CONVERGENCE WHILE MAINTAINING LOOP-FREE TOPOLOGY INFORMATION AT EVERY MOMENT. UNLIKE TRADITIONAL DISTANCE-VECTOR PROTOCOLS, EIGRP SENDS PARTIAL, BOUNDED UPDATES ONLY WHEN A NETWORK CHANGE OCCURS, AND ONLY TO THE ROUTERS THAT NEED THE INFORMATION, WHICH MAKES IT VERY EFFICIENT IN TERMS OF BANDWIDTH AND PROCESSOR USAGE.

EIGRP USES A COMPOSITE METRIC BASED ON BANDWIDTH, DELAY, RELIABILITY, AND LOAD, WITH BANDWIDTH AND DELAY BEING THE DEFAULT COMPONENTS. IT BUILDS THREE MAIN TABLES: THE NEIGHBOR TABLE, THE TOPOLOGY TABLE, AND THE ROUTING TABLE. THE DUAL ALGORITHM IDENTIFIES FEASIBLE SUCCESSORS (LOOP-FREE BACKUP PATHS), ALLOWING SUB-SECOND CONVERGENCE IF A PRIMARY ROUTE FAILS.

EIGRP FOR IPV6 PROVIDES THE SAME FUNCTIONALITY FOR IPV6 NETWORKS. IN ITS MODERN IMPLEMENTATION, IT RUNS AS A NAMED-MODE CONFIGURATION, WHERE THE ROUTING PROCESS IS DEFINED UNDER AN ADDRESS FAMILY AND ACTIVATED PER-INTERFACE RATHER THAN USING THE NETWORK COMMAND. EIGRP FOR IPV6 RETAINS THE SAME DUAL ALGORITHM, METRIC STRUCTURE, AND FAST CONVERGENCE PROPERTIES, BUT OPERATES ENTIRELY OVER IPV6 TRANSPORT, USING THE RESERVED IPV6 MULTICAST ADDRESS FF02::A FOR NEIGHBOR DISCOVERY AND UPDATES. BOTH CLASSIC EIGRP (IPV4) AND EIGRP FOR IPV6 CAN RUN CONCURRENTLY ON THE SAME ROUTER, EACH MAINTAINING SEPARATE NEIGHBOR AND TOPOLOGY TABLES FOR THEIR RESPECTIVE ADDRESS FAMILIES.

21.1. WHY EIGRP AND EIGRP FOR IPV6 MATTERS?

EIGRP MATTERS BECAUSE IT FILLS A CRITICAL GAP BETWEEN SIMPLE DISTANCE-VECTOR PROTOCOLS AND COMPLEX LINK-STATE PROTOCOLS. IT OFFERS:

RAPID CONVERGENCE: THANKS TO FEASIBLE SUCCESSORS, FAILOVER CAN OCCUR ALMOST INSTANTLY WITHOUT RECALCULATING THE ENTIRE TOPOLOGY.

EFFICIENT RESOURCE USAGE: PARTIAL, INCREMENTAL UPDATES REDUCE NETWORK OVERHEAD, MAKING EIGRP SUITABLE FOR LINKS WITH LIMITED BANDWIDTH, SUCH AS WAN CONNECTIONS.

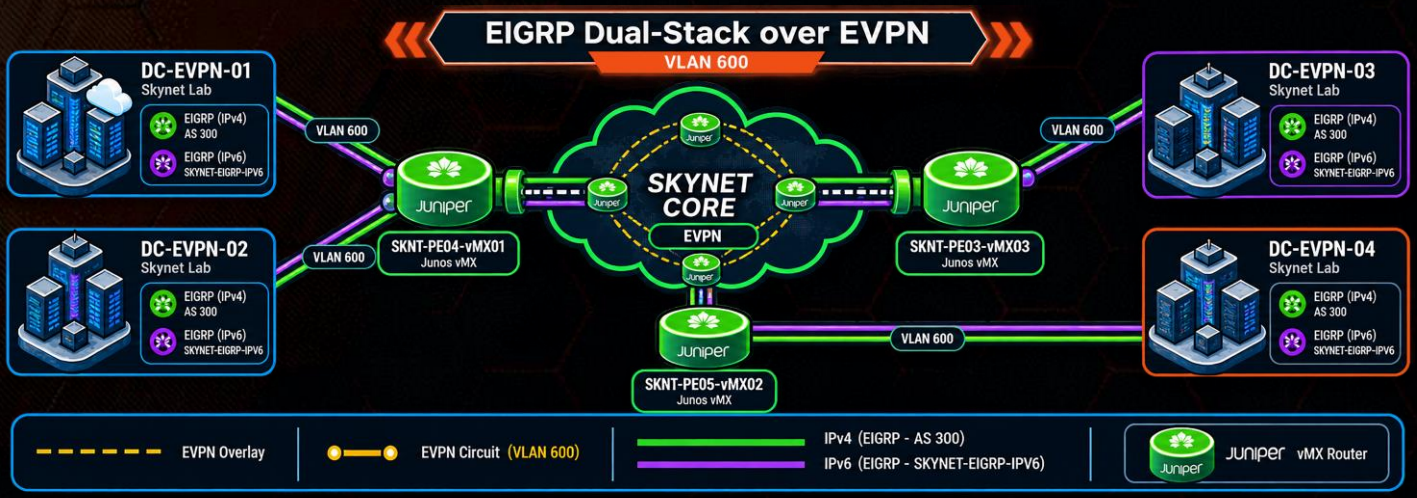
FLEXIBLE SUMMARIZATION AND UNEQUAL-COST LOAD BALANCING: EIGRP SUPPORTS MANUAL ROUTE SUMMARIZATION AT ANY INTERFACE AND THE UNIQUE ABILITY TO PERFORM UNEQUAL-COST LOAD BALANCING USING THE VARIANCE PARAMETER.

EASY INTEGRATION: AUTOMATIC PROTOCOL REDISTRIBUTION OCCURS BETWEEN EIGRP PROCESSES WHEN THEY SHARE THE SAME AUTONOMOUS SYSTEM NUMBER, AND ROUTE FILTERING IS STRAIGHTFORWARD WITH PREFIX-LISTS AND ROUTE-MAPS.

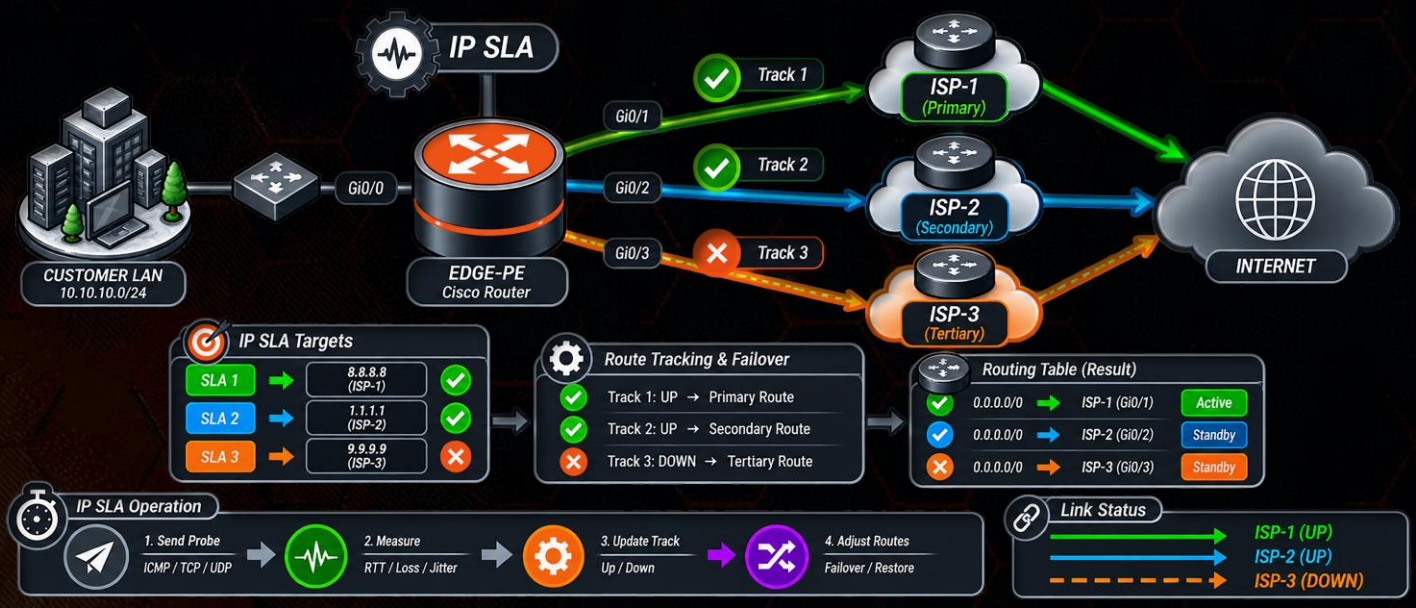
EIGRP FOR IPV6 EXTENDS THESE BENEFITS TO DUAL-STACK AND IPV6-ONLY ENVIRONMENTS. AS ORGANIZATIONS TRANSITION TO IPV6, HAVING A SINGLE, PROVEN ROUTING PROTOCOL FOR BOTH ADDRESS FAMILIES SIMPLIFIES NETWORK DESIGN, OPERATION, AND TROUBLESHOOTING. BECAUSE THE LOGICAL CONFIGURATION CLOSELY MIRRORS ITS IPV4 COUNTERPART, STAFF TRAINING AND MIGRATION BECOME LESS COMPLEX. IN MODERN DEPLOYMENTS, NAMED-MODE EIGRP CONSOLIDATES IPV4 AND IPV6 ROUTING UNDER ONE CONFIGURATION HIERARCHY, FURTHER REDUCING ADMINISTRATIVE OVERHEAD AND PROMOTING CONSISTENCY ACROSS THE NETWORK

21.2. CONCLUSION

EIGRP IS A MATURE, FAST, AND VERSATILE ROUTING PROTOCOL THAT HAS PROVEN ITS VALUE IN ENTERPRISE AND CAMPUS NETWORKS. ITS ABILITY TO PROVIDE LOOP-FREE, RAPID CONVERGENCE WITH MINIMAL OVERHEAD MAKES IT AN EXCELLENT CHOICE FOR MANY NETWORK DESIGNS. EIGRP FOR IPV6 ENSURES THAT THESE SAME ADVANTAGES ARE AVAILABLE IN NEXT-GENERATION NETWORKS, ENABLING A SEAMLESS TRANSITION AND UNIFIED ROUTING POLICY. UNDERSTANDING BOTH CLASSIC AND IPV6 EIGRP IS ESSENTIAL FOR DESIGNING RESILIENT, SCALABLE, AND EASY-TO-MANAGE MULTI-PROTOCOL ENVIRONMENTS.



22. IP SLA AND OBJECT TRACKING FOR DETERMINISTIC FAILOVER



22.1. OVERVIEW

IP SLA (SERVICE LEVEL AGREEMENT) AND OBJECT TRACKING ADD REAL-TIME REACHABILITY PROBING ON TOP OF THE STATIC ISP FALLBACK ROUTES ALREADY DOCUMENTED FOR THE THREE-ISP DESIGN. INSTEAD OF RELYING PURELY ON INTERFACE UP/DOWN STATE, THE ROUTERS ACTIVELY VERIFY END-TO-END REACHABILITY TOWARD THE NEXT-HOP ISP GATEWAY AND ONLY MOVE TRAFFIC WHEN THE PROBE ITSELF DETECTS A PROBLEM, SO THE TATA → VERIZON → HE FALLBACK CHAIN REACTS TO ACTUAL DATA-PLANE FAILURES, NOT JUST LINK STATE.

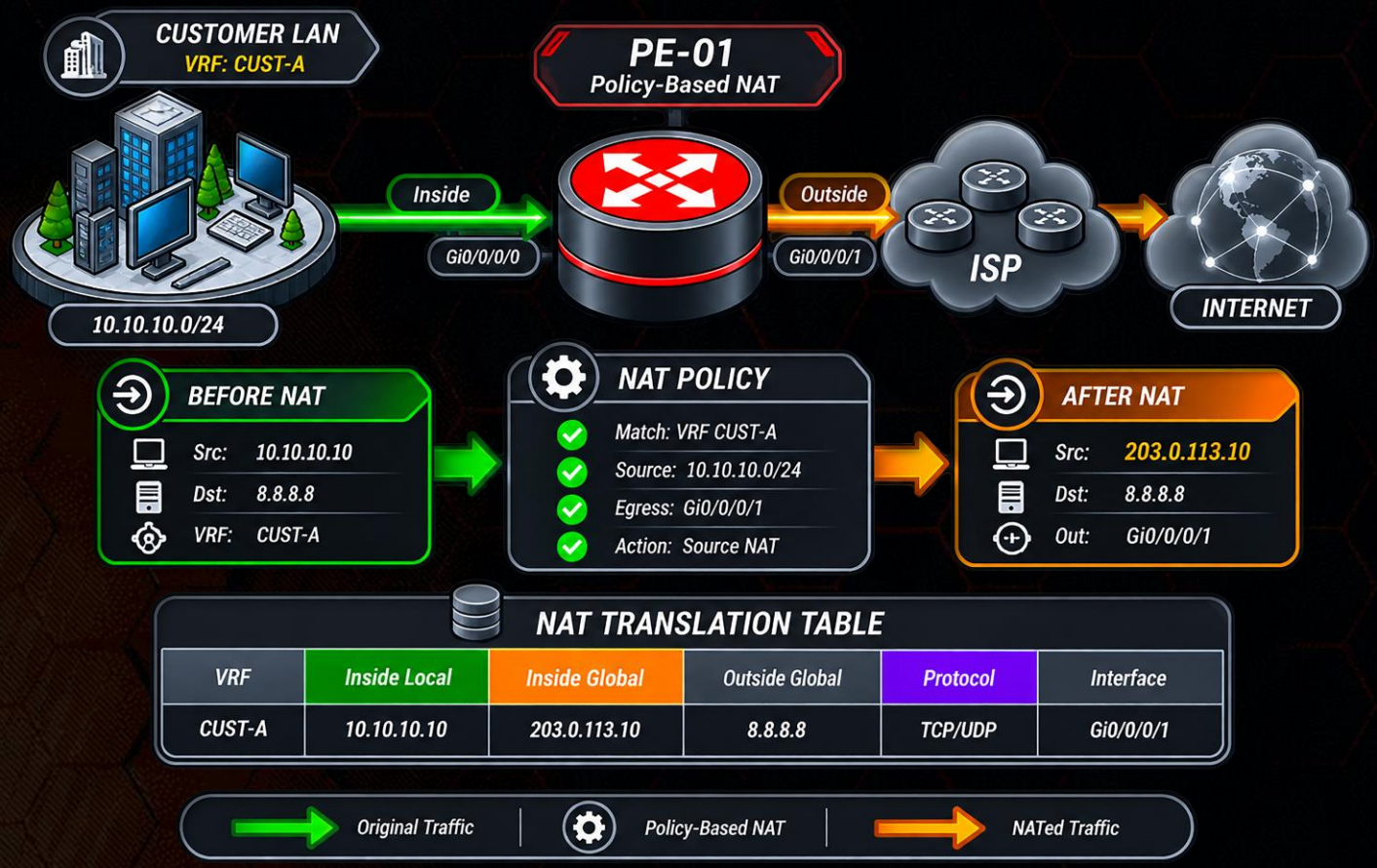
22.2. IMPLEMENTATION

- SKNT-PE02-VIOS RUNS TWO ICMP-ECHO PROBES, IP SLA 202 (TOWARD TATA PRIMARY 10.202.0.2 VIA GIGABITETHERNET0/1) AND IP SLA 204 (TOWARD TATA BACKUP 10.204.0.2 VIA GIGABITETHERNET0/2), EACH WITH A 1000 MS THRESHOLD AND A 5-SECOND FREQUENCY, SCHEDULED PERMANENTLY (“LIFE FOREVER START-TIME NOW”).
- TRACK OBJECTS 202 AND 204 WRAP THESE PROBES WITH A DELAY DOWN 10 / UP 10 DAMPENING TIMER TO PREVENT ROUTE FLAPPING, AND ARE REFERENCED DIRECTLY FROM THE INTER-AS-L3VPN VRF STATIC ROUTES TOWARD THE SPARKLE-ADVERTISED 172.19.10.1-172.19.50.1 TEST PREFIXES (E.G. “IP ROUTE VRF INTER-AS-L3VPN 172.19.10.1 255.255.255.255 10.202.0.2 GLOBAL TAG 500 NAME TATA-PRIMARY-17219-10 TRACK 202”).
- SKNT-PE07-POP02 RUNS IP SLA 10 (ICMP-ECHO TO 8.8.8.8 OUT GIGABITETHERNET0/0) TRACKED BY OBJECT 10, WHICH CONDITIONS THE POP02 DEFAULT ROUTE (“IP ROUTE 0.0.0.0 0.0.0.0 192.168.100.1 TRACK 10”) SO THE DEFAULT ROUTE IS WITHDRAWN AUTOMATICALLY IF UPSTREAM REACHABILITY IS LOST.
- ISP-TATA ITSELF CARRIES FOUR INDEPENDENT TRACK OBJECTS (1, 2, 5 AND 6) BOUND TO THEIR OWN IP SLA OPERATIONS, MIRRORING THE SAME REACHABILITY DISCIPLINE ON THE ISP SIDE OF THE PEERING.

22.3. CONCLUSION

IP SLA AND OBJECT TRACKING CLOSE THE GAP BETWEEN “INTERFACE IS UP” AND “PATH ACTUALLY WORKS,” GIVING THE LAB A REALISTIC, CARRIER-STYLE DETERMINISTIC FAILOVER MECHANISM THAT COMPLEMENTS THE COMMUNITY-BASED BGP FALLBACK POLICY ALREADY DOCUMENTED FOR THE THREE-ISP DESIGN.

23. POLICY-BASED NAT / PAT ON THE PROVIDER EDGE



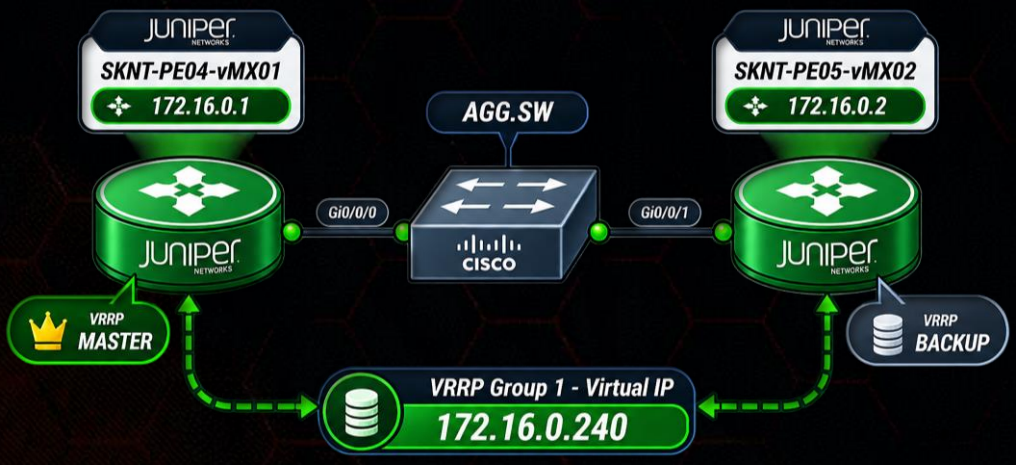
23.1. OVERVIEW

BEYOND THE ROUTING/MPLS SERVICE PLANE, SKNT-PE01-BORDER, SKNT-PE02-VIOS AND SKNT-PE07-POP02 PERFORM NETWORK ADDRESS TRANSLATION SO THAT INTERNAL AND VRF-ATTACHED TEST TRAFFIC CAN REACH ISP-FACING AND MANAGEMENT-FACING PREFIXES WITHOUT EXPOSING INTERNAL RFC1918 ADDRESSING DIRECTLY ON THE WAN LINKS.

23.2. IMPLEMENTATION

- SKNT-PE01-BORDER: “IP NAT INSIDE SOURCE LIST NAT INTERFACE GIGABITETHERNET0/1 OVERLOAD” HIDES HOME-WAN-FACING MANAGEMENT TRAFFIC BEHIND THE P01-ISP-FACING INTERFACE, USING STANDARD ACL NAT (DENY 10.206.0.0/30, PERMIT ANY) TO EXCLUDE THE HE-SECONDARY TRANSIT SUBNET FROM TRANSLATION. A SECOND, VRF-AWARE POLICY-BASED NAT STATEMENT, “IP NAT INSIDE SOURCE ROUTE-MAP RM-NAT-L3VPN-HE INTERFACE GIGABITETHERNET0/2 VRF INTER-AS-L3VPN OVERLOAD,” TRANSLATES ONLY INTER-AS-L3VPN VRF TRAFFIC THAT MATCHES EXTENDED ACL ACL-NAT-L3VPN-HE (192.168.111.0/24 DESTINED TO THE 172.19.10.1/20.1/30.1 TEST LOOPBACKS) WHEN IT EGRESSES TOWARD HE.
- SKNT-PE02-VIOS: TWO PARALLEL POLICY-BASED NAT STATEMENTS, RM-NAT-L3VPN-TATA-G01 AND RM-NAT-L3VPN-TATA-G02, PAT-OVERLOAD THE SAME INTER-AS-L3VPN VRF TRAFFIC OUT GIGABITETHERNET0/1 AND GIGABITETHERNET0/2 RESPECTIVELY, MATCHING THE SHARED ACL-NAT-L3VPN-TATA AND GIVING CE02’S TEST TRAFFIC DUAL, PER-LINK TRANSLATED PATHS ACROSS TATA’S TWO UPLINKS.
- SKNT-PE07-POP02 APPLIES A SIMPLE OVERLOAD NAT (“IP NAT INSIDE SOURCE LIST NAT-POP02 INTERFACE GIGABITETHERNET0/0 OVERLOAD”) TO HIDE THE POP02 PSEUDOWIRE/DMVPN TEST SEGMENTS BEHIND ITS MANAGEMENT/INTERNET-FACING INTERFACE.

24. VRRP FIRST-HOP REDUNDANCY (PE04/PE05 ACCESS GATEWAY)



24.1. OVERVIEW

THE JUNIPER VMX PAIR SKNT-PE04-VMX01 AND SKNT-PE05-VMX02 BOTH ATTACH TO SKNT-AGG.SW AND SHARE A COMMON LAYER-2 SEGMENT (172.16.0.0/24). VRRP PROVIDES A RESILIENT, VENDOR-STANDARD FIRST-HOP GATEWAY ACROSS THIS SHARED SEGMENT INSTEAD OF RELYING ON EITHER PE ALONE.

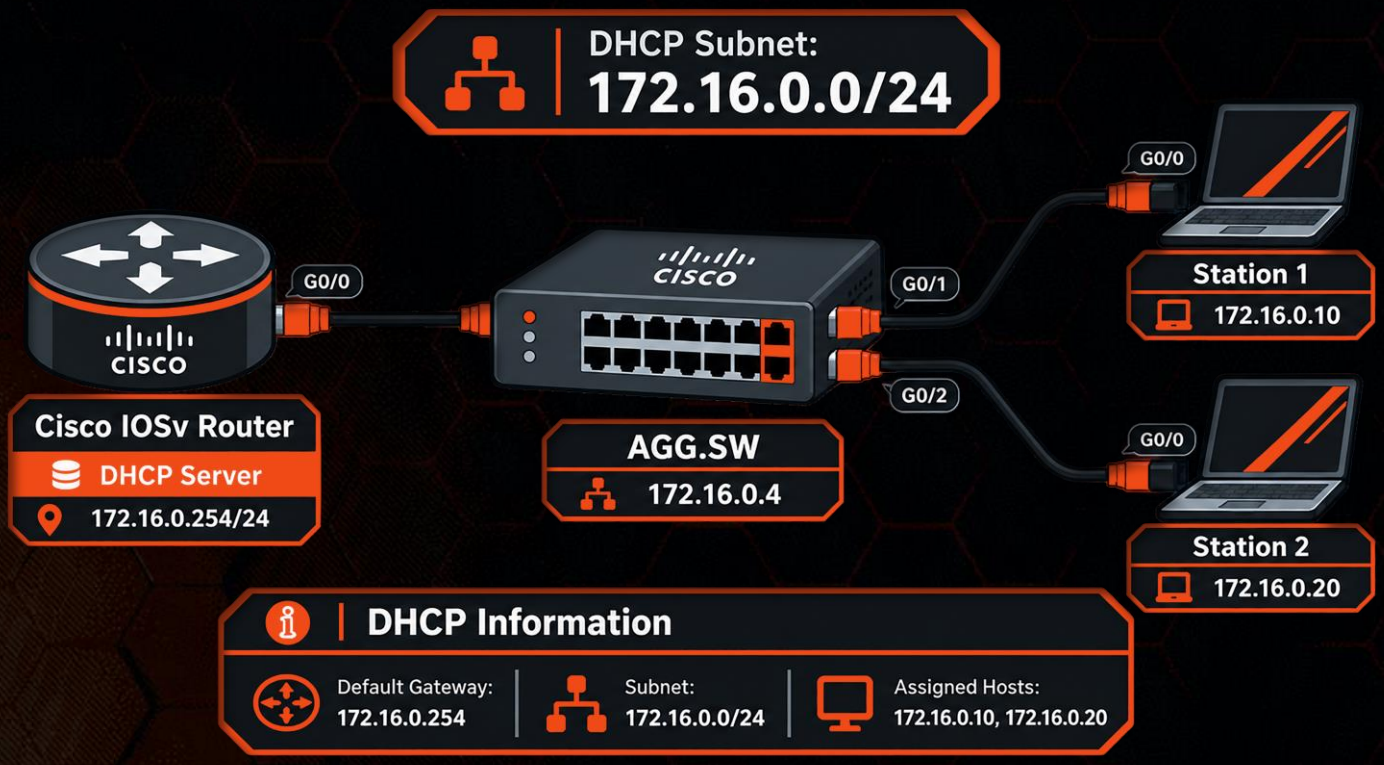
24.2. IMPLEMENTATION

- SKNT-PE04-VMX01, INTERFACE GE-0/0/0.0 (172.16.0.1/24), IS CONFIGURED AS VRRP GROUP 1 WITH VIRTUAL ADDRESS 172.16.0.240, PRIORITY 200, PREEMPT AND ACCEPT-DATA ENABLED, MAKING IT THE VRRP MASTER UNDER NORMAL CONDITIONS.
- SKNT-PE05-VMX02, INTERFACE GE-0/0/1.0 (172.16.0.2/24), JOINS THE SAME VRRP GROUP 1 WITH THE SAME VIRTUAL ADDRESS 172.16.0.240, DEFAULT PRIORITY, PREEMPT AND ACCEPT-DATA ENABLED, ACTING AS THE STANDBY/BACKUP GATEWAY.
- BECAUSE PREEMPT IS ENABLED ON BOTH ROUTERS, MASTERSHIP ALWAYS RETURNS TO SKNT-PE04-VMX01 ONCE IT RECOVERS, KEEPING THE HIGHER-PRIORITY JUNIPER PE AS THE PRIMARY ACCESS GATEWAY FOR THE AGG.SW-ATTACHED SEGMENT.

24.3. CONCLUSION

VRRP DEMONSTRATES A JUNIPER-NATIVE FIRST-HOP REDUNDANCY PATTERN THAT COMPLEMENTS THE LAB'S ROUTED RESILIENCY (OSPF, BGP, TRACKED STATIC ROUTES) WITH A LAYER 2/3 BOUNDARY FAILOVER MECHANISM.

25. DHCP SERVICES ON POP01 | POP02 | AGG.SW



25.1. OVERVIEW

SKNT-PE06-POP01 AND SKNT-PE07-POP02 EACH RUN A SMALL LOCAL DHCP SERVER TO AUTO-ADDRESS THE POINT-TO-POINT IPSEC VPN ENDPOINT SEGMENTS USED FOR THE POP-TO-POP ENCRYPTED OVERLAY TEST PATH, IN ADDITION TO THE MANUALLY ADDRESSED CORE/SERVICE INTERFACES DOCUMENTED ELSEWHERE. THE AGG.SW IS IN CHARGE TO SERVE THE DEVICES THE SUBNET OF THE PROVIDER LAN 172.16.0.0/24.

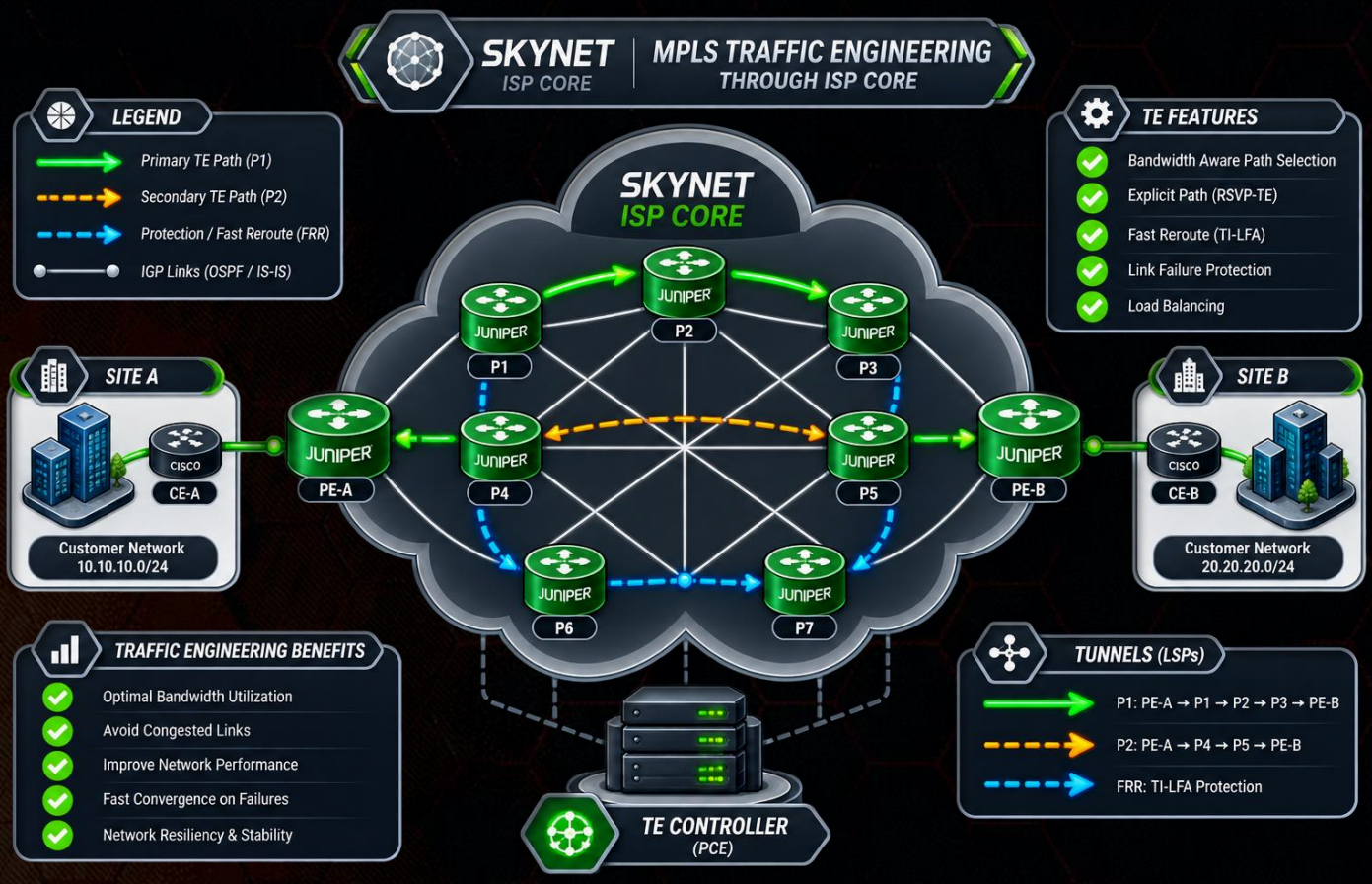
25.2. IMPLEMENTATION

- SKNT-PE06-POP01: “IP DHCP EXCLUDED-ADDRESS 192.151.151.2” RESERVES THE ROUTER’S OWN INTERFACE ADDRESS, AND POOL VPN1 (“NETWORK 192.151.151.0 255.255.255.252”, “DEFAULT-ROUTER 192.151.151.2”) SERVES THE /30 ATTACHED TO THE IPSEC-VPN01-EP01 INTERFACE (GIGABITETHERNET0/4).
- SKNT-PE07-POP02: THE MIRRORED CONFIGURATION EXCLUDES 192.152.152.2 AND POOL VPN2 SERVES 192.152.152.0/30 WITH DEFAULT-ROUTER 192.152.152.2 ON THE IPSEC-VPN01-EP02 INTERFACE.
- CE-01-PW01, CE-02-PW01, CE01-L3VPN-01 AND CE02-L3VPN-01 ALL DISABLE THE DHCP SERVICE DEFAULTS (“NO SERVICE DHCP”, “NO IP DHCP USE VRF CONNECTED”) SINCE THEY USE STATIC ADDRESSING, KEEPING THE DHCP SCOPE INTENTIONALLY ISOLATED TO THE TWO SMALL VPN TEST SEGMENTS.

25.3. CONCLUSION

DHCP IS USED SPARINGLY AND PURPOSEFULLY, ONLY WHERE IT SIMPLIFIES BRING-UP OF THE IPSEC VPN TEST ENDPOINTS, WHILE EVERY CUSTOMER-FACING AND CORE INTERFACE KEEPS DETERMINISTIC STATIC ADDRESSING FOR THE REST OF THE LAB.

26. MPLS TRAFFIC ENGINEERING (RSVP-TE) READINESS



26.1. OVERVIEW

MPLS TRAFFIC ENGINEERING (MPLS-TE) LETS OPERATORS STEER TRAFFIC THROUGH SPECIFIC NETWORK PATHS INSTEAD OF RELYING ONLY ON THE IGP'S SHORTEST PATH. IT CONSIDERS BANDWIDTH, LINK ATTRIBUTES, PRIORITIES, AND OPERATIONAL POLICIES WHEN BUILDING LABEL-SWITCHED PATHS. RSVP-TE IS THE SIGNALING PROTOCOL COMMONLY USED TO ESTABLISH AND MAINTAIN THESE ENGINEERED MPLS TUNNELS. RSVP RESERVES RESOURCES ALONG THE SELECTED PATH AND DISTRIBUTES THE LABELS REQUIRED FOR FORWARDING. TOGETHER, MPLS-TE AND RSVP IMPROVE BANDWIDTH UTILIZATION, SUPPORT FAST REROUTE, AND PROVIDE MORE PREDICTABLE SERVICE PERFORMANCE.

IN ADDITION TO THE LDP-SIGNALED LABEL TRANSPORT ALREADY DOCUMENTED FOR THE MPLS SERVICE PLANE, EVERY CISCO CORE/PE NODE (PE01-BORDER, P01-ISP, PE02-VIOS, PE06-POP01, PE07-POP02, PE08-XRV AND AGG.SW) AND EVERY JUNIPER VMX NODE (PE04-VMX01, PE05-VMX02, PE03-VMX03) ENABLES MPLS TRAFFIC ENGINEERING AND RSVP AT THE PROTOCOL LEVEL, GIVING THE BACKBONE RSVP-TE READINESS ALONGSIDE ITS LDP TRANSPORT.

26.2. IMPLEMENTATION

- CISCO/IOS-XRV NODES ENABLE "MPLS TRAFFIC-ENG TUNNELS" GLOBALLY AND PER-INTERFACE, SET "MPLS TRAFFIC-ENG ROUTER-ID LOOPBACK0," ADVERTISE TE LINK-STATE INFORMATION WITH "MPLS TRAFFIC-ENG AREA 0" UNDER OSPF, AND RESERVE RSVP BANDWIDTH PER LINK (E.G. "IP RSVP BANDWIDTH 500000" ON PE01-BORDER'S CORE-FACING INTERFACES, "IP RSVP BANDWIDTH 50000" ON THE OTHER CISCO CORE LINKS).
- JUNIPER VMX NODES ENABLE RSVP DIRECTLY ON THEIR CORE-FACING AND LOOPBACK INTERFACES ("SET PROTOCOLS RSVP INTERFACE GE-0/0/X.0", "SET PROTOCOLS RSVP INTERFACE LO0.X") AND ENABLE THE OSPF TRAFFIC-ENGINEERING EXTENSION ("SET PROTOCOLS OSPF TRAFFIC-ENGINEERING") SO TE LINK-STATE ADVERTISEMENTS ARE EXCHANGED CONSISTENTLY WITH THE CISCO SIDE OF THE CORE.

27. LAYER 2 HARDENING ON SKNT-AGG.SW



27.1. OVERVIEW

SKNT-AGG.SW AGGREGATES THE DC/EVPN AND EVC300-FACING TRUNK PORTS FROM THE FOUR PE ROUTERS (PE04-VMX01, PE05-VMX02, PE06-POP01, PE07-POP02) AS WELL AS THE ROUTED UPLINKS TO P01-ISP AND PE08-XRV. THE SWITCH RUNS RAPID-PVST+ WITH PORTFAST TO KEEP THIS AGGREGATION LAYER LOOP-SAFE AND FAST-CONVERGING.

27.2. IMPLEMENTATION

- “SPANNING-TREE MODE RAPID-PVST” AND “SPANNING-TREE EXTEND SYSTEM-ID” ARE CONFIGURED GLOBALLY, GIVING IEEE 802.1W-BASED RAPID CONVERGENCE WITH PER-VLAN SPANNING-TREE INSTANCES.
- ALSO IMPLEMENTED ROOT GUARD, BPDU GUARD, BPDU FILTER, DAI AND DHCP SNOOPING.
- ETHERNET0/0-0/3 (THE TRUNK LINKS TO PE04-VMX01, PE05-VMX02, PE06-POP01 AND PE07-POP02) ARE CONFIGURED AS 802.1Q TRUNKS WITH “SPANNING-TREE PORTFAST EDGE TRUNK,” AND ETHERNET1/0-1/1 (THE ROUTED, NO-SWITCHPORT UPLINKS TO P01-ISP AND PE08-XRV) CARRY “SPANNING-TREE PORTFAST EDGE” FOR IMMEDIATE FORWARDING-STATE TRANSITION, APPROPRIATE FOR POINT-TO-POINT PROVIDER LINKS WHERE STP LOOP DETECTION IS NOT THE PRIMARY CONCERN.

27.3. CONCLUSION

RAPID-PVST WITH PORTFAST ON EVERY AGGREGATION-LAYER PORT KEEPS SKNT-AGG.SW’S CONVERGENCE BEHAVIOR ALIGNED WITH THE FAST IGP/LDP/RSVP CONVERGENCE ALREADY ENGINEERED ELSEWHERE IN THE CORE, AVOIDING UNNECESSARY 802.1D-STYLE FORWARDING DELAY ON A SWITCH THAT CARRIES EVPN, EVC300 AND MPLS TE TRAFFIC.

28. UPDATED CONFIGURATION BASELINE (15-AUG-2026)

THIS SECTION IS THE AUTHORITATIVE CONFIGURATION-DERIVED INVENTORY FOR THE JULY 2026 REVISION. IT SUPERSEDES EARLIER DEVICE, ADDRESSING, ROLE, AND FEATURE INVENTORIES WHERE DIFFERENCES EXIST. THE TOPOLOGY WAS UPDATED ON PAGE 7 FROM THE SUPPLIED TOPOLOGY FILE.

28.1. DEVICE AND SERVICE SUMMARY

DEVICE	ROLE	PLATFORM	VERSION	ASN	DETECTED PROTOCOLS / SERVICES
SKNT-PE01-BORDER	Border PE and customer/upstream edge	Cisco IOS/IOS-XE/IOL	15.6	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, NAT/PAT, SSH Hardening
SKNT-P01-ISP	Core P router and route reflector	Cisco IOS/IOS-XE/IOL	15.6	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, SSH Hardening
SKNT-PE02-vIOS	Provider edge / service node	Cisco IOS/IOS-XE/IOL	15.6	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, NAT/PAT, IP SLA/Tracking, SSH Hardening
SKNT-PE04-vMX01	Provider edge / service node	Juniper Junos	14.1R1.10	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, EVPN, L2VPN/Pseudowire, QoS, VRRP/HSRP, SSH Hardening
SKNT-PE05-vMX02	Provider edge / service node	Juniper Junos	14.1R1.10	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, EVPN, L2VPN/Pseudowire, QoS, VRRP/HSRP, SSH Hardening
SKNT-PE03-vMX03	Provider edge / service node	Juniper Junos	14.1R1.10	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, EVPN, QoS, SSH Hardening
SKNT-AGG.SW	Aggregation and route-reflector node	Cisco IOS/IOS-XE/IOL	15.2	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, QoS, CoPP, DHCP, STP

					Hardening, SSH Hardening
SKNT-PE06-POP01	Provider edge / POP service node	Cisco IOS/IOS-XE/IOL	15.6	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, L2VPN/Pseudowire, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, SSH Hardening
SKNT-PE07-POP02	Provider edge / POP service node	Cisco IOS/IOS-XE/IOL	15.6	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, L2VPN/Pseudowire, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, NAT/PAT, IP SLA/Tracking, SSH Hardening
SKNT-PE08-XRv	Provider edge / service node	Cisco IOS XR	6.1.3	300	OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, QoS, CoPP
ISP-TATA	External upstream / transit provider	Cisco IOS/IOS-XE/IOL	15.4	500	BGP, MPLS, L3VPN/VRF, QoS, CoPP, IP SLA/Tracking, SSH Hardening
ISP-VERIZON	External upstream / transit provider	Cisco IOS/IOS-XE/IOL	15.4	200	BGP, QoS, CoPP, SSH Hardening
ISP-HE	External upstream / transit provider	Huawei VRP	vrpskynet.zip	400	BGP, SSH Hardening
ISP-SPARKLE	External upstream / transit provider	Cisco IOS/IOS-XE/IOL	15.4	777	BGP, QoS, CoPP, SSH Hardening
DC-eVPN-01	EVPN customer/server endpoint	Cisco IOS/IOS-XE/IOL	15.2	-	EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening
DC-eVPN-02	EVPN customer/server endpoint	Cisco IOS/IOS-XE/IOL	15.2	-	EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening
DC-eVPN-03	EVPN customer/server endpoint	Cisco IOS/IOS-XE/IOL	15.2	-	OSPFv2, EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening
DC-eVPN-04	EVPN customer/server endpoint	Cisco IOS/IOS-XE/IOL	15.2	-	EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening
CE-01-PW01	Pseudowire customer edge or endpoint	Cisco IOS/IOS-XE/IOL	15.4	-	OSPFv2, OSPFv3, MPLS, CoPP, SSH Hardening

CE-02-PW01	Pseudowire customer edge or endpoint	Cisco IOS/IOS-XE/IOL	15.4	-	OSPFv2, OSPFv3, MPLS, CoPP, SSH Hardening
CE01-L3VPN-01	L3VPN customer edge	Cisco IOS/IOS-XE/IOL	15.4	301	BGP, L3VPN/VRF, QoS, CoPP, SSH Hardening
CE02-L3VPN-01	L3VPN customer edge	Cisco IOS/IOS-XE/IOL	15.4	301	BGP, L3VPN/VRF, CoPP, SSH Hardening
EVC-DC-01	EVC300 customer endpoint	Cisco IOS/IOS-XE/IOL	15.2	-	VLAN300/EVC, DUAL-STACK GRE, IKEV2/IPSEC, RIPv2, RIPng, QoS, CoPP, LLDP
EVC-DC-02	EVC300 customer endpoint	Cisco IOS/IOS-XE/IOL	15.2	-	VLAN300/EVC, DUAL-STACK GRE, IKEV2/IPSEC, RIPv2, RIPng, CoPP, LLDP
PC1-PW2-JUNOS	Pseudowire customer edge or endpoint	Cisco IOS/IOS-XE/IOL	15.4	-	OSPFv2, OSPFv3, CoPP
PC2-PW2-JUNOS	Pseudowire customer edge or endpoint	Cisco IOS/IOS-XE/IOL	15.4	-	OSPFv2, OSPFv3, CoPP



28.2. SKNT-PE01-BORDER CONFIGURATION INVENTORY

ROLE: Border PE and customer/upstream edge
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.6
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, NAT/PAT, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	Internal-Transit	192.168.1.1/32	FC50::1/128
Loopback1	Provider-Lan	172.16.0.254/32	-
Loopback10	CE01-L3VPN-DC01	192.168.98.1/32	2001:DB8:98::1/128
Loopback20	CE01-L3VPN-DC02	192.168.98.2/32	2001:DB8:98::2/128
Loopback30	CE01-L3VPN-DC03	192.168.98.3/32	2001:DB8:98::3/128
Loopback40	CE01-L3VPN-DC04	192.168.98.4/32	2001:DB8:98::4/128
Loopback50	CE01-L3VPN-DC05	192.168.98.5/32	2001:DB8:98::5/128
Loopback300	DMVPN-ENDPOINT-PE01	10.254.250.1/32	-
Tunnel10	DMVPN-SPOKE-TO-P01	10.15.150.8/24	-
Tunnel20	IPv6-DMVPN-SPOKE-TO-P01	-	2001:DB8:150::8/64
Tunnel21	IPv6-P2P-GRE-TO-P01	-	2001:DB8:151::2/64
GigabitEthernet0/0	To-SKNT-P01-ISP	10.0.0.1/30	FC02::/127
GigabitEthernet0/1	To-Home-WAN	192.168.100.100/24	-
GigabitEthernet0/2	To-ISP-HE	10.206.0.1/30	FC99::1/127
GigabitEthernet0/3	NOT-USED	-	-
GigabitEthernet0/4	NOT-USED	-	-
GigabitEthernet0/5	To-CE01-L3VPN-01	192.168.111.1/24	2001:DB8:111::1/64
GigabitEthernet0/6	NOT-USED	-	-
GigabitEthernet0/7	NOT-USED	-	-

28.3. SKNT-P01-ISP CONFIGURATION INVENTORY

ROLE: Core P router and route reflector
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.6
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	Internal-Transit	192.168.1.2/32	FC51::1/128
Loopback1	Provider-Lan	172.16.0.3/32	-
Loopback300	DMVPN-ENDPOINT-HUB	10.254.252.1/32	-
Tunnel10	DMVPN-HUB	10.15.150.1/24	-
Tunnel20	IPv6-DMVPN-HUB	-	2001:DB8:150::1/64
Tunnel21	IPv6-P2P-GRE-TO-PE01	-	2001:DB8:151::1/64
Tunnel22	IPv6-P2P-GRE-TO-PE02	-	2001:DB8:152::1/64
Tunnel23	IPv6-P2P-GRE-TO-PE06	-	2001:DB8:153::1/64
Tunnel24	IPv6-P2P-GRE-TO-PE07	-	2001:DB8:154::1/64
GigabitEthernet0/0	To-SKNT-PE01-BORDER	10.0.0.2/30	FC02::1/127
GigabitEthernet0/1	To-SKNT-AGG.SW	10.0.0.29/30	FC24::1/127
GigabitEthernet0/2	To-SKNT-PE04-vMX01	10.0.0.5/30	FC01::/127
GigabitEthernet0/3	To-SKNT-PE02-vIOS	10.0.0.21/30	FC03::/127
GigabitEthernet0/4	To-SKNT-PE05-vMX02	10.0.0.9/30	FC04::/127
GigabitEthernet0/5	NOT-USED	-	-
GigabitEthernet0/6	NOT-USED	-	-
GigabitEthernet0/7	NOT-USED	-	-

28.4. SKNT-PE02-vIOS CONFIGURATION INVENTORY

ROLE: Provider edge / service node
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.6
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, NAT/PAT, IP SLA/Tracking, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	Internal-Transit	192.168.1.250/32	FC65::1/128
Loopback1	Provider-Lan	172.16.0.250/32	-
Loopback10	CE02-L3VPN-DC01	192.168.99.1/32	2001:DB8:99::1/128
Loopback20	CE02-L3VPN-DC02	192.168.99.2/32	2001:DB8:99::2/128
Loopback30	CE02-L3VPN-DC03	192.168.99.3/32	2001:DB8:99::3/128
Loopback40	CE02-L3VPN-DC04	192.168.99.4/32	2001:DB8:99::4/128
Loopback50	CE02-L3VPN-DC05	192.168.99.5/32	2001:DB8:99::5/128
Loopback300	DMVPN-ENDPOINT-PE02	10.254.251.1/32	-
Tunnel10	DMVPN-SPOKE-TO-P01	10.15.150.9/24	-
Tunnel20	IPv6-DMVPN-SPOKE-TO-P01	-	2001:DB8:150::9/64
GigabitEthernet0/0	To-SKNT-P01-ISP	10.0.0.22/30	FC03::1/127
GigabitEthernet0/1	To-ISP-TATA	10.202.0.1/30	FC23::1/127
GigabitEthernet0/2	To-ISP-TATA	10.204.0.1/30	FC68::1/127
GigabitEthernet0/3	NOT-USED	-	-
GigabitEthernet0/4	To-SKNT-PE03-vMX03	10.10.11.1/30	FC79::1/127
GigabitEthernet0/5	To-CE02-L3VPN-01	192.168.222.1/24	-
GigabitEthernet0/6	NOT-USED	-	-
GigabitEthernet0/7	NOT-USED	-	-

28.5. SKNT-PE04-vMX01 CONFIGURATION INVENTORY

ROLE: Provider edge / service node
PLATFORM: Juniper Junos | VERSION: 14.1R1.10
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, EVPN, L2VPN/Pseudowire, QoS, VRRP/HSRP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
ge-0/0/0.0	To-SKNT-AGG.SW	172.16.0.1/24, 172.16.0.1/24, 172.16.0.1/24, 172.16.0.1/24	fc07::/127
ge-0/0/1.0	To-SKNT-P01-ISP	10.0.0.6/30	fc01::1/127
ge-0/0/2.0	To-SKNT-PE06-POP01	10.0.0.13/30	fc05::/127
ge-0/0/3	To-DC-eVPN-02	-	-
ge-0/0/3.600	To-DC-eVPN-02	-	-
ge-0/0/4	To-DC-eVPN-01	-	-
ge-0/0/4.600	To-DC-eVPN-01	-	-
ge-0/0/5.0	To-ISP-HE	10.205.0.1/30	fc77::1/127
ge-0/0/6	To-EVC300-CUSTOMER	-	-
ge-0/0/6.300	EVC300-CUST-VLAN300-POP-PUSH	-	-
ge-0/0/7	To-PWPC-PE04	-	-
ge-0/0/8	NOT-USED	-	-
ge-0/0/9	NOT-USED	-	-
irb.600	L3-GW-VLAN600-vMX01	192.168.160.1/24	-
lo0.1	Internal-Transit	192.168.1.3/32	fc60::1/128

28.6. SKNT-PE05-vMX02 CONFIGURATION INVENTORY

ROLE: Provider edge / service node
PLATFORM: Juniper Junos | VERSION: 14.1R1.10
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, EVPN, L2VPN/Pseudowire, QoS, VRRP/HSRP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
ge-0/0/0.0	To-ISP-VERIZON	10.200.200.1/30	fc10::1/127
ge-0/0/1.0	To-SKNT-AGG.SW	172.16.0.2/24, 172.16.0.2/24, 12.16.0.2/24	fc07::1/127
ge-0/0/2.0	To-SKNT-P01-ISP	10.0.0.10/30	fc04::1/127
ge-0/0/3.0	To-SKNT-PE07-POP02	10.0.0.17/30	fc06::/127
ge-0/0/4	NOT-USED	-	-
ge-0/0/5	To-DC-eVPN-04	-	-
ge-0/0/5.600	To-DC-eVPN-04	-	-
ge-0/0/6	To-EVC300-CUSTOMER	-	-
ge-0/0/6.300	EVC300-CUST-VLAN300-POP-PUSH	-	-
ge-0/0/7	To-PWPC-PE05	-	-
ge-0/0/8	NOT-USED	-	-
ge-0/0/9	NOT-USED	-	-
irb.600	L3-GW-VLAN600-DC-eVPN	10.35.0.254/24	2001:db8:35:600::254/64
lo0.1	Internal-Transit	192.168.1.5/32	fc61::1/128

28.7. SKNT-PE03-vMX03 CONFIGURATION INVENTORY

ROLE: Provider edge / service node
PLATFORM: Juniper Junos | VERSION: 14.1R1.10
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, EVPN, QoS, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
ge-0/0/0	To-DC-eVPN-03	-	-
ge-0/0/0.600	To-DC-eVPN-03	-	-
ge-0/0/1	NOT-USED	-	-
ge-0/0/2.0	To-ISP-VERIZON	10.209.0.1/30	-
ge-0/0/3	NOT-USED	-	-
ge-0/0/4.0	To-SKNT-PE02-vIOS	10.10.11.2/30	fc79::/127
ge-0/0/5	NOT-USED	-	-
ge-0/0/6	NOT-USED	-	-
ge-0/0/7	NOT-USED	-	-
ge-0/0/8	NOT-USED	-	-
ge-0/0/9	NOT-USED	-	-
lo0.0	Internal-Transit	192.168.1.251/32, 172.16.0.251/32	fc66::1/128

28.8. SKNT-AGG.SW CONFIGURATION INVENTORY

ROLE: Aggregation and route-reflector node
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.2
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, QoS, CoPP, DHCP, STP Hardening, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	Internal-Transit	192.168.1.4/32	FC62::1/128
Loopback1	Provider-Lan	172.16.0.4/32	-
Ethernet0/0	To-SKNT-PE04-vMX01	-	-
Ethernet0/1	To-SKNT-PE05-vMX02	-	-
Ethernet0/2	To-SKNT-PE06-POP01	-	-
Ethernet0/3	To-SKNT-PE07-POP02	-	-
Ethernet1/0	To-SKNT-P01-ISP	10.0.0.30/30	FC24::/127
Ethernet1/1	To-SKNT-PE08-XRv	10.34.0.1/30	FC34::1/127
Ethernet1/2	NOT-USED	-	-
Ethernet1/3	NOT-USED	-	-

28.9. SKNT-PE06-POP01 CONFIGURATION INVENTORY

ROLE: Provider edge / POP service node
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.6
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, L2VPN/Pseudowire, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	Internal-Transit	192.168.1.100/32	FC63::1/128
Loopback1	Provider-Lan	172.16.0.100/32	-
Loopback10	Private-Subnet-POP01	10.151.151.1/30	-
Loopback300	DMVPN-ENDPOINT-PE06	10.254.253.1/32	-
Tunnel10	DMVPN-SPOKE-TO-P01	10.15.150.6/24	-
Tunnel20	IPv6-DMVPN-SPOKE-TO-P01	-	2001:DB8:150::6/64
Tunnel100	VPN-to-POP02	192.153.153.1/30	-
GigabitEthernet0/0	NOT-USED	-	-
GigabitEthernet0/1	To-SKNT-PE07-POP02	10.0.0.25/30	FC20::1/127
GigabitEthernet0/2	To-SKNT-PE04-vMX01	10.0.0.14/30	FC05::1/127
GigabitEthernet0/3	NOT-USED	-	-
GigabitEthernet0/4	To-IPSEC-VPN01-EP01	192.151.151.2/30	-
GigabitEthernet0/5	To-CE-01-PW01	-	-
GigabitEthernet0/6	NOT-USED	-	-
GigabitEthernet0/7	NOT-USED	-	-

28.10. SKNT-PE07-POP02 CONFIGURATION INVENTORY

ROLE: Provider edge / POP service node
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.6
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, L2VPN/Pseudowire, DMVPN/NHRP, IPsec/IKE, QoS, CoPP, DHCP, NAT/PAT, IP SLA/Tracking, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	Internal-Transit	192.168.1.200/32	FC64::1/128
Loopback1	Provider-Lan	172.16.0.200/32	-
Loopback20	Private-Subnet-POP02	10.152.152.1/30	-
Loopback300	DMVPN-ENDPOINT-PE07	10.254.254.1/32	-
Tunnel10	DMVPN-SPOKE-TO-P01	10.15.150.7/24	-
Tunnel20	IPv6-DMVPN-SPOKE-TO-P01	-	2001:DB8:150::7/64
Tunnel200	VPN-to-POP01	192.153.153.2/30	-
GigabitEthernet0/0	To-EVE-MGMT-Cloud	192.168.100.200/24	-
GigabitEthernet0/1	To-SKNT-PE06-POP01	10.0.0.26/30	FC20::/127
GigabitEthernet0/2	NOT-USED	-	-
GigabitEthernet0/3	To-SKNT-PE05-vMX02	10.0.0.18/30	FC06::1/127
GigabitEthernet0/4	To-IPSEC-VPN01-EP02	192.152.152.2/30	-
GigabitEthernet0/5	To-CE-02-PW01	-	-
GigabitEthernet0/6	NOT-USED	-	-
GigabitEthernet0/7	NOT-USED	-	-

28.11. SKNT-PE08-XRv CONFIGURATION INVENTORY

ROLE: Provider edge / service node
PLATFORM: Cisco IOS XR | VERSION: 6.1.3
AUTONOMOUS SYSTEM(S): 300
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, BGP, MPLS, LDP, RSVP-TE, L3VPN/VRF, QoS, CoPP

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	Internal-Transit	-	fc93::1/128
Loopback1	Provider-Lan	-	-
Loopback16	SKNT-IPV6-ISP-AGGREGATE-ANCHOR	-	fc16::101/128
MgmtEth0/0/CPU0/0	NOT-USED	-	-
GigabitEthernet0/0/0/0	To-ISP-SPARKLE	-	fc32::1/127
GigabitEthernet0/0/0/1	To-SKNT-AGG.SW	-	fc34::/127
GigabitEthernet0/0/0/2	NOT-USED	-	-

28.12. ISP-TATA CONFIGURATION INVENTORY

ROLE: External upstream / transit provider
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): 500
DETECTED PROTOCOLS / SERVICES: BGP, MPLS, L3VPN/VRF, QoS, CoPP, IP SLA/Tracking, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback1	Anuncio-eBGP1	172.19.10.1/32	FC11::1/128
Loopback2	Anuncio-eBGP2	172.19.20.1/32	FC12::1/128
Loopback3	Anuncio-eBGP3	172.19.30.1/32	FC13::1/128
Loopback4	Anuncio-eBGP4	172.19.40.1/32	FC14::1/128
Loopback5	Anuncio-eBGP5	172.19.50.1/32	FC15::1/128
Ethernet0/0	NOT-USED	-	-
Ethernet0/1	To-SKNT-PE02-vIOS	10.202.0.2/30	FC23::/127
Ethernet0/2	To-SKNT-PE02-vIOS	10.204.0.2/30	FC68::/127
Ethernet0/3	NOT-USED	-	-

28.13. ISP-VERIZON CONFIGURATION INVENTORY

ROLE: External upstream / transit provider
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): 200
DETECTED PROTOCOLS / SERVICES: BGP, QoS, CoPP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback1	Anuncio-eBGP1	172.19.10.1/32	FC11::1/128
Loopback2	Anuncio-eBGP2	172.19.20.1/32	FC12::1/128
Loopback3	Anuncio-eBGP3	172.19.30.1/32	FC13::1/128
Loopback4	Anuncio-eBGP4	172.19.40.1/32	FC14::1/128
Loopback5	Anuncio-eBGP5	172.19.50.1/32	FC15::1/128
Ethernet0/0	To-PE-05-vMX-02	10.200.200.2/30	FC10::/127
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	To-SKNT-PE03-vMX03	10.209.0.2/30	-
Ethernet0/3	NOT-USED	-	-

28.14. ISP-HE CONFIGURATION INVENTORY

ROLE: External upstream / transit provider
PLATFORM: Huawei VRP | VERSION: vrpskynet.zip
AUTONOMOUS SYSTEM(S): 400
DETECTED PROTOCOLS / SERVICES: BGP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Ethernet1/0/0	NOT-USED	-	-
Ethernet1/0/1	To-CE01-L3VPN-01	10.207.0.2/30	-
Ethernet1/0/2	To-SKNT-PE01-BORDER	10.206.0.2/30	FC99::/127
Ethernet1/0/3	NOT-USED	-	-
Ethernet1/0/4	NOT-USED	-	-
Ethernet1/0/5	To-PE-04-vMX-01	10.205.0.2/30	FC77::/127
Ethernet1/0/6	NOT-USED	-	-
Ethernet1/0/7	NOT-USED	-	-
Ethernet1/0/8	NOT-USED	-	-
Ethernet1/0/9	NOT-USED	-	-
LoopBack1	Anuncio-eBGP1	172.19.10.1/32	FC11::1/128
LoopBack2	Anuncio-eBGP2	172.19.20.1/32	FC12::1/128
LoopBack3	Anuncio-eBGP3	172.19.30.1/32	FC13::1/128
LoopBack4	Anuncio-eBGP4	172.19.40.1/32	FC14::1/128
LoopBack5	Anuncio-eBGP5	172.19.50.1/32	FC15::1/128
NULL0	AS-PREPEND-HE-2-SKYNET	-	-

28.15. ISP-SPARKLE CONFIGURATION INVENTORY

ROLE: External upstream / transit provider
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): 777
DETECTED PROTOCOLS / SERVICES: BGP, QoS, CoPP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback1	Anuncio-eBGP1	172.19.10.1/32	FC11::1/128
Loopback2	Anuncio-eBGP2	172.19.20.1/32	FC12::1/128
Loopback3	Anuncio-eBGP3	172.19.30.1/32	FC13::1/128
Loopback4	Anuncio-eBGP4	172.19.40.1/32	FC14::1/128
Loopback5	Anuncio-eBGP5	172.19.50.1/32	FC15::1/128
Ethernet0/0	To-SKNT-PE08-XRv	10.201.201.2/30	FC32::/127
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-

28.16. DC-eVPN-01 CONFIGURATION INVENTORY

ROLE: EVPN customer/server endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.2
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	DC-01-SERVER-01	172.31.35.1/32	2001:DB8:35:1::1/128
Loopback1	DC-eVPN-01-SERVER-01	172.31.35.5/32	2001:DB8:35:1::5/128
Loopback2	DC-eVPN-01-SERVER-02	172.31.35.6/32	2001:DB8:35:1::6/128
Loopback3	DC-eVPN-01-SERVER-03	172.31.35.7/32	2001:DB8:35:1::7/128
Loopback4	DC-eVPN-01-SERVER-04	172.31.35.8/32	2001:DB8:35:1::8/128
Loopback5	DC-eVPN-01-SERVER-05	172.31.35.9/32	2001:DB8:35:1::9/128
Ethernet0/0	To-SKNT-PE04-vMX01	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Vlan600	To-SKNT-PE04-vMX01	10.35.0.1/24	2001:DB8:35:600::1/64

28.17. DC-eVPN-02 CONFIGURATION INVENTORY

ROLE: EVPN customer/server endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.2
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	DC-02-SERVER-01	172.31.35.2/32	2001:DB8:35:1::2/128
Loopback1	DC-eVPN-02-SERVER-01	172.31.35.10/32	2001:DB8:35:1::10/128
Loopback2	DC-eVPN-02-SERVER-02	172.31.35.11/32	2001:DB8:35:1::11/128
Loopback3	DC-eVPN-02-SERVER-03	172.31.35.12/32	2001:DB8:35:1::12/128
Loopback4	DC-eVPN-02-SERVER-04	172.31.35.13/32	2001:DB8:35:1::13/128
Loopback5	DC-eVPN-02-SERVER-05	172.31.35.14/32	2001:DB8:35:1::14/128
Ethernet0/0	To-SKNT-PE04-vMX01	-	-
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Vlan600	To-SKNT-PE04-vMX01	10.35.0.2/24	2001:DB8:35:600::2/64

28.18. DC-eVPN-03 CONFIGURATION INVENTORY

ROLE: EVPN customer/server endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.2
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: OSPFv2, EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	DC-03-SERVER-01	172.31.35.3/32	2001:DB8:35:1::3/128
Loopback1	DC-eVPN-03-SERVER-01	172.31.35.15/32	2001:DB8:35:1::15/128
Loopback2	DC-eVPN-03-SERVER-02	172.31.35.16/32	2001:DB8:35:1::16/128
Loopback3	DC-eVPN-03-SERVER-03	172.31.35.17/32	2001:DB8:35:1::17/128
Loopback4	DC-eVPN-03-SERVER-04	172.31.35.18/32	2001:DB8:35:1::18/128
Loopback5	DC-eVPN-03-SERVER-05	172.31.35.19/32	2001:DB8:35:1::19/128
Ethernet0/0	To-SKNT-PE03-vMX03	-	-
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Vlan600	To-SKNT-PE03-vMX03	10.35.0.3/24	2001:DB8:35:600::3/64

28.19. DC-eVPN-04 CONFIGURATION INVENTORY

ROLE: EVPN customer/server endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.2
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: EVPN, CoPP, EIGRP/EIGRPv6, STP Hardening, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	DC-04-SERVER-01	172.31.35.4/32	2001:DB8:35:1::4/128
Loopback1	DC-eVPN-04-SERVER-01	172.31.35.20/32	2001:DB8:35:1::20/128
Loopback2	DC-eVPN-04-SERVER-02	172.31.35.21/32	2001:DB8:35:1::21/128
Loopback3	DC-eVPN-04-SERVER-03	172.31.35.22/32	2001:DB8:35:1::22/128
Loopback4	DC-eVPN-04-SERVER-04	172.31.35.23/32	2001:DB8:35:1::23/128
Loopback5	DC-eVPN-04-SERVER-05	172.31.35.24/32	2001:DB8:35:1::24/128
Ethernet0/0	To-SKNT-PE05-vMX02	-	-
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Vlan600	To-SKNT-PE05-vMX02	10.35.0.4/24	2001:DB8:35:600::4/64

28.20. CE-01-PW01 CONFIGURATION INVENTORY

ROLE: Pseudowire customer edge or endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, MPLS, CoPP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	CE-01-Lo0-PW	192.168.1.6/32	-
Ethernet0/0	To-CEPC01	10.40.40.254/24	2001:DB8:40::254/64
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Ethernet1/0	NOT-USED	-	-
Ethernet1/1	To-POP-02-PW-AC	10.10.20.1/30	2001:DB8:20::1/64
Ethernet1/2	NOT-USED	-	-
Ethernet1/3	NOT-USED	-	-
Loopback10-50	PW-SERVICE-01..05	10.201.1.1/32 ... 10.201.1.5/32	2001:DB8:201:1::1/128 ... ::5/128

28.21. CE-02-PW01 CONFIGURATION INVENTORY

ROLE: Pseudowire customer edge or endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, MPLS, CoPP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	CE-02-Lo0-PW	192.168.1.7/32	-
Ethernet0/0	To-CEPC02	10.50.50.254/24	2001:DB8:50::254/64
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Ethernet1/0	NOT-USED	-	-
Ethernet1/1	To-POP-01-PW-AC	10.10.20.2/30	2001:DB8:20::2/64
Ethernet1/2	NOT-USED	-	-
Ethernet1/3	NOT-USED	-	-
Loopback10-50	PW-SERVICE-01..05	10.202.1.1/32 ... 10.202.1.5/32	2001:DB8:202:1::1/128 ... ::5/128

28.22. CE01-L3VPN-01 CONFIGURATION INVENTORY

ROLE: L3VPN customer edge
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): 301
DETECTED PROTOCOLS / SERVICES: BGP, L3VPN/VRF, QoS, CoPP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Ethernet0/0	To-SKNT-PE01-BORDER	192.168.111.2/24	-
Ethernet0/1	To-ISP-HE	10.207.0.1/30	-
Ethernet0/2	To-CE01-CUSTOMER-LAN	192.168.245.1/30	-
Ethernet0/3	NOT-USED	-	-

28.23. CE02-L3VPN-01 CONFIGURATION INVENTORY

ROLE: L3VPN customer edge
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): 301
DETECTED PROTOCOLS / SERVICES: BGP, L3VPN/VRF, CoPP, SSH Hardening

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Ethernet0/0	To-SKNT-PE02-vIOS	192.168.222.2/24	-
Ethernet0/1	To-CE02-CUSTOMER-LAN	192.168.246.1/30	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-

28.24. EVC-DC-01 CONFIGURATION INVENTORY

ROLE: EVC300 CUSTOMER ENDPOINT PLATFORM: CISCO IOS/IOS-XE/IOL | VERSION: 15.2
AUTONOMOUS SYSTEM(S): - DETECTED PROTOCOLS / SERVICES: VLAN300/EVC, DUAL-STACK GRE, IKEV2/IPSEC, RIPV2, RIPNG, QOS, COPP, LLDP

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	EVC-DC-01-SERVER-01	10.16.16.1/32	-
Loopback1	EVC-DC-01-V4V6-01	192.168.227.1/32	FD00:16:16::1/128
Loopback2	EVC-DC-01-V4V6-02	192.168.227.2/32	FD00:16:16::2/128
Loopback3	EVC-DC-01-V4V6-03	192.168.227.3/32	FD00:16:16::3/128
Loopback4	EVC-DC-01-V4V6-04	192.168.227.4/32	FD00:16:16::4/128
Loopback5	EVC-DC-01-V4V6-05	192.168.227.5/32	FD00:16:16::5/128
Loopback300	SIMULATED-INTERNAL-LAN-DC1	172.16.172.1/32	FD00:172:16::1/128
Tunnel0	GRE over IKEv2/IPsec	10.177.177.1/30	-
Tunnel1	GRE over IPv6 protected by IKEv2/IPsec	-	2001:177:177::1/64
Ethernet0/0	To-SKNT-PE04-vMX01	-	-
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Vlan300	EVC300-Test-SVI	192.168.131.1/24	FC91::1/64

28.25. EVC-DC-02 CONFIGURATION INVENTORY

ROLE: EVC300 CUSTOMER ENDPOINT PLATFORM: CISCO IOS/IOS-XE/IOL | VERSION: 15.2
AUTONOMOUS SYSTEM(S): - DETECTED PROTOCOLS / SERVICES: VLAN300/EVC, DUAL-STACK GRE, IKEV2/IPSEC, RIPV2, RIPNG, COPP, LLDP

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	EVC-DC-02-SERVER-01	10.17.17.1/32	-
Loopback1	EVC-DC-02-V4V6-01	192.168.228.1/32	FD00:17:17::1/128
Loopback2	EVC-DC-02-V4V6-02	192.168.228.2/32	FD00:17:17::2/128
Loopback3	EVC-DC-02-V4V6-03	192.168.228.3/32	FD00:17:17::3/128
Loopback4	EVC-DC-02-V4V6-04	192.168.228.4/32	FD00:17:17::4/128
Loopback5	EVC-DC-02-V4V6-05	192.168.228.5/32	FD00:17:17::5/128
Loopback300	SIMULATED-INTERNAL-LAN-DC2	172.16.173.1/32	FD00:173:16::1/128
Tunnel0	GRE over IKEv2/IPsec	10.177.177.2/30	-
Tunnel1	GRE over IPv6 protected by IKEv2/IPsec	-	2001:177:177::2/64
Ethernet0/0	To-SKNT-PE05-vMX02	-	-
Ethernet0/1	NOT-USED	-	-
Ethernet0/2	NOT-USED	-	-
Ethernet0/3	NOT-USED	-	-
Vlan300	EVC300-Test-SVI	192.168.131.2/24	FC91::2/64

28.26. PC1-PW2-JUNOS CONFIGURATION INVENTORY

ROLE: Pseudowire customer edge or endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, CoPP

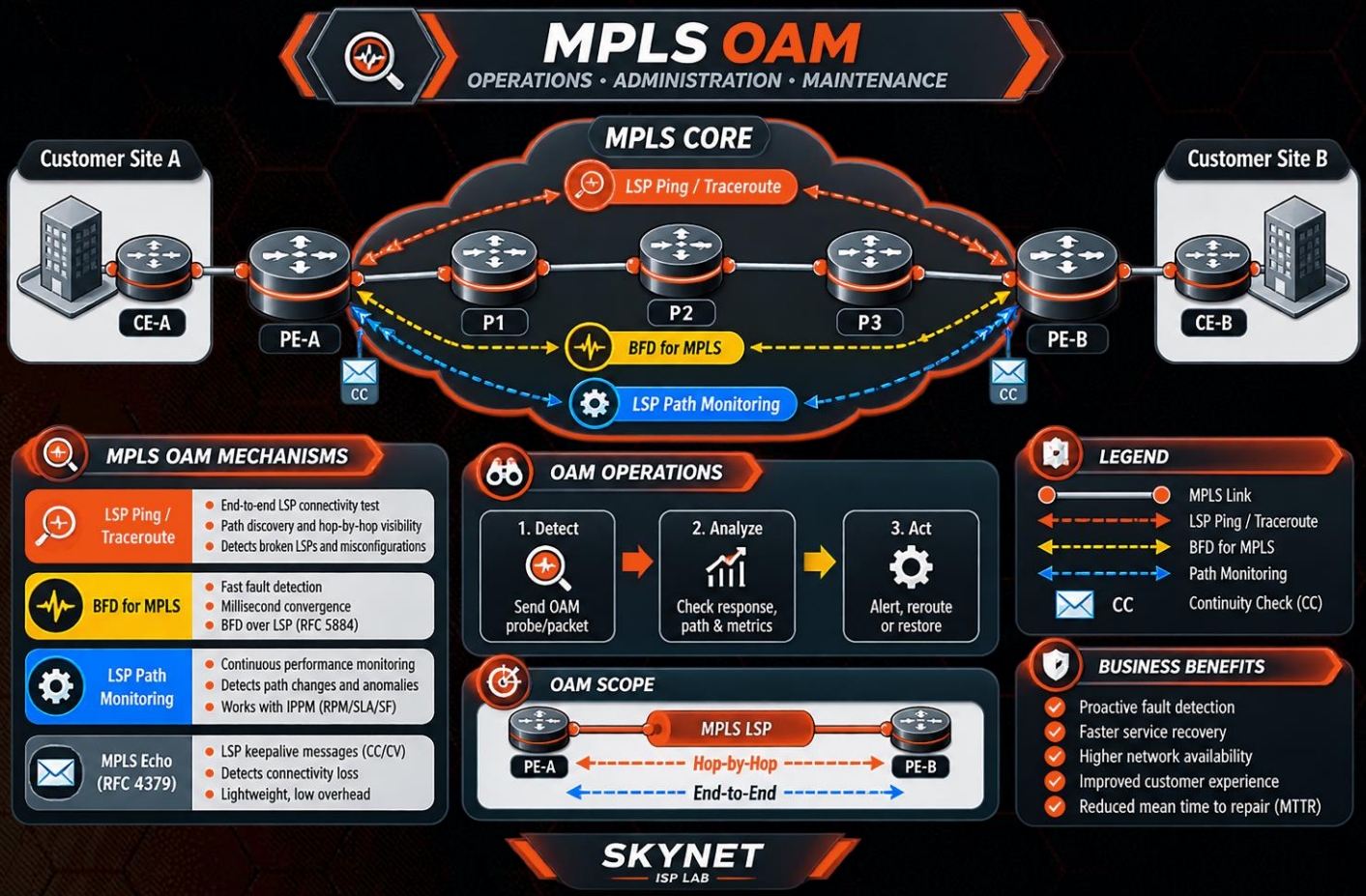
INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	PW-PC-01 Source Address	192.168.167.1/32	-
Loopback1	DC1-PW2-SRVR-01	192.168.171.1/32	FD00:2026:7:10::101/128
Loopback2	DC1-PW2-SRVR-02	192.168.171.2/32	FD00:2026:7:10::102/128
Loopback3	DC1-PW2-SRVR-03	192.168.171.3/32	FD00:2026:7:10::103/128
Loopback4	DC1-PW2-SRVR-04	192.168.171.4/32	FD00:2026:7:10::104/128
Loopback5	DC1-PW2-SRVR-05	192.168.171.5/32	FD00:2026:7:10::105/128
Ethernet0/0	To-SKNT-PE04-vMX01-PW	192.168.169.1/24	FD00:2026:7:10:1::1/127

28.27. PC2-PW2-JUNOS CONFIGURATION INVENTORY

ROLE: Pseudowire customer edge or endpoint
PLATFORM: Cisco IOS/IOS-XE/IOL | VERSION: 15.4
AUTONOMOUS SYSTEM(S): -
DETECTED PROTOCOLS / SERVICES: OSPFv2, OSPFv3, CoPP

INTERFACE	DESCRIPTION / PEER	IPV4 ADDRESSING	IPV6 ADDRESSING
Loopback0	PW-PC-02 Source Address	192.168.167.2/32	-
Loopback1	DC2-PW2-SRVR-01	192.168.172.1/32	FD00:2026:7:10::201/128
Loopback2	DC2-PW2-SRVR-02	192.168.172.2/32	FD00:2026:7:10::202/128
Loopback3	DC2-PW2-SRVR-03	192.168.172.3/32	FD00:2026:7:10::203/128
Loopback4	DC2-PW2-SRVR-04	192.168.172.4/32	FD00:2026:7:10::204/128
Loopback5	DC2-PW2-SRVR-05	192.168.172.5/32	FD00:2026:7:10::205/128
Ethernet0/0	To-SKNT-PE05-vMX02-PW	192.168.169.2/24	FD00:2026:7:10:1::2/127

29. MPLS OPERATIONS, ADMINISTRATION AND MAINTENANCE (OAM)



VALIDATION DATE: 02-AUG-2026. THIS SECTION DOCUMENTS THE MPLS OAM IMPLEMENTATION FOR THE EXISTING SKYNET AS300 MPLS TRANSPORT DOMAIN. THE DESIGN USES ONLY NATIVE ROUTER CAPABILITIES AND DOES NOT ADD MONITORING SERVERS, COLLECTORS, DASHBOARDS, TELEMETRY PLATFORMS, OR ADDITIONAL EVE-NG NODES.

29.1. OBJECTIVE AND IMPLEMENTATION SCOPE

THE OBJECTIVE IS TO VERIFY THAT AN LDP FORWARDING EQUIVALENCE CLASS (FEC) HAS A VALID LABEL-SWITCHED PATH, THAT TRANSIT LABEL OPERATIONS ARE CORRECT, AND THAT THE REMOTE ROUTER CAN TERMINATE AND RESPOND TO MPLS ECHO REQUESTS. THE IMPLEMENTATION COVERS THE MPLS TRANSPORT LOOPBACKS AND DOES NOT EXTEND OAM INTO CUSTOMER CE DEVICES OR NON-MPLS EXTERNAL ISP NODES.

SCOPE ITEM	IMPLEMENTATION
TRANSPORT DOMAIN	AS300 MPLS/LDP PE AND P ROUTERS
ACTIVE TEST TYPES	ON-DEMAND MPLS LSP PING AND MPLS LSP TRACEROUTE
PERSISTENT CHANGE	MPLS OAM ENABLED ONLY ON SKNT-PE08-XRV
OTHER MPLS NODES	EXISTING NATIVE OAM COMMANDS AND RESPONDER BEHAVIOR; NO NEW PERSISTENT CONFIGURATION
EXCLUDED INFRASTRUCTURE	NO NEW DEVICES, SERVERS, PROBES, COLLECTORS, OR EXTERNAL MONITORING TOOLS

29.2. WHY ONLY IOS-XRV REQUIRED NEW CONFIGURATION?

MPLS OAM INITIATOR, RESPONDER, AND FORWARDING FUNCTIONS ARE PLATFORM-SPECIFIC. CISCO IOSV AND JUNIPER VMX EXPOSE MANUAL LSP PING AND TRACEROUTE THROUGH THEIR EXISTING MPLS/LDP SOFTWARE. IOS XR USES A GLOBAL LSP VERIFICATION SUBSYSTEM THAT MUST BE EXPLICITLY ENABLED WITH MPLS OAM. THE AGGREGATION IOL L2/L3 IMAGE CAN FORWARD LABELED PACKETS AND RESPOND TO MPLS ECHO REQUESTS, BUT ITS SOFTWARE IMAGE CANNOT ORIGINATE LSP PING.

PLATFORM	OAM BEHAVIOR
CISCO IOS-XRV - SKNT-PE08-XRV	REQUIRES GLOBAL MPLS OAM; SUPPORTS INITIATOR AND RESPONDER ROLES
CISCO IOSV/IOL ROUTERS	NO EXTRA GLOBAL OAM CONFIGURATION; MANUAL PING MPLS IPV4 IS AVAILABLE WHEN INCLUDED IN THE IMAGE
JUNIPER VMX	NO EXTRA CONFIGURATION FOR MANUAL PING MPLS LDP AND TRACEROUTE MPLS LDP
SKNT-AGG.SW IOL L2/L3	MPLS FORWARDER AND ECHO RESPONDER; LSP PING INITIATION IS NOT SUPPORTED BY THE IMAGE
CUSTOMER AND EXTERNAL NODES	OUTSIDE THE PROVIDER MPLS TRANSPORT OAM SCOPE

29.3. SKNT-PE08-XRV PERSISTENT MPLS OAM CONFIGURATION

THE FOLLOWING IOS XR CONFIGURATION ACTIVATES THE MPLS LSP VERIFICATION SERVER. NO PERIODIC PROBE, SCHEDULER, OR BACKGROUND TEST IS CONFIGURED.

```
configure
 mpls oam
commit
end
```

THE CONFIGURATION WAS VERIFIED WITH SHOW RUNNING-CONFIG MPLS OAM. THE OAM CLIENT REGISTRATION DISPLAY SHOWED L2VPN_MGR, MPLS_LDP, BFD, BGP, AND TE_CONTROL REGISTERED WITH THE LSP VERIFICATION SERVER. GLOBAL COLLABORATORS IP ARM, IM, NETIO, AND RIB REPORTED UP.

29.4. STANDARD OPERATIONAL COMMANDS

PLATFORM	ON-DEMAND COMMANDS
IOS XR	PING MPLS IPV4 <FEC>/32; TRACEROUTE MPLS IPV4 <FEC>/32; SHOW MPLS OAM CLIENT; SHOW MPLS OAM COUNTERS GLOBAL; SHOW MPLS OAM COUNTERS PACKET
CISCO IOSV/IOL	PING MPLS IPV4 <FEC>/32; TRACEROUTE MPLS IPV4 <FEC>/32; SHOW MPLS LDP NEIGHBOR; SHOW MPLS FORWARDING-TABLE
JUNIPER VMX	PING MPLS LDP <FEC>/32; TRACEROUTE MPLS LDP <FEC>/32; SHOW LDP SESSION; SHOW ROUTE TABLE MPLS.0
AGG IOL LIMITATION	USE SHOW IP ROUTE, SHOW MPLS LDP BINDINGS, AND SHOW MPLS FORWARDING-TABLE; DO NOT EXPECT THE IMAGE TO ORIGINATE LSP PING

29.5. PE08-XRV OUTBOUND LSP PING VALIDATION

SKNT-PE08-XRV ORIGINATED MPLS ECHO REQUESTS TO EVERY REMOTE MPLS TRANSPORT LOOPBACK. ALL FORTY-FIVE PROBES RECEIVED VALID REPLIES. NO FEC MISMATCH, NO LABEL ENTRY, PREMATURE TERMINATION, MALFORMED REQUEST, OR TIMEOUT RETURN CODES WERE OBSERVED.

REMOTE FEC	DEVICE	RESULT
192.168.1.1/32	SKNT-PE01-BORDER	5/5 SUCCESS - AVG 14 MS
192.168.1.2/32	SKNT-P01-ISP	5/5 SUCCESS - AVG 8 MS
192.168.1.3/32	SKNT-PE04-VMX01	5/5 SUCCESS - AVG 14 MS
192.168.1.4/32	SKNT-AGG.SW	5/5 SUCCESS - OAM TERMINATION BASELINE 108-136 MS
192.168.1.5/32	SKNT-PE05-VMX02	5/5 SUCCESS - AVG 12 MS
192.168.1.100/32	SKNT-PE06-POP01	5/5 SUCCESS - AVG 12 MS
192.168.1.200/32	SKNT-PE07-POP02	5/5 SUCCESS - AVG 12 MS
192.168.1.250/32	SKNT-PE02-VIOS	5/5 SUCCESS - AVG 10 MS
192.168.1.251/32	SKNT-PE03-VMX03	5/5 SUCCESS - AVG 18 MS

29.6. BIDIRECTIONAL AND MULTIVENDOR VALIDATION

REVERSE-DIRECTION TESTS CONFIRMED THAT PE08 ALSO OPERATES AS AN MPLS ECHO RESPONDER FOR CISCO IOSV AND JUNIPER VMX INITIATORS.

TEST	COMMAND / DIRECTION	RESULT
CISCO IOSV TO IOS-XRV	SKNT-PE01-BORDER -> 192.168.1.101/32	5/5 SUCCESS; MIN/AVG/MAX 8/13/32 MS
JUNIPER VMX TO IOS-XRV	SKNT-PE04-VMX01 -> 192.168.1.101/32	5 TRANSMITTED; 5 RECEIVED; 0% LOSS
FINAL RESULT	JUNOS LDP OAM THROUGH CISCO IOSV/IOL TO IOS XR	PASSED

29.7. JUNOS MPLS LSP TRACEROUTE EVIDENCE

THE JUNOS LSP TRACEROUTE FROM SKNT-PE04-VMX01 TO PE08 IDENTIFIED THE COMPLETE LABEL-SWITCHED PATH AND THE PENULTIMATE-HOP POP OPERATION.

TTL	LABEL	ADDRESS / NODE	STATUS AND INTERPRETATION
1	11007	10.0.0.5 - SKNT-P01-ISP	SUCCESS - FIRST CISCO TRANSIT LSR
2	4010	10.0.0.30 - SKNT-AGG.SW	SUCCESS - AGG LABEL FOR THE PE08 LOOPBACK FEC
3	3	10.34.0.2 - SKNT-PE08-XRV	EGRESS - IMPLICIT-NULL / PENULTIMATE-HOP POP COMPLETED

LABEL 3 IS THE RESERVED IMPLICIT-NULL VALUE. SKNT-AGG.SW REMOVES THE TRANSPORT LABEL BEFORE DELIVERING THE PACKET TO PE08. THE JUNOS PROTOCOL FIELD DISPLAYED UNKNOWN FOR SOME CISCO REPLIES, BUT THE SUCCESS AND EGRESS PROBE STATES CONFIRM CORRECT MPLS FORWARDING AND FEC TERMINATION.

29.8. SKNT-AGG.SW FORWARDING AND IMAGE LIMITATION

AGG.SW WAS VALIDATED AS A CORRECT MPLS FORWARDER AND OAM RESPONDER. ITS OSPF ROUTE TO PE08 USES NEXT HOP 10.34.0.2 ON ETHERNET1/1. LDP ASSIGNS LOCAL LABEL 4010 FOR 192.168.1.101/32, WHILE PE08 ADVERTISES IMPLICIT-NULL. THE LFIB ENTRY POPS LABEL 4010 AND FORWARDS THE PACKET TO 10.34.0.2.

CHECK	OBSERVED RESULT
OSPF ROUTE	192.168.1.101/32 VIA 10.34.0.2 ON ETHERNET1/1, METRIC 31
LDP SESSION	PEER 192.168.1.101:0 IN OPER STATE
LABEL BINDING	LOCAL LABEL 4010; REMOTE PE08 LABEL IMPLICIT-NULL
LFIB	4010 POP LABEL -> ETHERNET1/1 -> 10.34.0.2; BYTE COUNTER INCREASING
INITIATOR CAPABILITY	PING MPLS RETURNS: % LSP PING NOT SUPPORTED ON THIS SYSTEM

THE AGG LSP PING RESPONSE TIME WAS HIGHER THAN THE OTHER FEC RESULTS, BUT TWO NORMAL ICMP TESTS EACH RETURNED 20/20 REPLIES WITH 1/2/9 MS MINIMUM/AVERAGE/MAXIMUM DELAY. AGG CPU UTILIZATION WAS EFFECTIVELY ZERO AT THE TIME OF VALIDATION. THIS PROVES THE HIGH LSP ECHO ROUND-TRIP VALUE IS A VIRTUAL-IOL CONTROL-PLANE OAM TERMINATION CHARACTERISTIC, NOT A TRANSPORT FORWARDING DEFECT OR LINK-LATENCY PROBLEM.

29.9. RETURN-CODE INTERPRETATION

CODE	MEANING	OPERATIONAL RESPONSE
!	SUCCESS	THE FEC AND LABEL-SWITCHED PATH VALIDATED
.	TIMEOUT	CHECK CONTROL PLANE, RETURN PATH, FILTERING, AND RESPONDER SUPPORT
F / f	NO FEC MAPPING OR FEC MISMATCH	COMPARE IGP ROUTE, LDP BINDINGS, AND LFIB PROGRAMMING
N	NO LABEL ENTRY	VERIFY THE FEC EXISTS IN THE LOCAL AND REMOTE LABEL TABLES
P / p	LABEL PROTOCOL OR PREMATURE LSP TERMINATION	CHECK MPLS ENABLEMENT, PHP, AND THE TRANSIT HOP
M / m	MALFORMED REQUEST OR UNSUPPORTED TLV	CHECK PLATFORM INTEROPERABILITY AND SOFTWARE SUPPORT
R / X / x	TRANSIT OR UNEXPECTED RETURN CONDITION	USE MPLS TRACEROUTE TO LOCALIZE THE AFFECTED HOP

29.10. OPERATIONAL ACCEPTANCE AND FINAL STATUS

ACCEPTANCE ITEM	FINAL STATUS
IOS XR OAM ACTIVATION	PASSED
PE08 TO ALL MPLS LOOPBACKS	PASSED - 45/45 REPLIES
IOSV PE01 TO IOS-XRV PE08	PASSED
JUNOS PE04 TO IOS-XRV PE08	PASSED
JUNOS MPLS LSP TRACEROUTE	PASSED
P01 AND AGG LABEL SWITCHING	PASSED
PENULTIMATE-HOP POPPING	PASSED
MULTIVENDOR MPLS OAM INTEROPERABILITY	PASSED

29.11. ROUTINE USE AND TROUBLESHOOTING PROCEDURE

• USE SKNT-PE08-XRV AS THE PRIMARY CENTRAL OAM PROBE WHEN A BROAD CORE CHECK IS REQUIRED. RUN MANUAL TESTS ONLY WHEN VALIDATING A CHANGE, INVESTIGATING A FAULT, OR RECORDING AN ACCEPTANCE RESULT.

• BEGIN WITH THE IGP ROUTE, THEN VERIFY THE LDP BINDING AND THE MPLS FORWARDING ENTRY BEFORE RUNNING LSP PING. THIS SEPARATES ROUTING, LABEL DISTRIBUTION, AND DATA-PLANE PROBLEMS.

• RUN MPLS TRACEROUTE ONLY WHEN LSP PING FAILS, RETURNS A NON-SUCCESS CODE, OR WHEN THE EXACT LABEL-SWITCHED PATH MUST BE DOCUMENTED.

• MPLS ECHO IS CONTROL-PLANE OAM AND VIRTUAL IMAGE SCHEDULING CAN INFLATE THE RESPONSE TIME.

• DO NOT ALTER WORKING LDP OR MPLS CONFIGURATION TO TRY TO ENABLE A COMMAND THAT THE IMAGE DOES NOT IMPLEMENT.

29.12. CONCLUSION

THE SKYNET MULTIVENDOR MPLS/LDP BACKBONE NOW HAS A DOCUMENTED AND VERIFIED OAM PROCEDURE. IOS XR OAM IS EXPLICITLY ENABLED ON SKNT-PE08-XRV, WHILE CISCO IOSV AND JUNIPER VMX USE THEIR NATIVE MANUAL OAM COMMANDS WITHOUT ADDITIONAL PERSISTENT CONFIGURATION. SKNT-AGG.SW IS A VERIFIED LABEL FORWARDER AND MPLS ECHO RESPONDER WITH AN IMAGE-SPECIFIC INITIATOR LIMITATION. END-TO-END LSP PING, REVERSE-DIRECTION TESTING, AND JUNOS LSP TRACEROUTE ALL PASSED, CONFIRMING CORRECT MULTIVENDOR FEC MAPPING, LABEL SWITCHING, AND PENULTIMATE-HOP POPPING.

30. DIRECT CE01-ISP-HE EBGP MIGRATION

VALIDATION DATE: 10-AUG-2026. THIS SECTION RECORDS THE COMPLETED MIGRATION OF THE EXISTING 10.207.0.0/30 CE01-L3VPN-01 ↔ ISP-HE LINK FROM STATIC-ROUTE-ONLY USE TO A STRICTLY FILTERED DIRECT EBGP SESSION BETWEEN AS301 AND AS400. THE CHANGE WAS DESIGNED TO PRESERVE THE EXISTING SKYNET THREE-ISP FALLBACK AND TO AVOID MODIFYING PE01/PE02 PROVIDER-SELECTION LOGIC.

30.1. OBJECTIVE, SCOPE AND NON-GOALS

- CE01-L3VPN-01 (AS301) AND ISP-HE (AS400) WERE THE ONLY ROUTERS CONFIGURED FOR THE NEW DIRECT EBGP SESSION.
- THE EXISTING CE01 ↔ SKNT-PE01-BORDER AS301/AS300 SESSION AND CE01 STATIC DEFAULT VIA 192.168.111.1 WERE PRESERVED.
- THE EXISTING TATA, VERIZON AND HE PROVIDER-FALLBACK POLICIES INSIDE SKYNET AS300 WERE NOT REDESIGNED.
- BFD WAS NOT INTRODUCED. THE EXISTING NQA OBJECT ON ISP-HE WAS RETAINED BECAUSE OTHER TRACKED STATIC ROUTES STILL DEPEND ON IT.

30.2. FINAL PEERING AND POLICY DESIGN

ITEM	CE01-L3VPN-01	ISP-HE	FINAL BEHAVIOR
AS / LINK	AS301 / 10.207.0.1	AS400 / 10.207.0.2	DIRECT EBGP OVER 10.207.0.0/30
HE PREFIXES	IMPORT ONLY FIVE 172.19.X.X /32S	EXPORT ONLY FIVE 172.19.X.X /32S	NO DEFAULT OR UNRELATED ROUTES
CE PREFIX	EXPORT ONLY 192.168.245.0/30	IMPORT ONLY 192.168.245.0/30	NO TRANSIT LEAKAGE
PREFERENCE	LOCAL-PREF 50 ON DIRECT HE PATH	AS-PATH PREPEND 400 400 ADDITIVE	SKYNET AS300 PATH REMAINS PRIMARY AT LP100
ORIGIN TAG	COMMUNITY 300:400 ADDITIVE	-	DIRECT HE ROUTES REMAIN IDENTIFIABLE IN CE RIB/POLICY
MAX PREFIX	10, WARNING-ONLY	STRICT PREFIX-LIST / ROUTE-POLICY	BOUNDED DIRECT-PEER ROUTE SCALE

30.3. CE01-L3VPN-01 IMPLEMENTED CONFIGURATION

THE EXISTING RM-CE01-TO-PE OUTBOUND POLICY WAS FIRST ATTACHED TO THE AS300 PEER. VALIDATION CONFIRMED THAT CE01 CONTINUED TO ADVERTISE ONLY 192.168.245.0/30 TOWARD SKNT-PE01-BORDER, PROVIDING AN ANTI-TRANSIT BOUNDARY BEFORE THE DIRECT HE PEER WAS ACTIVATED.

```
ip prefix-list PL-CE01-FROM-HE-DIRECT seq 10 permit 172.19.10.1/32
ip prefix-list PL-CE01-FROM-HE-DIRECT seq 20 permit 172.19.20.1/32
ip prefix-list PL-CE01-FROM-HE-DIRECT seq 30 permit 172.19.30.1/32
ip prefix-list PL-CE01-FROM-HE-DIRECT seq 40 permit 172.19.40.1/32
ip prefix-list PL-CE01-FROM-HE-DIRECT seq 50 permit 172.19.50.1/32
ip prefix-list PL-CE01-TO-HE-DIRECT seq 10 permit 192.168.245.0/30
!
route-map RM-CE01-FROM-HE-DIRECT permit 10
 match ip address prefix-list PL-CE01-FROM-HE-DIRECT
 set local-preference 50
 set community 300:400 additive
route-map RM-CE01-FROM-HE-DIRECT deny 100
!
route-map RM-CE01-TO-HE-DIRECT permit 10
 match ip address prefix-list PL-CE01-TO-HE-DIRECT
route-map RM-CE01-TO-HE-DIRECT deny 100
!
router bgp 301
 address-family ipv4 vrf INTER-AS-L3VPN
  neighbor 10.207.0.2 remote-as 400
  neighbor 10.207.0.2 description To-ISP-HE-DIRECT-AS400
  neighbor 10.207.0.2 activate
  neighbor 10.207.0.2 route-map RM-CE01-FROM-HE-DIRECT in
  neighbor 10.207.0.2 route-map RM-CE01-TO-HE-DIRECT out
  neighbor 10.207.0.2 maximum-prefix 10 warning-only
  neighbor 192.168.111.1 route-map RM-CE01-TO-PE out
```

30.4. ISP-HE IMPLEMENTED CONFIGURATION

THE HUAWEI PEER WAS BUILT WITH PEER IGNORE AS A SAFETY LOCK, COMMITTED AND AUDITED WITH BOTH ROUTE-POLICY BINDINGS PRESENT, THEN ACTIVATED BY REMOVING PEER IGNORE. THIS PREVENTED ROUTE EXCHANGE UNTIL THE FILTERS WERE VERIFIED.

```
ip ip-prefix CE01-AS301-IN-V4 index 10 permit 192.168.245.0 30
ip ip-prefix HE-AS400-TO-CE01-V4 index 10 permit 172.19.10.1 32
ip ip-prefix HE-AS400-TO-CE01-V4 index 20 permit 172.19.20.1 32
ip ip-prefix HE-AS400-TO-CE01-V4 index 30 permit 172.19.30.1 32
ip ip-prefix HE-AS400-TO-CE01-V4 index 40 permit 172.19.40.1 32
ip ip-prefix HE-AS400-TO-CE01-V4 index 50 permit 172.19.50.1 32
#
route-policy IMPORT-FROM-CE01-AS301 permit node 10
 if-match ip-prefix CE01-AS301-IN-V4
route-policy IMPORT-FROM-CE01-AS301 deny node 100
#
route-policy EXPORT-TO-CE01-AS301 permit node 10
 if-match ip-prefix HE-AS400-TO-CE01-V4
 apply as-path 400 400 additive
route-policy EXPORT-TO-CE01-AS301 deny node 100
#
bgp 400
 peer 10.207.0.1 as-number 301
 peer 10.207.0.1 description To-CE01-L3VPN-01-AS301
 ipv4-family unicast
  peer 10.207.0.1 enable
  peer 10.207.0.1 route-policy IMPORT-FROM-CE01-AS301 import
  peer 10.207.0.1 route-policy EXPORT-TO-CE01-AS301 export
```

30.5. STATIC-TO-BGP RETURN-PATH CUTOVER

AFTER THE AS301/AS400 SESSION WAS ESTABLISHED, ISP-HE WAS VERIFIED TO RECEIVE 192.168.245.0/30 FROM CE01 BEFORE ANY FORWARDING CHANGE. THE LEGACY TRACKED STATIC FOR ONLY THAT CUSTOMER PREFIX WAS THEN REMOVED. THE OTHER TRACKED STATIC ROUTES VIA 10.207.0.1, INCLUDING 192.168.98.0/24 AND 192.168.111.0/24, AND THE NQA TEST INSTANCE SKYNET CE01L3VPNLINK WERE RETAINED.

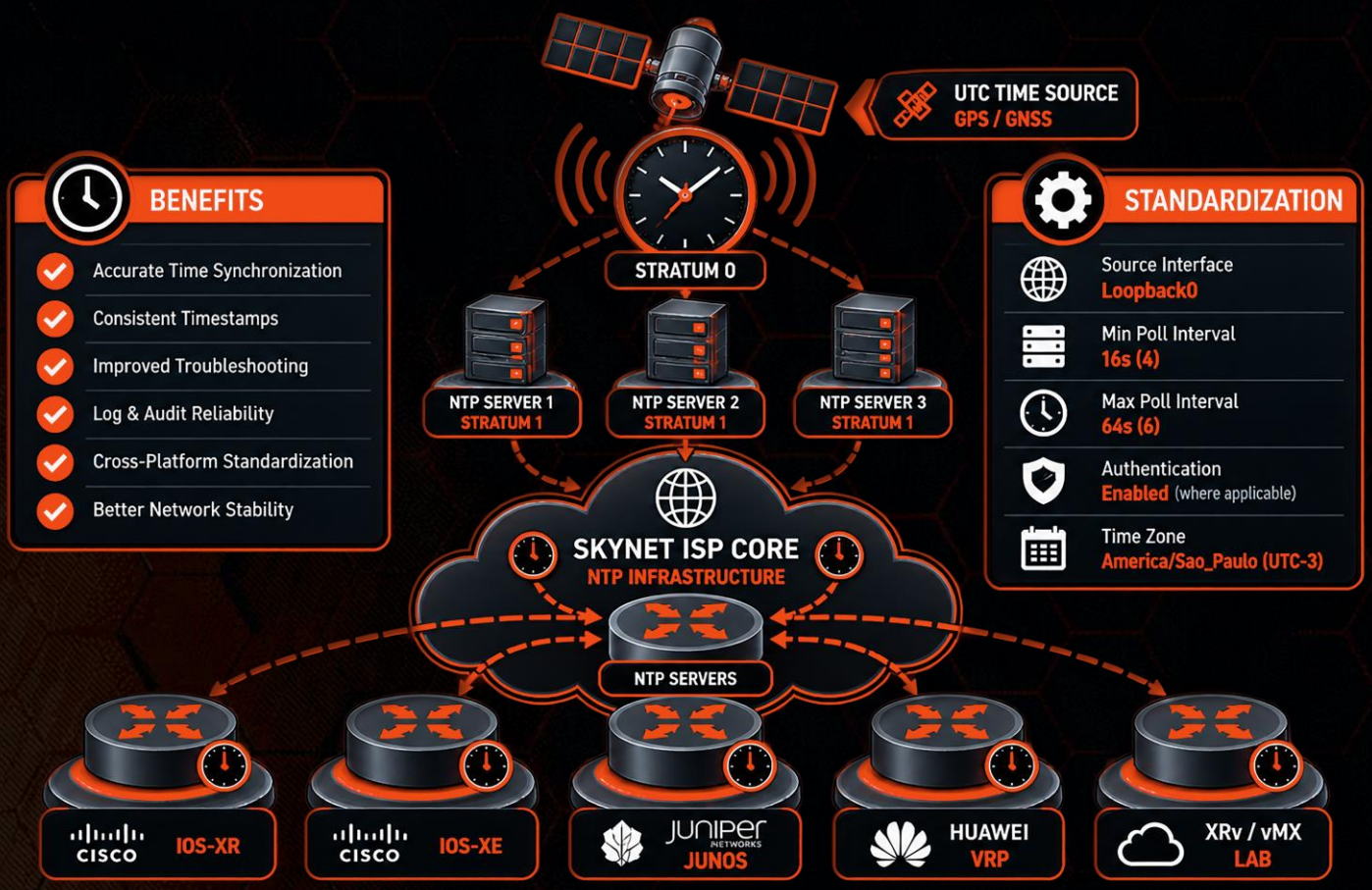
```
undo ip route-static 192.168.245.0 255.255.255.252 10.207.0.1
commit
```

30.6. VALIDATION EVIDENCE AND FINAL STATE

VALIDATION ITEM	ACCEPTED RESULT
DIRECT BGP SESSION	CE01 10.207.0.2 / AS400 AND ISP-HE 10.207.0.1 / AS301 ESTABLISHED.
HE → CE01 ROUTES	EXACTLY FIVE 172.19.10.1/32 THROUGH 172.19.50.1/32 ROUTES; DIRECT COPY ARRIVES AT CE01 WITH LP50, COMMUNITY 300:400 AND AS-PATH 400 400 400.
SKYNET PRIMARY	THE SAME FIVE PREFIXES VIA 192.168.111.1 / AS300 REMAIN BEST AT LP100.
CE01 → HE EXPORT	EXACTLY 192.168.245.0/30; NO SKYNET OR DEFAULT ROUTE LEAKAGE.
CE01 → PE01 EXPORT	EXACTLY 192.168.245.0/30; ANTI-TRANSIT POLICY CONFIRMED.
HE RETURN ROUTE	192.168.245.0/30 INSTALLED AS EBGP, PREFERENCE 255, NEXT HOP 10.207.0.1.
DATA PLANE	HE 10.207.0.2 → CE01 192.168.245.1: 5/5, 0% LOSS. CE01 → HE DIRECT LINK ALSO 5/5.
FAILURE TESTS	TATA, VERIZON, AND COMBINED TATA+VERIZON OUTAGE TESTS WERE COMPLETED SUCCESSFULLY; THE EXISTING SKYNET FAILURE LOGIC REMAINED OPERATIONAL.

FINAL ACCEPTANCE: THE MIGRATION IS COMPLETE WITH NO ROLLBACK REQUIRED. THE DIRECT CE01-HE EBGP PATH IS AVAILABLE AS A CONTROLLED BACKUP FOR THE HE TEST PREFIXES, WHILE THE EXISTING SKYNET AS300 PATH REMAINS PRIMARY. CE01’S STATIC DEFAULT VIA 192.168.111.1 WAS NOT CHANGED, SO THIS MIGRATION DOES NOT CONVERT THE DIRECT HE LINK INTO A GENERIC INTERNET DEFAULT-ROUTE FAILOVER.

31. NTP FINE TUNING AND STANDARDIZATION



ALIDATION DATE: 10-AUG-2026. NTP WAS FINE-TUNED TO REDUCE SINGLE-SOURCE DEPENDENCY AND IMPROVE TIMESTAMP CONSISTENCY FOR ROUTING EVENTS, SYSLOG, CHANGE VALIDATION AND MULTIVENDOR TROUBLESHOOTING. THE ADJUSTMENT IS OPERATIONAL ONLY AND DOES NOT CHANGE BGP, MPLS OR CUSTOMER FORWARDING POLICY.

31.1. STANDARDIZED REFERENCE SET

ROLE	NTP SERVER	USAGE
PREFERRED	200.160.7.186	PRIMARY REFERENCE ON UPDATED CISCO / IOS XR NODES.
BACKUP 1	186.192.158.147	REDUNDANT PUBLIC TIME SOURCE.
BACKUP 2	200.20.186.76	SECOND REDUNDANT PUBLIC TIME SOURCE.

31.2. CISCO IOS / IOS-XE / IOL FINAL PATTERN

```
ntp source Loopback0
ntp update-calendar
ntp server 200.160.7.186 prefer
ntp server 186.192.158.147
ntp server 200.20.186.76
```

THE UPDATED BASELINE SHOWS THIS THREE-SERVER MODEL ON SKYNET CISCO NODES SUCH AS SKNT-PE01-BORDER AND SKNT-PE02-VIOS. ISP-TATA, ISP-VERIZON AND ISP-SPARKLE ALSO USE THE SAME PREFERRED/REDUNDANT SERVER ORDER. LOOPBACK0 IS USED AS THE NTP SOURCE ON PROVIDER NODES WHERE CONFIGURED, PROVIDING A STABLE MANAGEMENT/CONTROL-PLANE IDENTITY.

31.3. CISCO IOS XR FINAL PATTERN

```
ntp
 server 186.192.158.147
 server 200.20.186.76
 server 200.160.7.186 prefer
 source Loopback0
 update-calendar
```

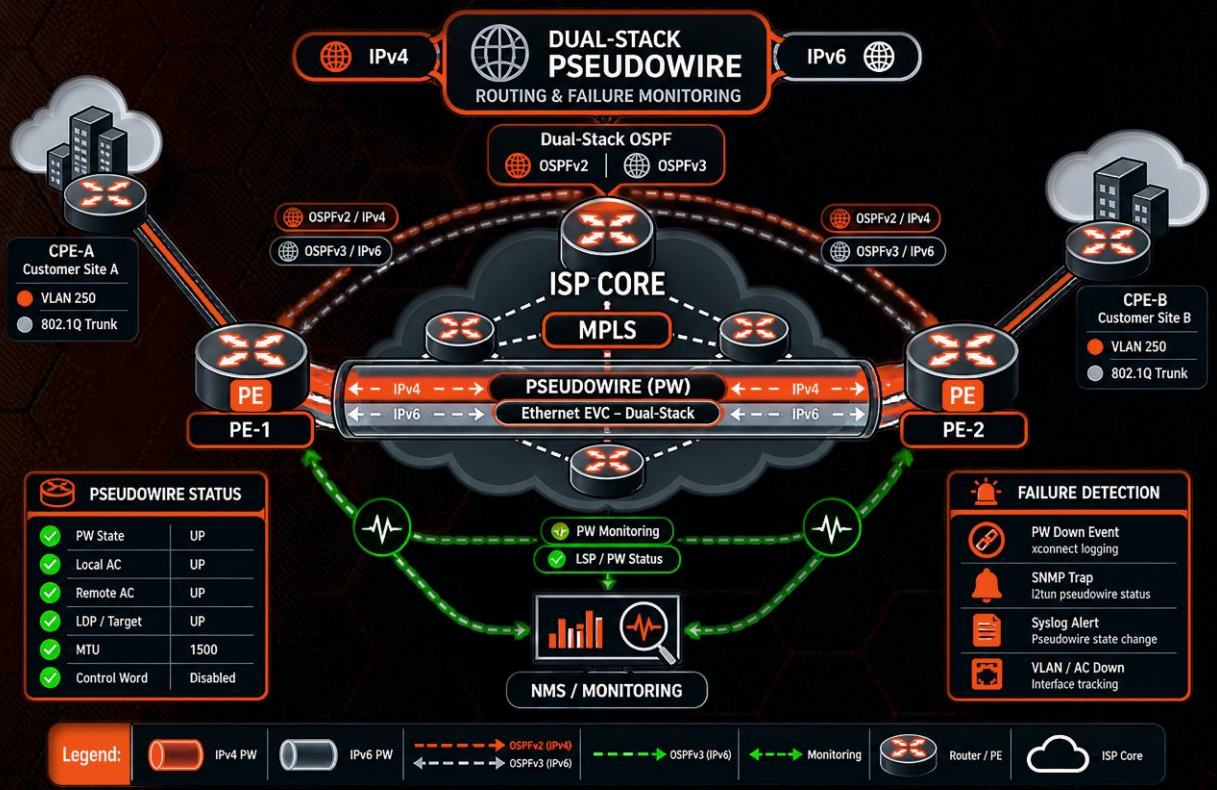
SKNT-PE08-XRV USES THE IOS XR NTP HIERARCHY WITH THE SAME THREE REFERENCES, THE SAME PREFERRED SERVER, LOOPBACK0 AS SOURCE, AND CALENDAR UPDATE. THIS KEEPS TIME-SELECTION INTENT CONSISTENT ACROSS IOS AND IOS XR WHILE RESPECTING PLATFORM SYNTAX.

31.4. OPERATIONAL OUTCOME

- THREE INDEPENDENT TIME REFERENCES ARE AVAILABLE INSTEAD OF A SINGLE PUBLIC SOURCE.
- THE PREFERRED SERVER IS CONSISTENT WHERE THE FINE-TUNED BASELINE WAS APPLIED.
- LOOPBACK-SOURCED NTP IS USED ON PROVIDER ROUTERS WHERE SUPPORTED/CONFIGURED, IMPROVING SOURCE STABILITY.
- CONSISTENT CLOCKS IMPROVE BGP/OSPF/MPLS EVENT CORRELATION, SYSLOG REVIEW, CHANGE-WINDOW EVIDENCE AND FAILURE-TEST ANALYSIS.

FINAL ACCEPTANCE: NTP FINE TUNING IS DOCUMENTED AS A SEPARATE OPERATIONAL STANDARDIZATION ITEM. IT DOES NOT ALTER THE THREE-ISP FALLBACK, CE01-HE EBGP POLICY, OR CUSTOMER DATA-PLANE ROUTING.

32. DUAL-STACK PSEUDOWIRE ROUTING AND FAILURE MONITORING



VALIDATION DATE: 15-AUG-2026. VC-ID 1 BETWEEN SKNT-PE06-POP01 AND SKNT-PE07-POP02 WAS ENHANCED AT THE CUSTOMER EDGE WITH DUAL-STACK OSPF ROUTING, NEW IPV4/IPV6 SERVICE LOOPBACKS, VERIFIED VCCV/MPLS PSEUDOWIRE OAM, AND EXPLICIT PSEUDOWIRE FAILURE SYSLOG MONITORING. THE WORK PRESERVES THE EXISTING MPLS/LDP TRANSPORT AND QOS DESIGN.

32.1. SERVICE PATH AND ADDRESSING

THE CISCO VPWS SERVICE REMAINS CE-01-PW01 -> SKNT-PE06-POP01 -> MPLS/LDP VC-ID 1 -> SKNT-PE07-POP02 -> CE-02-PW01. THE EXISTING IPV4 TRANSIT 10.10.20.0/30 AND IPV6 TRANSIT 2001:DB8:20::/64 ARE USED AS THE DUAL-STACK OSPF ADJACENCY NETWORKS.

ENDPOINT / FUNCTION	IPV4	IPV6
CE-01-PW01 E1/1	10.10.20.1/30	2001:DB8:20::1/64
CE-02-PW01 E1/1	10.10.20.2/30	2001:DB8:20::2/64
CE-01 LOOPBACK10-50	10.201.1.1-10.201.1.5/32	2001:DB8:201:1::1-::5/128
CE-02 LOOPBACK10-50	10.202.1.1-10.202.1.5/32	2001:DB8:202:1::1-::5/128
CE-01 CUSTOMER LAN	10.40.40.0/24	2001:DB8:40::/64
CE-02 CUSTOMER LAN	10.50.50.0/24	2001:DB8:50::/64

32.2. DUAL-STACK OSPF IMPLEMENTATION OVER VC-ID 1

OSPFV2 PROCESS 1 AND OSPFV3 PROCESS 1 FORM AREA 0 POINT-TO-POINT ADJACENCIES ACROSS ETHERNET1/1 ON THE TWO CE ROUTERS. PASSIVE-INTERFACE DEFAULT PREVENTS UNWANTED NEIGHBOR FORMATION ON CUSTOMER LANS AND LOOPBACKS, WHILE ETHERNET1/1 IS EXPLICITLY NON-PASSIVE. THE FORMER IPV4 STATIC ROUTES WERE REMOVED AFTER OSPFV2 LEARNED THE REMOTE CUSTOMER LAN, REMOTE CE LOOPBACK0, AND ALL NEW SERVICE LOOPBACKS. OSPFV3 CARRIES THE CORRESPONDING IPV6 CUSTOMER AND LOOPBACK PREFIXES.

32.3. VCCV / MPLS PSEUDOWIRE OAM VALIDATION

VCCV WAS ALREADY NEGOTIATED BY THE EXISTING AToM PSEUDOWIRE. BOTH ENDS ADVERTISE CONTROL-CHANNEL TYPES CW [1] AND RA [2] WITH LSPV [2] CONNECTIVITY VERIFICATION. THE CONTROL WORD IS ACTIVE THROUGH AUTOSENSE. BIDIRECTIONAL MPLS PSEUDOWIRE PING COMPLETED AT 100%, SO NO ADDITIONAL MPLS OAM CONFIGURATION WAS REQUIRED.

```
SKNT-PE06-POP01# ping mpls pseudowire 192.168.1.200 1
SKNT-PE07-POP02# ping mpls pseudowire 192.168.1.100 1

Expected healthy result: !!!!!  Success rate is 100 percent
```

32.4. PSEUDOWIRE FAILURE MONITORING

PSEUDOWIRE STATUS SIGNALING IS ENABLED/SUPPORTED ON BOTH PE NODES. XCONNECT STATUS LOGGING WAS ENABLED SO VC-ID 1 STATE CHANGES ARE WRITTEN TO SYSLOG. AN ADMINISTRATIVE SHUTDOWN OF PE06 GIGABITETHERNET0/5 CORRECTLY GENERATED LOCAL AC ERROR AND VC ADMIN-DOWN EVENTS. PE06 WITHDREW ITS LOCAL PW LABEL WHILE THE LDP PEER SESSION REMAINED UP, AND PE07 REPORTED THE REMOTE LABEL WITHDRAWAL. DURING THE OUTAGE, PW PING RETURNED QQQQQ BECAUSE NO OPERATIONAL PW FEC WAS AVAILABLE. AFTER RESTORING THE AC, VC-ID 1 RETURNED TO UP AUTOMATICALLY AND VCCV RETURNED TO 100%. THE PE06 LOCAL VC LABEL WAS DYNAMICALLY REALLOCATED FROM 6000 TO 6019, WHICH IS NORMAL MPLS LABEL BEHAVIOR.

32.5. PE06 / PE07 MONITORING CONFIGURATION

```
configure terminal
!
pseudowire-class <EXISTING-PW-CLASS>
 status
!
xconnect logging pseudowire status
!
snmp-server enable traps l2tun pseudowire status
!
end
write memory
```

ON PE06 THE EXISTING CLASS IS CE01-TO-CE02; ON PE07 IT IS CE02-TO-CE01. THE SNMP TRAP CAPABILITY IS ENABLED, BUT DELIVERY TO AN EXTERNAL NMS STILL REQUIRES A VALID SNMP-SERVER HOST DESTINATION.

32.6. FAILURE LOGGING EVIDENCE AND ACCEPTANCE

VALIDATION ITEM	ACCEPTED RESULT
DUAL-STACK OSPF	OSPFV2 AND OSPFV3 NEIGHBORS FULL BETWEEN CE-01-PW01 AND CE-02-PW01 OVER ETHERNET1/1 / VC-ID 1.
STATIC ROUTE REMOVAL	FORMER IPV4 STATIC CUSTOMER/REMOTE-CE ROUTES REMOVED; REMOTE PREFIXES LEARNED BY OSPFV2.
VCCV	BIDIRECTIONAL PING MPLS PSEUDOWIRE COMPLETED 5/5, 100%.
AC FAILURE DETECTION	PE06 VC-ID 1 ENTERED ADMIN DOWN; LOCAL PW LABEL WITHDRAWN WHILE THE LDP PEER SESSION REMAINED OPERATIONAL.
REMOTE PE REACTION	PE07 VC-ID 1 DOWN WITH LAST ERROR REMOTE PEER WITHDREW LABEL.
SYSLOG	%XCONNECT-5-PW_STATUS REPORTED VC STATE DOWN, LOC AC ERR AND VC STATE ADMIN DOWN.
RECOVERY	VC-ID 1 RETURNED TO UP AUTOMATICALLY; PE06 LOCAL LABEL REALLOCATED TO 6019; VCCV RETURNED TO 100%.

FINAL ACCEPTANCE: THE CISCO PSEUDOWIRE SERVICE NOW HAS DUAL-STACK DYNAMIC ROUTING AT THE CUSTOMER EDGE, DOCUMENTED SERVICE LOOPBACKS, VERIFIED VCCV OAM, EXPLICIT PW STATUS SYSLOG MONITORING, AND VALIDATED AC-FAILURE/RECOVERY BEHAVIOR WITHOUT CHANGING THE SINGLE-PATH MPLS/LDP TRANSPORT DESIGN.



FULL LAB DOWNLOAD LINK (OVF FORMAT). LAB IS READY AND FULLY OPERATIONAL.

ATTENTION* - DON'T FORGET TO READ THE "README.TXT" FOR VM ACCESS IP CHANGE, EVE-VM HAS IT'S IP ADDRESS STATICALLY CONFIGURED FOR MY HOME LAN, YOU HAVE TO CHANGE IT FOR GET ACCESS TO WEB GUI AND DEVICES.

https://tinyurl.com/Skynet-Lab

http://bit.ly/Skynet-Lab